

Urban and Regional Planning, M.Sc. Dual Degree
Master Thesis

**Transforming Urban Mobility in Bogotá:
Implementation and Integration of a Rail-based
Mass Transit System and the 15-Minute City.
A Study at the Local Planning Unit (UPL) Scale.**

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To God, to my family, and to my city.

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Santiago Silvera Malagón
Bogotá, 2026

Resumen

La movilidad urbana en Bogotá se ha caracterizado históricamente por elevados tiempos de viaje y una marcada segregación socioespacial, lo que limita el acceso equitativo a oportunidades urbanas, especialmente para la población periférica. Esta investigación evalúa el impacto potencial de la Primera Línea del Metro de Bogotá y de una eventual Red Férrea Regional, así como de estrategias de uso del suelo orientadas a la proximidad, sobre la accesibilidad urbana y la Calidad de Vida Social.

Se adopta un enfoque metodológico cuantitativo basado en *spatial big data*, mediante la simulación de siete escenarios progresivos que comparan una línea base (2023) con distintas combinaciones de expansión de infraestructura de transporte y redistribución de equipamientos. El análisis se apoya en el cálculo de matrices de accesibilidad acumulativa y en la estimación del Índice de Alta Calidad de Vida Social (HQSL).

Los resultados muestran que, si bien la infraestructura férrea reduce significativamente los tiempos de viaje y la brecha centro-periferia, su implementación aislada es insuficiente para garantizar la equidad territorial. El escenario con mejores resultados se alcanza únicamente a través de la sinergia entre una red férrea metropolitana y una política de distribución de equipamientos a escala de Unidad de Planeación Local. En consecuencia, se concluye que el modelo más eficiente para Bogotá no es una autarquía local estricta, sino una estructura híbrida de “15+30 minutos”, que combina proximidad peatonal para servicios básicos y de cuidado con accesibilidad rápida en transporte masivo para empleo y educación superior.

Palabras clave

Accesibilidad Urbana, Transporte público, 15-Minute City, Índice HQSL, Equidad Espacial, Bogotá.

Abstract

Urban mobility in Bogotá has historically been characterised by high commuting times and marked socio-spatial segregation, limiting equitable access to urban opportunities, particularly for the population in peripheral areas. This research assesses the potential impact of the Bogotá Metro Line 1 and a prospective Regional Rail Network (Regiotram), as well as proximity-oriented land-use strategies, on urban accessibility and Social Quality of Life.

A quantitative methodological approach based on spatial big data is adopted, simulating seven progressive scenarios that compare a baseline (2023) with distinct combinations of transport infrastructure expansion and the redistribution of urban facilities. The analysis is supported by the calculation of cumulative accessibility matrices and the estimation of the High Quality of Societal Life (HQSL) Index.

The results indicate that, whilst rail infrastructure significantly reduces travel times and the centre-periphery gap, its isolated implementation is insufficient to guarantee territorial equity. The optimal scenario is achieved solely through the synergy between a metropolitan rail network and a policy for distributing facilities at the Local Planning Unit scale. Consequently, it is concluded that the most efficient model for Bogotá is not strict local autarky, but a hybrid "15+30-minute" structure, combining pedestrian proximity for basic and care services with rapid mass transit accessibility for employment and higher education..

Keywords

Urban Accessibility, Public Transport, 15-Minute City, HQSL Index, Spatial Equity, Bogotá.

Zusammenfassung

Die urbane Mobilität in Bogotá ist historisch durch lange Reisezeiten und eine ausgeprägte sozioräumliche Segregation gekennzeichnet, wodurch der gleichberechtigte Zugang zu urbanen Chancen insbesondere für die peripheren Bevölkerungsgruppen eingeschränkt wird. Diese Arbeit untersucht die potenziellen Auswirkungen der Ersten Linie der Metro von Bogotá sowie eines möglichen regionalen Schienennetzes in Kombination mit nutzungsorientierten Strategien der räumlichen Nähe auf die urbane Erreichbarkeit und die soziale Lebensqualität.

Es wird ein quantitativer, auf *Spatial Big Data* basierender methodischer Ansatz verfolgt, der sieben progressive Szenarien simuliert und eine Basislinie aus dem Jahr 2023 mit verschiedenen Kombinationen aus dem Ausbau der Verkehrsinfrastruktur und der Umverteilung urbaner Einrichtungen vergleicht. Die Analyse stützt sich auf die Berechnung kumulativer Erreichbarkeitsmatrizen sowie auf die Schätzung des Index für Hohe Soziale Lebensqualität (HQSL).

Die Ergebnisse zeigen, dass schienengebundene Verkehrsinfrastruktur zwar wesentlich zur Reduzierung der Reisezeiten und zur Verringerung der Zentrum–Peripherie-Differenzen beiträgt, jedoch allein nicht ausreicht, um räumliche Gerechtigkeit zu gewährleisten. Die besten Ergebnisse werden nur durch die Kombination eines metropolitenen Schienennetzes mit einer Strategie zur Verteilung von Einrichtungen auf der Ebene der Lokalen Planungseinheiten erzielt. Daraus folgt, dass das effektivste Modell für Bogotá nicht eine strikte lokale Autarkie ist, sondern eine hybride „15+30-Minuten“-Struktur, die fußläufige Nähe zu grundlegenden und sorgerelevanten Dienstleistungen mit einem schnellen öffentlichen Verkehrszugang zu Arbeitsplätzen und Hochschulbildung verbindet.

Schlagwörter

Urbane Erreichbarkeit, Öffentlicher Personennahverkehr (ÖPNV), 15-Minuten-Stadt, HQSL-Index, Räumliche Gerechtigkeit, Bogotá

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Appendix A: Script Repository on GitHub.
Appendix B: Classification Table of Facilities.
Appendix C: Schematic Map of Public Transportation.

List of Abbreviations and Glossary

Big Data	Large volumes of data
EM2023	2023 Mobility Survey
GPKG	GeoPackage (open and standard geospatial file format)
GTFS	General Transit Feed Specification
HQSL	High Quality of Societal Life
LOC	Locality
MNZ	Block (urban block)
O-D	Origin-Destination
OSM	OpenStreetMap
POT	Territorial Planning Plan
R5R	Rapid Realistic Routing on Real-world and Reimagined networks (routing engine)
Shape	Vector layers / vector data
UPL	Local Planning Unit
UTAM	Mobility Analysis Territorial Unit
ZAT	Transport Analysis Zone

1

Introduction

1.1. Context and Problem Statement

Bogotá, D.C., the capital of Colombia and the core of a metropolitan area exceeding 10.3 million inhabitants (DANE, 2018), faces a mobility crisis that has become one of the main obstacles to its development and social well-being. This congestion is not a subjective perception, but a measurable reality that ranks the city at the top of various global indices.

The top ten traffic delay times

1. London	156 hr
2. Chicago	155 hr
3. Paris	138 hr
4. Boston	134 hr
5. Bogota	122 hr
6. Palermo	121 hr
7. Toronto	118 hr
8. New York	117 hr
9. Monterrey	116 hr
10. Philadelphia	114 hr

Figure 1. The top ten traffic delay times. Author's own elaboration, based on data from: (INRIX, 2023)

Average hours lost by drivers due to peak hour traffic

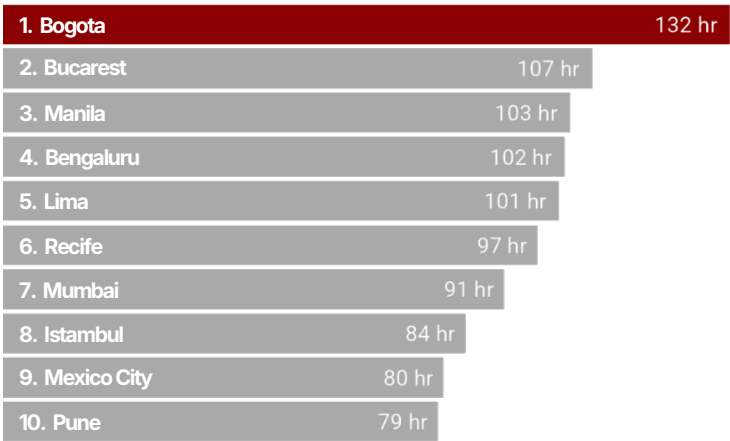


Figure 2. Average hours lost by drivers due to peak hour traffic. Author's own elaboration, based on data from: (Financial Times, 2023)

As illustrated by analyses from the INRIX Global Traffic Scorecard (INRIX, 2023) and TomTom (Financial Times, 2023) in **Figure 1** and **Figure 2**, Bogotá is consistently ranked as one of the most congested cities in the world. According to these data, the average driver in Bogotá loses up to 132 hours per year in traffic (Financial Times, 2023), a time that drastically exceeds that of other Latin American and global megacities.

This paralysis has a direct impact on urban competitiveness—increasing logistics costs and reducing productivity—but its most severe impact is measured in the quality of life of its inhabitants. Low average travel speeds and prolonged commuting times become a daily "tax" that subtracts hours from rest, family life, education, and leisure, generating high levels of stress and also affecting public health.

Congestion level and average speed per hour, during one week in Bogotá

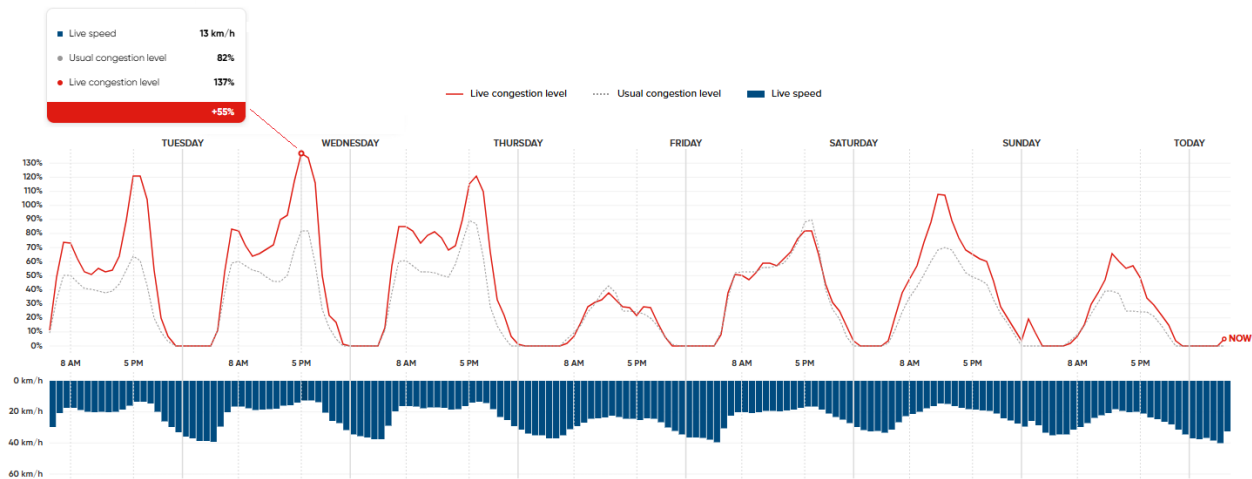


Figure 3. Congestion level and average speed per hour, during one week in Bogotá. Source: (TomTom, 2024).

The origin of this problem is structural and is grounded in two historical imbalances that form the core of this research:

- **An Insufficient Transport System:** Bogotá lacks a rail-based mass transit system (such as a metro, suburban trains, commuter rail, trams, etc.) to function as the high-capacity backbone indispensable for a metropolis of its scale. Mobility in Bogotá relies on a bus-based system (SITP and Transmilenio) which, whilst extensive, is vulnerable to surface congestion and presents capacity limitations for high-demand corridors. Furthermore, the low quality of public transport service has led to a migration of users towards private vehicles, especially motorcycles (Secretaría Distrital de Movilidad, Bogotá, DC., 2023), which further aggravates the situation for surface-level public transport.
- **A Monocentric Urban Model:** City planning has perpetuated a concentration of facilities, services, and main sources of employment in the central-eastern zone of the city.

The combination of these two factors compels the majority of citizens, especially those residing in the peripheries—where the lower-income population is concentrated (DANE, 2018) (Secretaría Distrital de Movilidad, Bogotá, DC., 2023)—to travel long distances to access opportunities for study, employment, and basic services.

The result is a city model that generates profound inequality in accessibility and imposes excessive travel times as a daily barrier to social and economic development.

1.2. Rationale

The diagnosis presented in the preceding section (1.1) demonstrates that Bogotá's urban development and mobility model has reached a point of unsustainability. The reliance on a surface-based transport system, operating at the limit of its capacity and competing for saturated road space, generates negative externalities that extend beyond mere congestion. This situation perpetuates a cycle of social inequity, as it is the lower-income populations, usually located in the urban peripheries, who suffer the longest commuting times, limiting their effective access to educational, employment, and quality service opportunities offered by the city (Fernandez, 2023).

Given this panorama, a paradigm shift in mobility and urban planning policies becomes imperative. This research is justified by the need to explore a simultaneous dual transformation:

- a. **The Transformation of Mobility via the Rail System.** The implementation of rail-based mass transit systems (such as the First Metro Line currently under development, and prospective future lines of metro, urban/suburban or regional rail) is a necessary condition to restructure the city's mobility. These systems offer the high capacity, speed, and reliability that the city requires to move millions of people, reducing dependence on individual and surface transport. However, a transport system, no matter how efficient, does not resolve territorial imbalances on its own.
- b. **The Transformation of the Territory via the City of Proximity.** Herein lies the second justification for this study. Building a metro or a train does not guarantee that citizens will cease making long journeys if their basic needs (education, health, commerce, leisure) are not located close to their homes. The concept of the "15-Minute City" or "City of Proximity" offers the theoretical framework to reconfigure the territory, promoting a polycentric model where essential facilities and services are accessible on foot or by bicycle (Moreno, 2023).

The synergy between these two transformations is key. The true opportunity for Bogotá lies not only in constructing a metro, but in using the new stations as catalysts to create urban centralities and proximity nodes. Nevertheless, the vision for the city must go far beyond the implementation of facilities and micro-centralities in the immediate vicinity of the mass transit system; rather, it must aim to distribute facilities to—and ensure their accessibility for—territories without direct access to the future rail mass transit network.

Therefore, this work is justified because it addresses a critical gap, analysing not only the impact of transport infrastructure in isolation, but also its strategic integration with an urban planning model (proximity) across different scenarios. Evaluating this synergy at the Local Planning Unit (UPL) scale (Secretaría Distrital de Planeación, Bogotá, DC., 2023) also allows for the generation of diagnoses and recommendations with a level of detail applicable to public policy, identifying which UPLs benefit the most, which require greater facility interventions, and how this dual strategy can effectively reduce inequality in accessibility.

| 1.3. Research Question

Taking into account the issues of congestion and territorial imbalance, added to the need for a comprehensive intervention, this thesis seeks to answer the following research question:

To what extent do the implementation of rail-based mass transit systems—such as the First Metro Line and a possible rail network—and the adoption of the City of Proximity model, applied across different scenarios and scales (at the station node and Local Planning Unit levels), modify accessibility to urban facilities and the structure of mobility in Bogotá?

| 1.4. Objectives

To answer the research question, the following objectives are proposed:

1.4.1. General Objective

To evaluate the impact on accessibility and mobility in the UPLs of Bogotá, derived from the implementation of combined scenarios of rail-based mass transit and proximity planning (focused on stations and at the UPL scale).

1.4.2. Specific Objectives

- 1.4.2.1. To diagnose the current state of population distribution, mobility, and the supply of facilities, as well as levels of accessibility and proximity (via the High Quality of Societal Life Index (Moreno, 2023)) in Bogotá, with an emphasis on the UPL scale.
- 1.4.2.2. To structure and model the intervention scenarios, including the rail-based mass transit networks (First Metro Line and proposed Rail Network), and the strategies for the implantation of facilities for the city of proximity (at station nodes and at the UPL scale).
- 1.4.2.3. To quantify and compare the change in accessibility to facilities for each of the proposed intervention scenarios.
- 1.4.2.4. To analyse the differential impact of each scenario, identifying the synergies between transport infrastructure and proximity models, to determine the most effective strategies for reducing inequality in accessibility.

| 1.5. Scope and Delimitation

This research defines its scope within the following parameters:

Geographic Scope: The study focuses on the consolidated urban area of Bogotá, D.C. The Local Planning Units (UPL) constitute the primary scale for analysis and the aggregation of results, although more detailed scales are also utilised, such as Blocks (Manzanas - MNZ), Hexagonal grids (HEX) with a 500m radius, and Traffic Analysis Zones (Zonas de Análisis de Transporte - ZAT). These latter scales are used for the graphic representation of information and the analysis of accessibility calculations.

Geographic Delimitation: The rural areas of the Capital District are excluded, given that their mobility dynamics, density, and provision of facilities respond to territorial logics that are fundamentally different from urban ones.

Thematic Scope: The analysis focuses on accessibility to facilities (defined in **Chapter 4**). The evaluation of proximity is based on the High Quality of Societal Life Index from the 15-Minute City concept (Moreno, 2023).

Model Delimitations: The study concentrates on the potential transformation of accessibility based on changes in the transport network and the location of services. It is not the objective of this thesis to conduct a cost analysis, an economic impact evaluation (such as land value capture or gentrification), nor a detailed design of the facility interventions or rail lines.

Furthermore, it is important to clarify that, due to the high complexity of obtaining consolidated official information from the authorities, the exercise of consulting the facility inventory and the subsequent accessibility analysis based upon it constitutes a fully synthetic exercise (based on OpenStreetMap queries). This exercise is replicable with official information once it becomes openly available from the institutions responsible.

| 1.6. Structure of the Document

To respond to the proposed objectives, this thesis is structured into eight chapters:

Chapter 1: Introduction. Presents the context of mobility in Bogotá, the rationale for the study, the research question, the objectives, and the scope of this work.

Chapter 2: Theoretical and Conceptual Framework. Develops the theoretical foundations upon which the thesis is built, including sustainable urban mobility, rail-based transport systems, the City of Proximity concept, and the measurement of accessibility.

Chapter 3: Methodology. Explains how the research was conducted, detailing the study area, data sources (demographic, Mobility Survey, facilities inventory), and the modelling and analysis tools utilised.

Chapter 4: Diagnosis of the Current State. Establishes the baseline (Scenario 1), presenting a detailed analysis of the demographics, mobility patterns, distribution of facilities, and current levels of accessibility and proximity in the UPLs of Bogotá.

Chapter 5: Intervention Scenarios. Describes in detail the configuration of each of the proposed scenarios, explaining the alignment of the rail networks (First Metro Line and Rail Network). Furthermore, the methodology for the implantation of facilities for future scenarios is presented.

Chapter 6: Results and Comparative Analysis. The modelling results for each scenario are presented and systematically compared against the baseline (Scenario 1) in terms of accessibility and High Quality of Social Life.

Chapter 7: Discussion and Implications. Interprets the results obtained, discussing the synergy between the improvement in transport supply and proximity, and drawing implications for public policy and urban planning in Bogotá.

Chapter 8: Conclusions and Future Work. Concisely answers the research question, summarises the main findings and contributions of this study, and suggests lines for future research.

2

Theoretical and Conceptual Framework

This chapter establishes the theoretical foundations that underpin the research. It articulates the key concepts linking transport infrastructure, territorial planning, and urban quality of life, defining the analytical framework for evaluating the transformation of mobility in Bogotá.

| 2.1. Sustainable Urban Mobility

Sustainable urban mobility is defined as the capacity to satisfy the travel needs of society in an equitable, efficient, and safe manner, whilst simultaneously minimising environmental impact and resource consumption (Banister, 2008). In the context of Latin American megacities like Bogotá, characterised by rapid urban expansion and high motorisation rates, achieving this sustainability can become a paramount challenge.

The transition towards sustainable urban mobility requires a portfolio of coordinated policies and strategies acting on multiple fronts. There is no single solution. Success depends on the implementation of a coherent set of interventions that mutually reinforce their effects. A useful conceptual framework for organising these interventions is "**Avoid-Shift-Improve**", which provides a logical hierarchy for action (Banco de Desarrollo de América Latina y El Caribe, 2024).

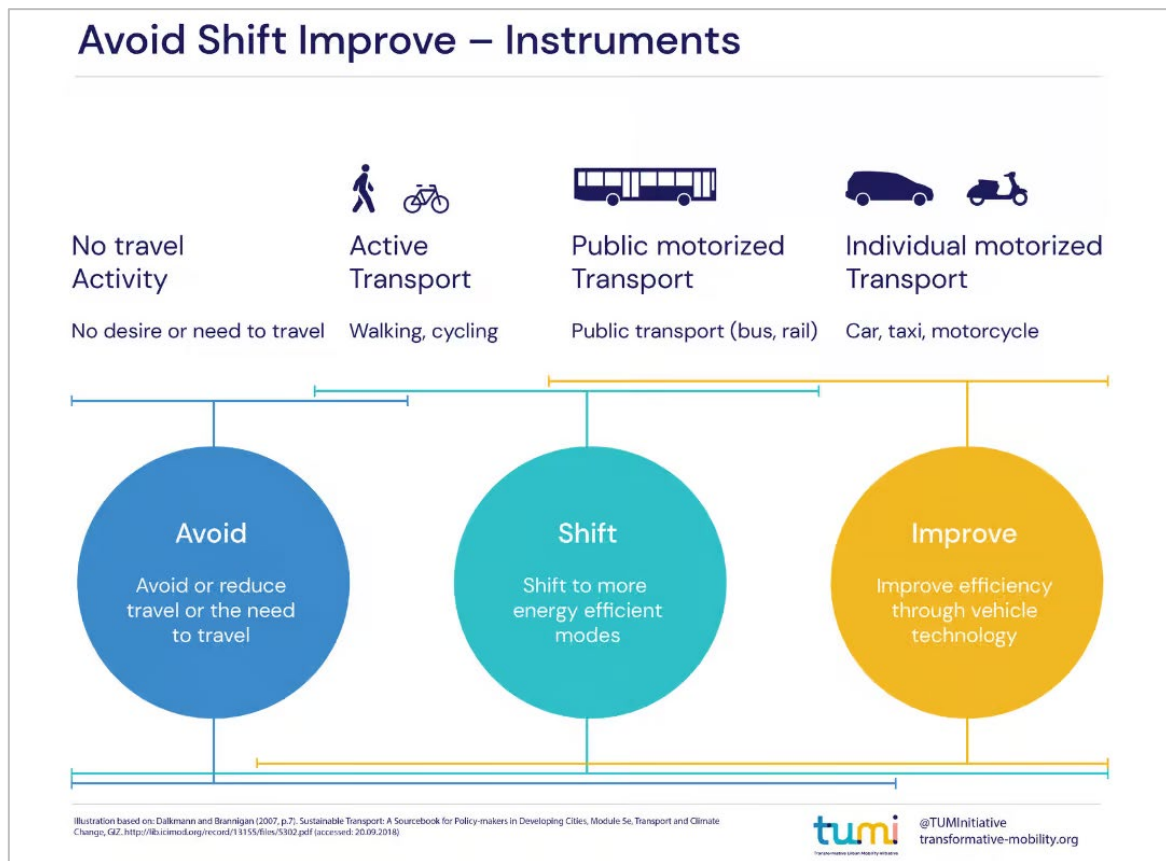


Figure 4. Avoid-Shift-Improve. Source: (TUMI Initiative, 2025).

2.1.1. Mitigation Framework

This framework structures policies according to their primary objective, creating a hierarchy of ambition and long-term effectiveness (Banco de Desarrollo de América Latina y El Caribe, 2024) (Bussi & Abriani, 2020):

- Avoid/Reduce:** This is the most strategic and fundamental level. Its objective is to reduce the need for motorised journeys and shorten the distances of necessary trips. The main tool to achieve this is integrated land-use and transport planning (Bussi & Abriani, 2020). By promoting compact cities with mixed uses (residential, commercial, labour) and high density, more people are enabled to access their daily destinations by walking, cycling, or making short public transport trips. This is the same principle as the city of proximity and the 15-Minute City by Carlos Moreno (Moreno, 2023). Transit-Oriented Development (TOD) is a paradigmatic example of this strategy (Consulting, et al., 2021).
- Shift/Maintain:** This level focuses on encouraging a modal shift from the least efficient and most polluting modes (private vehicles, diesel-based public transport, etc.) towards more sustainable modes. Strategies here seek to make public transport, walking, and cycling more attractive, safer, and convenient than the private car (Bussi & Abriani, 2020) (Moreno, 2023). This is achieved

through investments in infrastructure for these modes, demand management policies, and regulatory measures.

- c. **Improve:** This level seeks to optimise efficiency and reduce the environmental impact of all transport modes, including those vehicles that will continue to circulate. Strategies include the adoption of cleaner vehicular technologies (such as electromobility), improvements in fuel efficiency, intelligent traffic management to reduce congestion and idling, and the optimisation of logistics operations (Bussi & Abriani, 2020) (Banco de Desarrollo de América Latina y El Caribe, 2024).

Whilst all three strategies are necessary, the framework suggests a priority. The greatest gains in long-term sustainability stem from "Avoid", followed by "Shift". "Improve" is essential, but if unaccompanied by the other two, it risks perpetuating an unsustainable mobility model, despite the use of cleaner vehicles (e.g., congestion caused by electric cars or the excessive land use for private vehicles).

2.1.2. The Role of Rail-Based Mass Transit

Within the sustainability paradigm, rail-based mass transit systems (such as metros, light rail, and commuter rail) are globally recognised as the backbone of mobility in high-density metropolises. Their justification lies not only in their capacity to move large volumes of passengers but in their structuring effects on the territory (Cervero, 1998).

Unlike bus-based systems (BRT, Bus Rapid Transit)—which, whilst having been a valuable solution for Bogotá (Transmilenio), operate on the surface and, even with segregated lanes, are susceptible to congestion—rail systems offer strategic advantages:

- **High Capacity and Speed:** Rail systems mobilise a significantly higher number of passengers per hour per direction, doing so at higher and more reliable commercial speeds by operating on exclusive, segregated rights-of-way without level crossings. Although this aspect is often overlooked, boarding and alighting times in rail vehicles are considerably lower than in bus-based systems due to the greater number of doors per trainset, translating into a faster and more efficient operation.
- **Reliability and Efficiency:** Their operation is unaffected by vehicular congestion, weather (to a large extent), or surface traffic accidents, guaranteeing constant travel times.
- **Induction of Urban Development:** The permanence of their infrastructure (tracks and stations) generates certainty in the real estate market, attracting investment and densification around

stations. This Transit-Oriented Development (TOD) is a key tool for urban, suburban, and even regional planning.

- **Environmental Sustainability:** With few exceptions, urban rail transport systems operate on electric power, drastically reducing Greenhouse Gas (GHG) emissions and particulate matter (such as PM2.5) compared to diesel or gas bus fleets. Even when comparing electric rail systems versus electric bus fleets, rail maintains advantages regarding the non-emission of particles derived from friction wear between tyres and the road surface. Energy efficiency is also superior in rail systems regarding energy loss due to rolling resistance from pneumatic tyres versus steel wheels with negligible deformation.

The shift from a BRT-based mobility model to one incorporating a rail network (as proposed by this thesis) is not merely a leap in capacity, but a qualitative transformation that allows the city to reconfigure its main flows and liberate surface space for other sustainable transport modes, such as cycling or walking.

| 2.2. The City of Proximity

Whilst rail-based mass transit systems (discussed in 2.1.2) are the solution for long-distance and interurban journeys, the "City of Proximity" addresses the local scale and daily life. This paradigm arises as a direct response to the functionalist (or modernist) urban planning model that dominated the 20th century, which segregated urban functions in such a way that created zones exclusively for residence, others for work, others for shopping, others for leisure, etc. This city model has also been present in Bogotá and its Land Use Plans (Planes de Ordenamiento Territorial - POT) up to the present day. This model of functional segregation is, to a large extent, the root cause of city congestion, as it generates forced mobility for almost any daily activity (Jacobs, 1961).

The "15-Minute City" and "30-Minute Territory" Models

The concept of the "15-Minute City" and "30-Minute Territory", developed and disseminated by Carlos Moreno from the ETI Chair at IAE Paris-Sorbonne, proposes a new approach to address the contemporary challenges of urban development. Its central premise is "hyper-proximity", understood as the possibility of accessing essential urban services, activities, and functions within a timeframe considered reasonable on foot or by bicycle. This emphasis on proximity implies a paradigm shift vis-à-vis traditional urban models, usually structured around motorised mobility and functional segregation (Moreno, 2023).

In essence, the 15-Minute City proposes a polycentric model of urban organisation, in which neighbourhoods function as complete and self-sufficient units to resolve the population's daily needs: Living, working, supplying, caring, learning, and enjoying (Moreno, 2023).

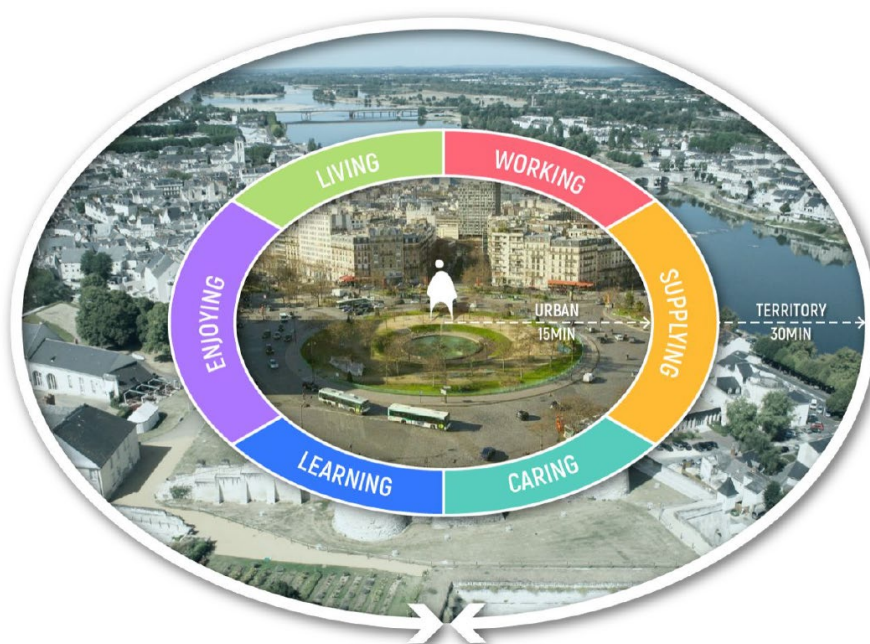


Figure 5. The 6 social functions of the 15-Minute City and 30-Minute Territory model. Source: (Moreno, 2023).

1. **Living:** Decent housing conditions and safety in the residential environment.
2. **Working:** Proximity to employment sources or workspaces (coworking), reducing forced commuting.
3. **Supplying:** Access to fresh food, pharmacies, and essential commerce.
4. **Caring:** Access to preventative and curative health services (hospitals, clinics) and care services.
5. **Learning:** Access to educational (schools, universities) and cultural infrastructure (libraries).
6. **Enjoying:** Access to public spaces, parks, sport, culture, and leisure.

It is, therefore, a vision of a sustainable city that integrates policies, urban projects, and digital tools with the aim of improving well-being, strengthening social cohesion, and reducing dependence on motorised transport.

Moreno posits that this approach is sustained by the convergence of three concepts:

- **Chronourbanism.** Incorporates the temporal dimension into urban planning, analysing how flows, displacements, and time usage shape urban form and social interactions. Its objective is to consider time as a design input as relevant as space.
- **Chronotopy.** Refers to the evolution of the usage of places over time. In dense cities, chronotopy invites a rethinking of the rhythms and sequences of space usage to promote multifunctionality and the optimal use of available urban land.
- **Topophilia.** Describes the emotional and affective bond between people and their environment. This concept underscores how attachment to places contributes to a sense of belonging, local identity, and well-being.

The health emergency caused by COVID-19 and the international agenda established by the Sustainable Development Goals (SDGs) (United Nations, s.f.) accelerated the global adoption of the 15-Minute City, positioning it as a relevant strategy in the face of environmental, social, and economic crises. Diverse networks and local governments, including the C40 Cities (a group within which Bogotá is found), have incorporated this vision to orient urban reconfiguration towards more resilient, sustainable models centred on quality of life (Moreno, 2023).

Taken together, the 15-Minute City and 30-Minute Territory models propose a new vision or way of relating time, space, and daily life, oriented towards building desirable, viable, and sustainable cities, where residents reclaim control of their time and neighbourhood life regains centrality. The objective is not to isolate people in their neighbourhoods or in micro-cities as some have misinterpreted, but to offer the option that, as far as possible, the use of a motorised transport mode is not necessary for daily life. This reduces congestion, fosters social cohesion, promotes public health (by incentivising walking and cycling), and revitalises the local economy.

In the context of this thesis, the City of Proximity is analysed in two ways: As a strategy to densify services around new rail stations (creating nodes of high accessibility) and as a policy to balance the distribution of facilities across all UPLs, as will be measured via the High Quality of Societal Life (HQSL) Index, an index that will be addressed subsequently.

| 2.3. Introduction to Urban Accessibility

Accessibility is the central concept that articulates the relationship between the transport system and land use. Historically, Hansen defined accessibility as the "potential of opportunities for interaction", establishing that the value of a place depends not only on its intrinsic characteristics but on its connectivity with other places (Hansen, 1959).

In contemporary literature, the most accepted definition is the one proposed by Geurs and van Wee, who describe accessibility as "the extent to which land-use and transport systems enable individuals (or groups of individuals) to reach activities or destinations by means of a [combination of] transport modes" (Geurs & van Wee, 2012). This definition transcends the simplistic view of road infrastructure to focus on the ultimate goal of displacement, which is to reach one or more "opportunities" (employment, education, health, etc.), which are associated with facilities. Furthermore, according to Pereira and Herszenhut in their book *Introduction to urban accessibility: a practical guide with R*—from which concepts were drawn for the development of this thesis, as will be seen later in this document—urban accessibility, understood as "macro-accessibility", focuses on the structural issues of urban development and how the arrangement of transport corridors impacts the capacity to access opportunities (Pereira & Herszenhut, 2024).

2.3.1. Distinction between Mobility and Accessibility

It is fundamental to establish the theoretical distinction between mobility and accessibility.

- **The Mobility Approach:** Focuses on the efficient movement of vehicles and people. Its traditional metrics are speed, vehicular flow, and congestion reduction. Planning for mobility assumes that "travelling further and faster" is always better, which has led to dispersed urban sprawl, the construction of infrastructure for traffic, and automobile dependence (Handy, 2005).
- **The Accessibility Approach:** Transport demand is a "derived demand"; that is, people do not travel for the pleasure of moving, but to participate in activities. Therefore, the objective of planning should not be to increase speed (mobility-transport), but to reduce the need for long journeys by bringing origins and destinations closer together, or by improving sustainable transport links.

This distinction theoretically justifies the hypothesis of this thesis, which suggests that the solution for Bogotá is not merely increasing the capacity to "move more people" (mobility-transport), but restructuring the city so that opportunities are within the citizens' reach (accessibility), whether through the

implementation of the Metro and the rail-based mass transit network (speed) or the City of Proximity (closeness).

2.3.2. Components of Accessibility

To operationalise this concept, four theoretical components are identified that determine accessibility levels (Geurs & van Wee, 2012), which are addressed in this study:

- a. **Transport Component:** Describes the transport system (supply), considering the time, cost, and effort to travel. In this thesis, this component changes between scenarios (Current Network vs. Metro Network vs. Rail Network).
- b. **Land-Use Component:** Reflects the quantity, quality, and spatial distribution of opportunities. This is where the proposal for the adoption of the "City of Proximity" for Bogotá intervenes, implementing facilities in deficit within the UPLs.
- c. **Temporal Component:** Considers the time constraints of users (e.g., service operating hours, travel times, etc.).
- d. **Individual Component:** Recognises that needs and capabilities vary among people (income, age, gender, occupation, etc.).

2.3.3. Accessibility and Social Equity

Recently, accessibility has been closely linked to exclusion and social vulnerability, as demonstrated by the Bogotá researcher Leidy Fernández in her Master's thesis *"Metodología de evaluación multidimensional de la accesibilidad a través del transporte público como expresión de la vulnerabilidad social en Bogotá"* (Fernandez, 2023). The lack of adequate accessibility leads to "transport poverty", preventing vulnerable populations from fully participating in society and accessing, for example, decent employment, higher education, or healthcare (Lucas, 2012). This problem is significant, as it leads to cycles that aggravate the situation of those who need access to facilities to improve their living conditions and achieve social mobility.

By evaluating accessibility at the Local Planning Unit (UPL) scale, this research aligns with the approach of distributive justice in transport, assessing the gains in terms of accessibility with the new infrastructures.

2.3.4. Cumulative Opportunities vs. Gravity-based Models

There are multiple ways to measure accessibility. Gravity-based models weight opportunities according to a decay function (a distant opportunity is worth less than a nearby one). According to Levinson et al. (2006), although gravity models are theoretically precise, cumulative measures (e.g., "how many jobs can I reach in 30 minutes") are superior for public communication and political decision-making, as they are easily interpretable by planners and the public (Levinson & El-Geneidy, 2006). Pereira and Herszenhut validate the use of modern computational tools (such as **r5r**, used in this thesis) to calculate any of these measures efficiently across large urban networks (Pereira & Herszenhut, 2024).

To quantify accessibility, this thesis utilises Cumulative Opportunities indicators, also known as contour measures or isochrones. This measure estimates the number of opportunities (e.g., number of jobs or schools) that can be reached from a specific origin within a specific time limit or travel cost, also called a cut-off threshold (Pereira & Herszenhut, 2024).

Mathematically, it is defined as:

$$A_i = \sum_{j=1}^n O_j \times f(c_{ij})$$

Where O_j is the opportunity at destination j , and $f(c_{ij})$ is a binary function that is 1 (one) when the travel cost is lower than the established threshold (e.g., a time of 30 minutes), and 0 (zero) if it is greater than that threshold.

The spatial representation of these thresholds generates Isochrones, which are lines or areas connecting points accessible within the same travel time. This indicator is widely used by international transport agencies due to its ease of interpretation and communication for public policy. Likewise, it is complemented by Minimum Travel Cost measures, which indicate, for example, the time necessary to reach the nearest hospital or school. In the figure below, two isochrones of 15 and 30 minutes are represented, with a common point of origin.

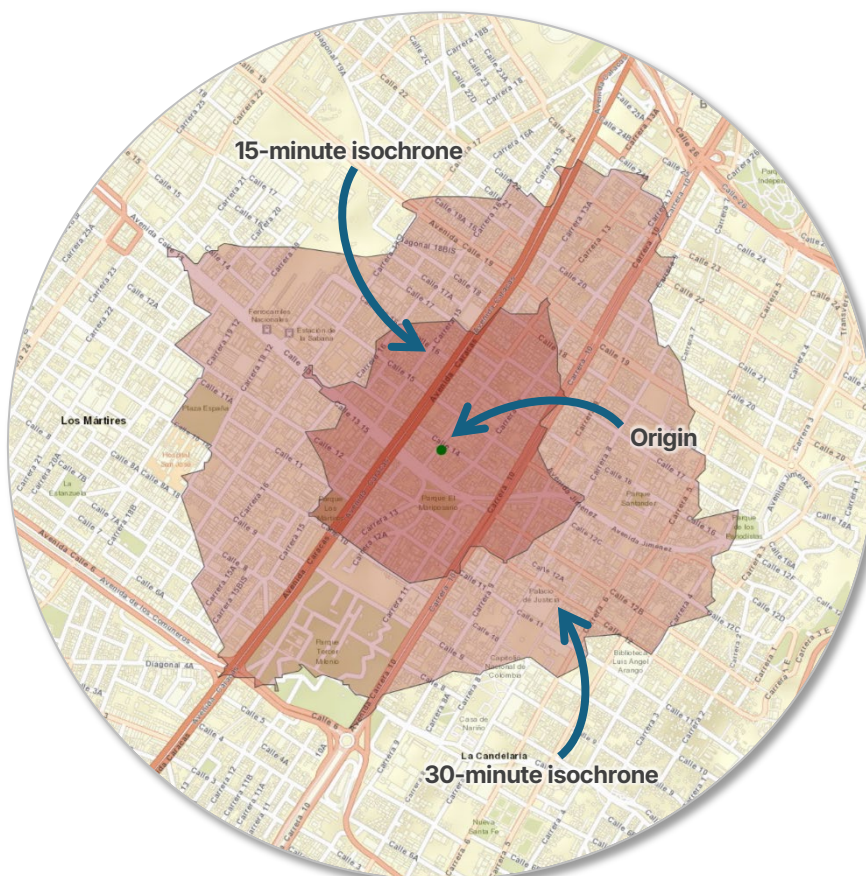


Figure 6. Isochrones of 15 and 30 minutes walking, with a random point of origin in the city of Bogotá. Source: Author's own elaboration.

In simple terms, an isochrone is a spatial representation showing how far it is possible to travel from a point of origin within a specific time, according to a specific transport mode and mobility conditions. For the specific case of this thesis, its utility lies in the fact that it allows for the identification of which facilities, services, or activities are contained within that temporal limit. That is, all elements found within the area delimited by the isochrone are considered accessible from the point of origin within the time defined for its construction. This facilitates the evaluation of the real accessibility of a territory and the comparison of the available supply under different mobility scenarios.

2.4. High Quality of Societal Life (HQSL) Index

Whilst accessibility measures the potential to reach destinations, urban quality of life evaluates the value these destinations contribute to human well-being. In a sustainable city, the objective of planning is not merely to optimise flows, but to ensure that the urban environment satisfies fundamental human needs in an equitable and efficient manner. To operationalise the proximity concept and evaluate the performance of the Local Planning Units (UPL) in Bogotá, this research adopts the framework of the High Quality of Societal Life (HQSL) Index.

Both the "15-Minute City" and the "30-Minute Territory" are not merely general ideas, but are grounded in an ontology, that is, a well-organised conceptual map. This ordered conceptual system defines the key elements of these urban proximity models, starting with the six basic social functions (see **Chapter 2.2**) and subsequently expanding them into more specific parts to enable their analysis or planning.

 Living	 Working	 Supplying	 Caring	 Learning	 Enjoying
Housing Energy Waste management Transport Utility infrastructure/ Service infrastructure	Urban environment Access Diversity Services	Food Other goods/ Other supplies Public utilities	Festivities/Public celebrations Culture Leisure Associations/Public organizations	Access Availability Performance Orientation	Access to Healthcare Prevention Emergency Living environment Well-being Sport/Physical activity Pollution

Table 1. Components and infrastructure integrated into the six social functions. Source: Author's own elaboration with data from (Moreno, 2023).

2.4.1. Composite Indicators

Based on the ontology of social functions and their respective subdivisions—which may present overlaps—it is possible to categorise the necessary infrastructures to determine the High Quality of Societal Life Index. According to Moreno (2023), this approach facilitates a territorial diagnosis that allows for the visualisation of both the deficits and the level of access to basic services in a specific area. Such analysis is achieved by articulating the social functions with three fundamental axes or components, also called Composite Indicators: Well-being, Sociability, and Sustainability.

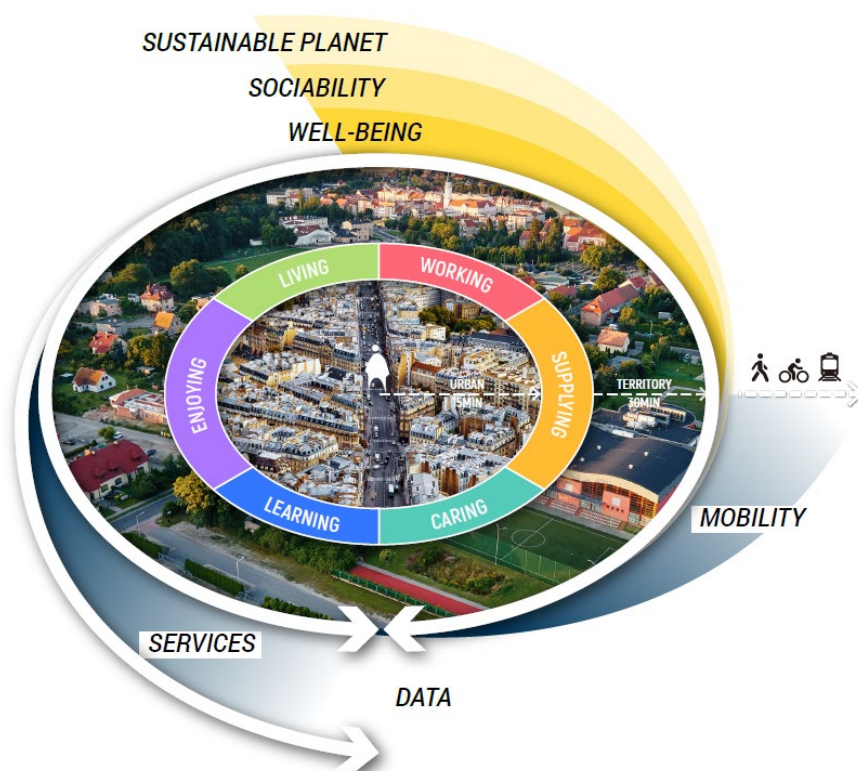


Figure 7. The six Social Functions articulated with the three Composite Indicators: Well-being, Sociability, and Sustainability. Source: (Moreno, 2023).

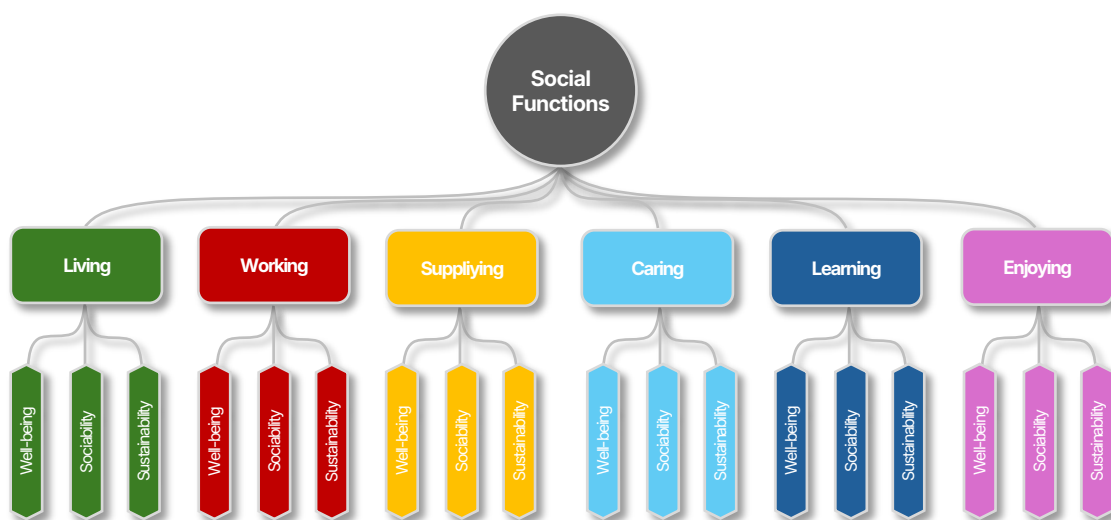


Figure 8. Integration of Composite Indicators into Social Functions. Source: Author's own elaboration based on (Moreno, 2023)

2.4.2. The Proximity ScoreCard

To visualise and compare the performance of different zones of the city, the ScoreCard or Territorial ScoreCard methodology is utilised. This tool allows for the translation of complex geospatial data into informative and intuitive visual diagnoses that can aid decision-making.

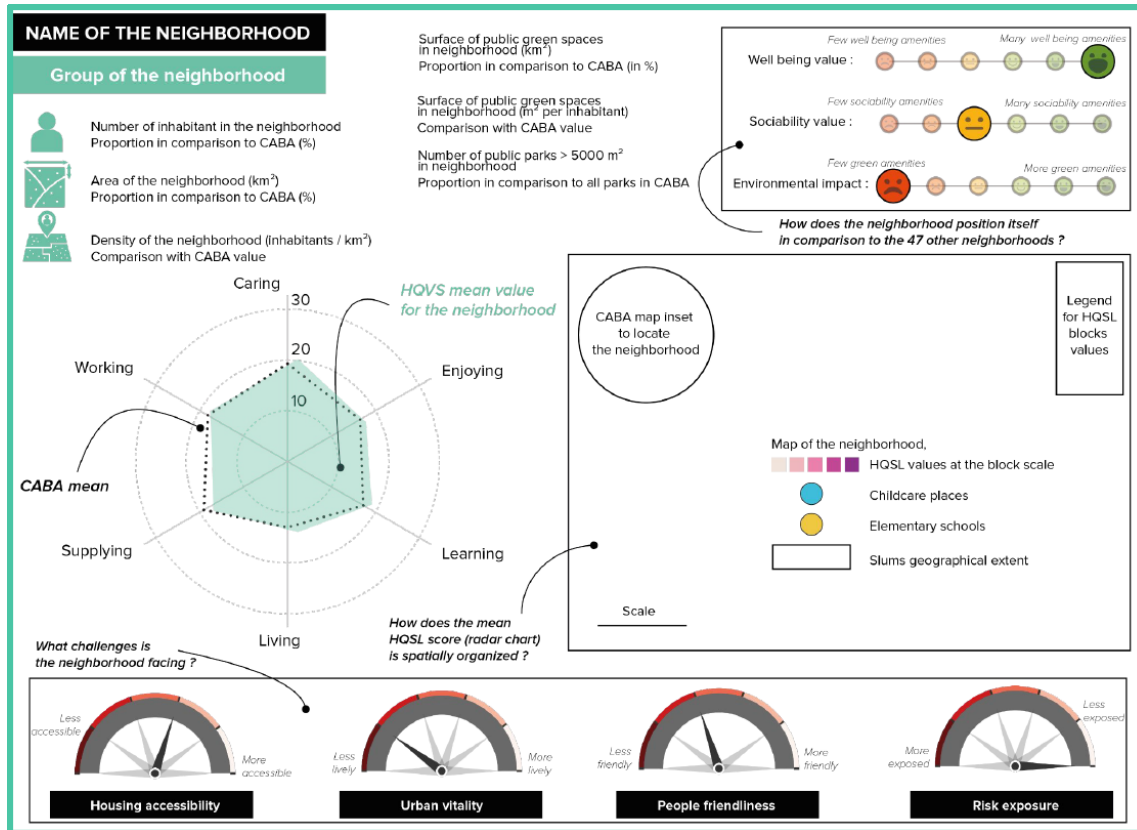


Figure 9. Example of a ScoreCard. Source: Excerpt from (Chaire ETI - Sorbonne Université, 2023)

The ScoreCard integrates a set of key indicators that allow for the systematic comparison of the characteristics of the study areas. Likewise, it incorporates visual representations of the accessibility patterns associated with each of the Social Functions, alongside comparative analyses of the state of Well-being, Sociability, and Sustainability. Finally, the instrument includes a reference map of the area, which facilitates the spatial contextualisation of the results..

Radar Chart

The analysis of the Social Functions is typically represented via a Radar Chart, where each axis represents one of the six functions. From the chart, several deductions can be made, such as the following:

- **Balance vs. Imbalance:** A regular and broad hexagonal shape indicates that the analysis area is "complete" or has high accessibility to facilities, and where all needs are covered locally.
- **Distortion:** An irregular shape (e.g., an axis highly developed in "Living" but null in "Working") identifies "dormitory neighbourhoods" or specialised zones that force motorised mobility or medium- to long-distance journeys, generating transport dependence and reducing quality of life.

Comparison of Composite Indicators

The ScoreCard also makes a comparison of the Composite Indicators between different areas, with the aim of establishing the current situation of the neighbourhood regarding Well-being, Sociability, and Sustainability.

Geospatial Representation of the HQSL Index

Another important element found within the ScoreCard is the geospatial representation of the study area and the HQSL Index. The construction of this index will be explained in greater detail in the following chapter: **Methodology**.

In the context of this thesis, the HQSL Index serves not only as a diagnosis of the current state but as the metric of success to evaluate the proposed intervention that has the greatest impact in terms of accessibility, and thus demonstrate how the integration of rail transport and a better distribution of facilities can expand the "radar area" of the UPLs, transforming deficit territories into nodes of high quality of life.

3

Methodology

This chapter details the methodological approach, information sources, and computational tools utilised to model, simulate, and evaluate mobility and accessibility scenarios in Bogotá. The methodological structure was designed to integrate the analysis of spatial big data with transport planning tools based on open specifications such as the General Transit Feed Specification (GTFS). Likewise, the methodological framework for the City of Proximity and the creation of the High Quality of Societal Life (HQSL) Index (Moreno, 2023) was adopted.

3.1. General Methodological Approach

The research adopts a quantitative and spatial approach, structured in a sequential workflow of three main phases:

Diagnosis and Baseline (Scenario 1): Geospatial processing and analysis of the city, census data, and mobility surveys to characterise current demand and mobility flows, as well as Origin-Destination matrices, alongside a georeferenced inventory of facilities to characterise the supply and trip attraction centres. Current accessibility indicators and HQSL Indices are calculated.

Scenario Modelling (Scenarios 2-7): Generation of synthetic transport networks (GTFS format) for the proposed rail projects and simulation of land-use strategies (densification/implementation of facilities) under the City of Proximity model.

Comparative Evaluation: Calculation of accessibility metrics and the HQSL for each scenario, allowing for the contrast of impacts in terms of coverage, equity, and efficiency at the Local Planning Unit (UPL) scale.



The methodological development of this thesis—spatial processing, data science techniques, modelling, and the generation of maps and graphs—was conducted using scripts implemented in the R language, ensuring the reproducibility of the calculations. These scripts are available in the Tesis_MPMU GitHub repository, described in **Appendix A**, and accessible by **clickig here**, via the attached QR code, or directly at: https://github.com/ssilveram/Tesis_MPMU



3.2. Study Area and Spatial Delimitation

The study area corresponds to the consolidated urban land of Bogotá D.C., administratively organised under the scheme of Local Planning Units (UPL), as stipulated in the current Land Use Plan (*Plan de Ordenamiento Territorial - POT*) (Decree 555 of 2021) (Secretaría Distrital de Planeación, 2021).

3.2.1. Spatial Analysis Units

Due to the different administrative scales and multiple geographic divisions for territorial analysis and mobility behaviour in Bogotá, it was determined that to ensure precision in accessibility calculations and relevance for public policy, it was necessary to utilise four main spatial aggregation scales:

- **Block (*Manzana* - MNZ):** Used as the minimum unit for accessibility calculation. The Block level is also determined as the minimum area unit in the National Population and Housing Census (DANE, 2018).
- **Hexagons (H3):** Used as the minimum unit for grouping facilities and service supply, in order to have a homogeneous area unit that would clearly show differences in facility and population densities. The hexagon diameter was determined at 500m, to ensure that a walk of more than 15 minutes does not imply changing more than one contiguous hexagon.
- **Traffic Analysis Zones (*Zonas de Análisis de Transporte* - ZAT):** These are the smallest area units (highest resolution) utilised in the 2023 Mobility Survey (Secretaría Distrital de Movilidad, Bogotá, DC., 2023), which served to determine the Origin-Destination (O-D) matrices of said survey and for model calibration.

- **Local Planning Unit (UPL):** Although at the date of this document's publication, the city of Bogotá had administrative divisions different from UPLs called Localities (*Localidades*), it is expected that with the implementation of the 2021 Land Use Plan (POT), UPLs will replace Localities as the city's main administrative units. For this reason, UPLs were the primary units for the aggregation and presentation of results, allowing for the diagnosis of the territorial balance of the "city of proximity" taking into account this imminent administrative change (Secretaría Distrital de Planeación, 2021).

Within the spatial aggregation units, other secondary administrative or support classifications (areas) included in some of the data sources were also processed. The complete list of spatial aggregation units taken into account for the development of this thesis is shown below:

- Bogotá (BOG)¹
- Localities (LOC)^{1,2}
- Local Planning Units (UPL)¹
- Territorial Unit of Mobility Analysis (UTAM)³
- Traffic Analysis Zone (ZAT)³
- Sectors (SECT)^{1,2}
- Sections (SECC)^{1,2}
- Hexagon (HEX)⁴
- Block (*Manzana*) (MNZ)^{1,2}

The analysis, processing, and standardisation of all the city's geographic information, including these secondary units, can be consulted in the Script: `1_Empty_Shapes.R` in the GitHub repository.



For the reproducibility of this work, it is recommended that original data be downloaded from official sources (DANE, *Secretaría Distrital de Movilidad*, etc.) and saved in the relative paths indicated in the Script..

¹ (Alcaldía de Bogotá, 2019)

² (DANE, 2018)

³ (Secretaría Distrital de Movilidad, Bogotá, DC., 2023)

⁴ Author's own elaboration. The script can be consulted in: `1_Empty_Shapes.R`

3.2.2. Exclusion of Rural Areas

The study is delimited exclusively to the urban perimeter. The rural areas of the Capital District (such as the Sumapaz locality and the rural fringes of the Usme or Ciudad Bolívar localities, as well as the rural areas of the Cerros Orientales, Tunjuelito UPLs, etc.) are excluded from the analysis. This methodological decision responds to the fact that mobility dynamics on rural land—characterised by low densities, long distances, and informal or non-motorised transport—require accessibility metrics and intervention strategies substantially different from those of the "city of proximity" and rail-based mass transit model designed for dense urban environments.

The application of this criterion was performed on the generated UPL shapefile, using only a single line of code in many of the working scripts. The filter takes the form:

```
# Select no rural UPLs:
Lista_No_Rural <- Shape_UPLs |> filter(ZONA_UPL != "Rural") |> as_tibble() |> select(COD_UPL)
Shape_Interes <- Shape_UPLs |> filter(COD_UPL %in% Lista_No_Rural$COD_UPL) |> ...
```

R Code Snippet 1. Fragment of code for the filter to discard rural areas of the UPLs. Source: Author's own elaboration.

3.3. Data Sources and Processing

The construction of the model is fed by three primary data sources, which were subjected to processes of cleaning, spatial structuring, and analysis:

- Demographic information of the population.
- 2023 Mobility Survey of the Capital District of Bogotá.
- Synthetic inventory of facilities and services of the city.

3.3.1. Preliminary Geospatial Analysis and Generation of Vector Layers

The National Population and Housing Census (DANE, 2018), in addition to demographic and socioeconomic information, contains a series of vector layers for the georeferencing of the information collected in the surveys. However, at the date of the census (2018), the Local Planning Units (UPL) were not yet defined—they were defined in the 2021 Land Use Plan (POT). Due to this, it was necessary to relate the existing Census information with that of the POT through which the UPLs will be adopted for geospatial analysis. The processing and cleaning of this information were absolutely necessary to initiate the analyses, in order to work with each of the vector layers completely clean and interrelated in a

coherent manner with the different data sources. The detail of the processing of these vector layers can be consulted in the Script: `1_Empty_Shapes.R` in the GitHub repository.

Schematically, the treatment given to each of the vector layers and their associated data is shown below:

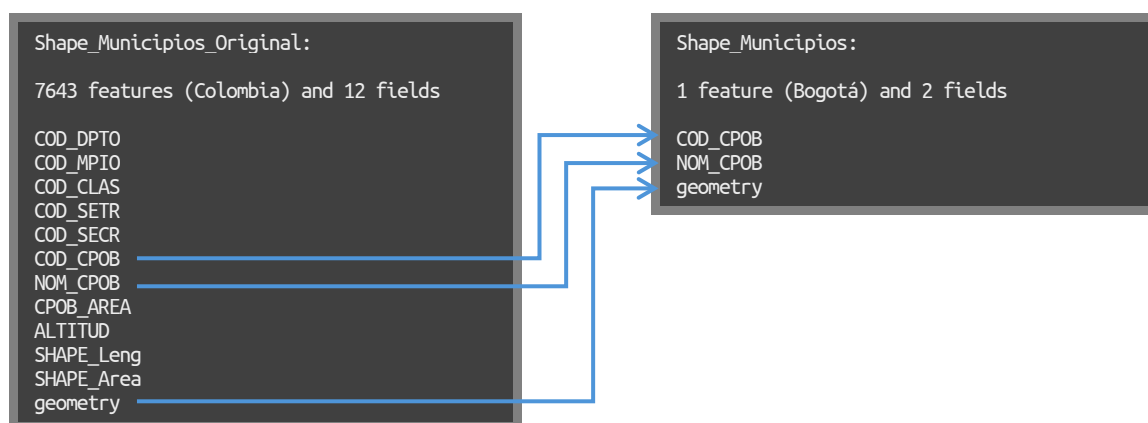


Figure 10. Schematic of the processing of Municipalities vector layers. Source: Author's own elaboration.

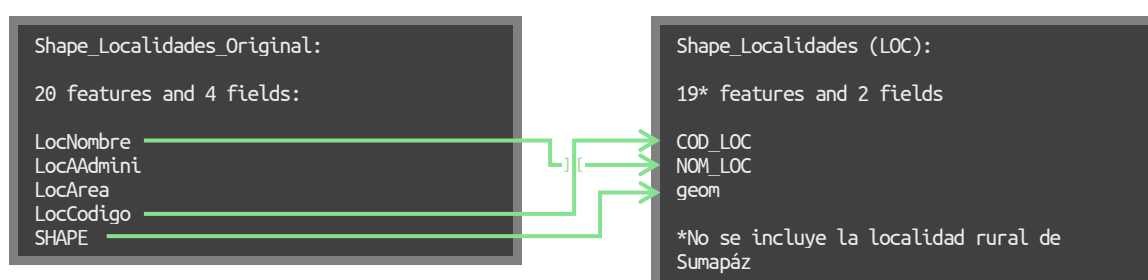


Figure 11. Schematic of the processing of Localities (LOC) vector layers. Source: Author's own elaboration.

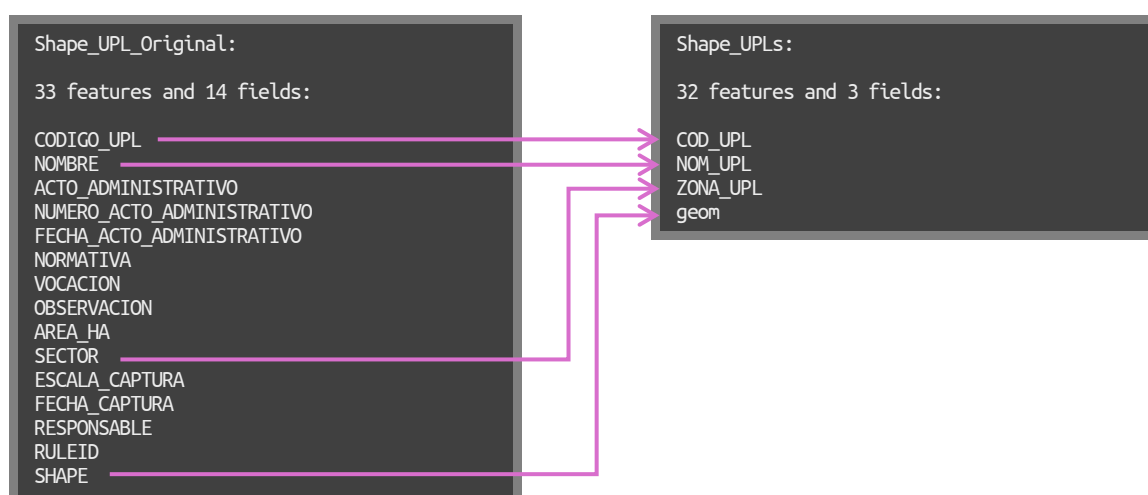


Figure 12. Schematic of the processing of Local Planning Units (UPL) vector layers. Source: Author's own elaboration.

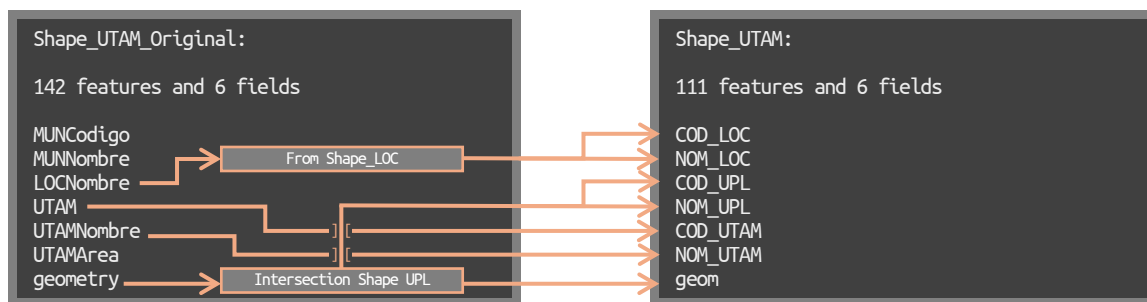


Figure 13. Schematic of the processing of Territorial Units of Mobility Analysis (UTAM) vector layers. Source: Author's own elaboration.

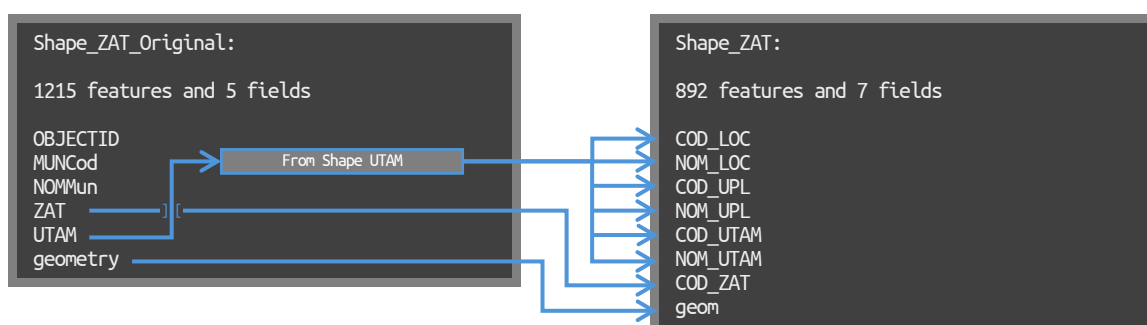


Figure 14. Schematic of the processing of Traffic Analysis Zones (ZAT) vector layers. Source: Author's own elaboration.

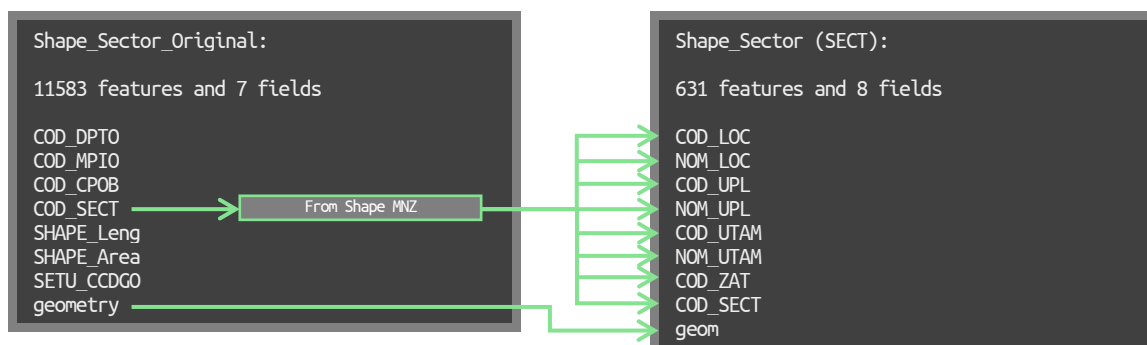


Figure 15. Schematic of the processing of Sectors (SECT) vector layers. Source: Author's own elaboration.

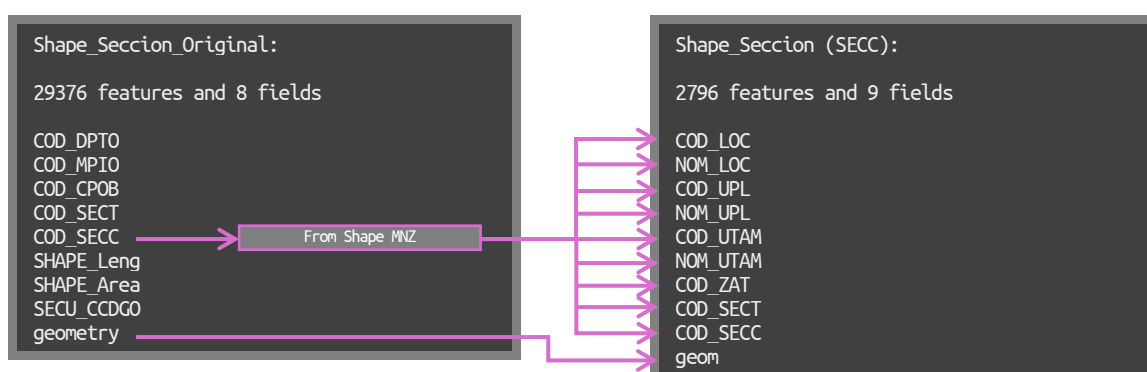


Figure 16. Schematic of the processing of Sections (SECC) vector layers. Source: Author's own elaboration.

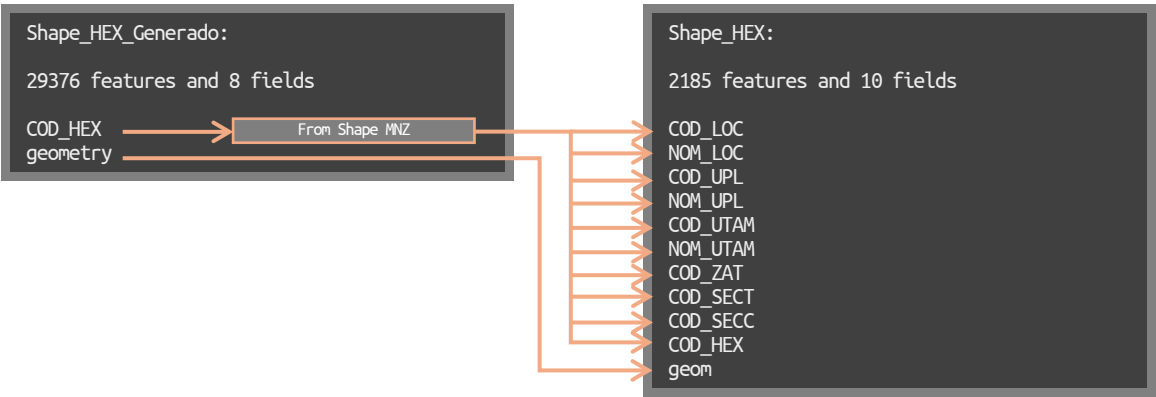


Figure 17. Schematic of the processing of Hexagons (HEX) vector layers. Source: Author's own elaboration.

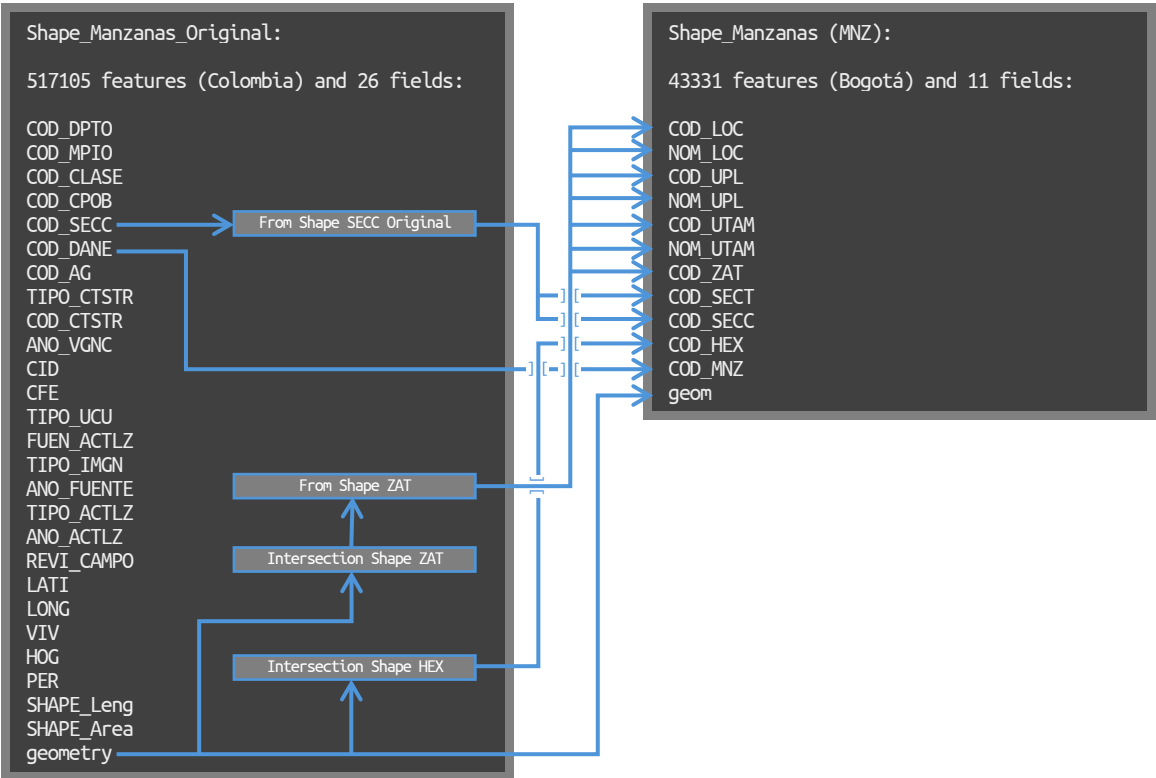


Figure 18. Schematic of the processing of Blocks (MNZ) vector layers. Source: Author's own elaboration.

Generally, the vector layers or shapes remained interrelated (similar to a relational database) by means of the variables corresponding to each administrative or geographic level. With this foundation, subsequent analyses for the development of this work were proceeded with.

# Manzanas	Hexágonos	Secciones	Sectores	ZAT	UTAM	UPL	LOC
# LOC	LOC	LOC	LOC	LOC	LOC	UPL	LOC
# UPL	UPL	UPL	UPL	UPL	UPL		
# UTAM	UTAM	UTAM	UTAM	UTAM	UTAM		
# ZAT	ZAT	ZAT	ZAT	ZAT			
# SECT	SECT	SECT	SECT				
# SECC	SECC	SECC					
# HEX	HEX						
# MNZ							

R Code Snippet 2. Relationship between the different administrative or geographic levels of the resulting vector layers. Source: Author's own elaboration.

The different spatial analysis units are shown below. This is to place the reader in context for when reference is made to these units subsequently.

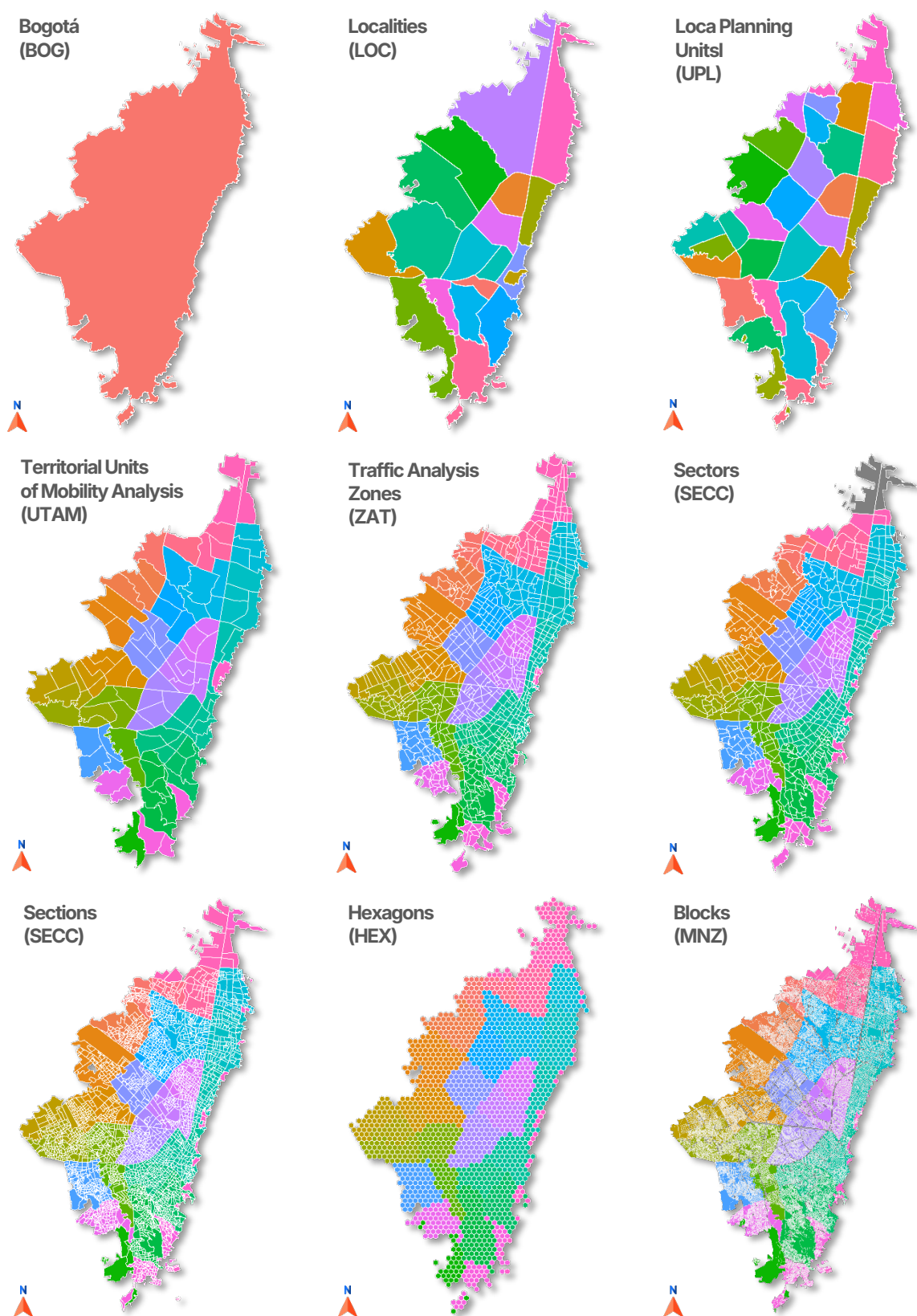


Figure 19. Graphic representation (maps) of the Spatial Analysis Units of Bogotá. Source: Author's own elaboration with data from (Alcaldía de Bogotá, 2019), (DANE, 2018), (Secretaría Distrital de Movilidad, Bogotá, DC., 2023)

3.3.2. Demographic Information

Information from the National Population and Housing Census (DANE, 2018) was utilised, which contains survey details at the person level. In this way, the population can be characterised and classified by sex, age, occupation, and educational level, among others. In addition to the above, individual information per person contains grouping data by family and block (*manzana*). This latter grouping corresponds to and coincides in coding with that existing in the Block vector layers accompanying the survey data. Thus, the individual information of persons and families is georeferenced with a maximum resolution of the Block level.

Processing of Demographic Information

Through spatial analysis techniques (Script `2_Population.R`), population and population density maps were generated at the different administrative levels of the city. Additionally, because the smallest level for map creation is that of the block, databases were created at the household, family, and person levels. The latter, with a level of detail including information by gender (male/female distribution), age ranges, occupation, and whether the person has reduced mobility. This detail of information at the citizen level allowed for the propagation of that same information (gender, occupations, reduced mobility information, etc.) in a grouped manner to all superior levels. This enabled the identification not only of where people are, but where the populations most vulnerable to transport exclusion are located.

```
>head(Censo_Personas)
  COD_LOC COD_UPL (...) COD_HEX          COD_MNZ          COD_VIVIENDA
1      13      32 (...)   1327 1100110000000051110309 1100110000000051110309_1
2      13      32 (...)   1327 1100110000000051110309 1100110000000051110309_1
3      13      32 (...)   1327 1100110000000051110309 1100110000000051110309_1
4      13      32 (...)   1327 1100110000000051110309 1100110000000051110309_1
5      13      32 (...)   1292 1100110000000051110111 1100110000000051110111_1
6      13      32 (...)   1292 1100110000000051110111 1100110000000051110111_1
...      ...      ...      ...      ...      ...
7130646    11      8 (...)   1900 1100110000000091190314 1100110000000091190314_1

  COD_FAMILIA COD_PERSONA SEXO  EDAD  LUG_NAC  DISCAPACIDAD  OFICIO
1          60400           1   M 10-14  Bogota          No Estudia
2          60400           2   F 45-49  Otro           No Trabaja
3          60400           3   F 20-24  Bogota          No Estudia
4          60400           4   M 50-54  Otro           No Trabaja
5          60404           5   F 55-59  Bogota          No Trabaja
6          60404           6   F 75-79  Otro           No Hogar
...      ...      ...      ...      ...      ...
7130646    902442737      7130646   F 20-24  Otro           No Estudia
```

R Code Snippet 3. Sample of the structure of the resulting data table of the population of Bogotá. Source: Author's own elaboration.

3.3.3. Mobility Survey (EM2023)

One of the most important data sources for the analysis of displacement times and accessibility in the city is the 2023 Mobility Survey of Bogotá (Secretaría Distrital de Movilidad, Bogotá, DC., 2023). This served as the basis for the characterisation of trips in the current scenario, and for the calibration of the model with which the other scenarios were evaluated. generally, the Mobility Survey contains important information that can serve for decision-making and for the implementation of public policies and projects regarding mobility in Bogotá, such as those developed and proposed in this work.

Processing of Mobility Survey Data

The survey is based on two main forms from which important data of both a qualitative and quantitative order are obtained. For the exercise of this thesis, exclusively qualitative information will be taken into account:

- **Interception Survey (EI):** This data table is filled out by officials of the National Administrative Department of Statistics (DANE), intercepting people on the street and asking a series of questions related to the way they move around the city.
- **Household Survey (EH):** In the household survey, DANE officials interview families about how they move around the city. Being a survey taken physically in households, it is conducted with more time and detail than the Interception survey.

The master table of trips was structured (Script 3_EM2023.R) by linking key attributes: origin, destination, trip purpose (work, study, care, etc.), transport mode (walking, cycling, car, public transport, etc.), gender, socioeconomic stratum, and reported travel time. This master table, with standardised data, allowed analyses to be performed in a simpler and more reliable manner.

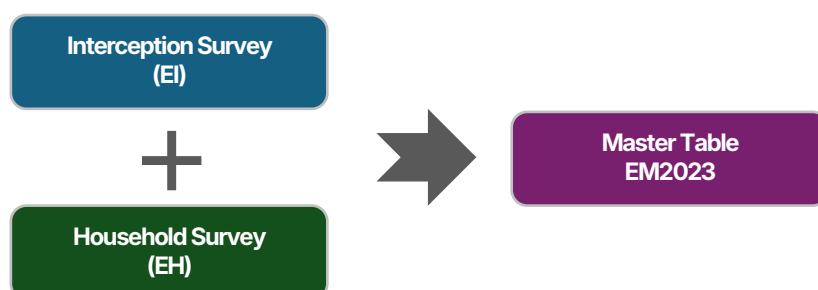


Figure 20. Unification scheme of tables from the Interception and Household Surveys that are part of the 2023 Mobility Survey. Source: Author's own elaboration.

Analysis of the Mobility Survey

The 2024 Mobility Survey allows for the generation of the O-D Matrix of the "Baseline", the identification of the current modal split, the distribution of trips by gender and socioeconomic stratum, the determination of the main trip purposes of the Bogotá population, and the calculation of average travel times perceived by users—vital inputs for calibrating transport models.

Thanks to the fact that the minimum geographic unit with which data are recorded in the Mobility Survey is the ZAT, survey data can be related to maps from that level and superior administrative levels. However, for the purposes of this work, the analysis based on the EM2023 will be performed exclusively at the ZAT level and its aggregation by UPL.

Taking into account the information from the O-D Matrix, the pattern of trip flows was visually represented on the city map, so that it is easier for the reader of this document to understand not only where trips originate and are destined, but where the greatest flow of trips moves in Bogotá. For this, the thread technique was used, whereby each trip is represented by a semi-transparent thread (normally with an opacity of around 5%). The accumulation of threads in the different corridors of the city allows one to see approximately where Bogotanos travel.

The code for the creation of this map and the diagrams accompanying the analysis of the information supplied by the EM2023 can be found in the Script `3_EM2023.R` on GitHub.

3.3.4. Facilities and Services Inventory

To evaluate proximity, it is necessary to have an inventory of georeferenced urban facilities. Due to the high complexity of acquiring an official inventory from District authorities, the Bogotá Chamber of Commerce, and other city sources, it was decided to generate and consolidate a synthetic inventory as a georeferenced database of urban facilities, taking OpenStreetMap (OSM) and information from the National Population and Housing Census as sources. The processing of information and consolidation of the facility inventory for the current scenario can be seen in the Script `4_1_Amenities.R` in the GitHub repository.



Note: The script is named `4_1` at the beginning because it is stage 4 of the process, but concerns the facilities of Scenario 1. Scenarios 3, 4, and 6 of the same stage were named as `4_3`, `4_4`, and `4_6` respectively.

Classification of Facilities

City facilities are part of three large classification groups:

- Classification by category.
- Classification by social function.
- Classification by composite indicators.

Except for the first (category), the rest of the classifications are not mutually exclusive; that is, a type of facility may belong to more than one social function or composite indicator, but only to one type of category. Naturally, each type of facility must have a classification by category, social function, and composite indicator.

Prior to performing facility queries, the types to be considered for the accessibility calculation and HQSL index estimation were defined in accordance with the characteristics of the urban context of Bogotá, including only those effectively present in the city. The classification of facilities queried in OSM, organised by Social Function, Composite Indicator, and Category, is presented in **Appendix B**. Some facilities were queried under more than one OSM classification due to differences in their keys and values, which were omitted for the sake of brevity. The query procedure can be reviewed in Script `4_1_Amenities.R`, whilst the corresponding function is declared in Script `0_Initialization.R`.

Supply Indicators

With the results of the OSM query and a subsequent consistency analysis of the information using techniques for the treatment of high volumes of data (Big Data) in R, facility point maps were generated by category and type, and facility densities were calculated per unit area using the vector layers by Hexagon and UPL.

Likewise, the previous calculations served to represent, via graphs and maps, facilities per every 10,000 inhabitants at the UPL level, allowing for the a priori identification of zones with service deficits ("service deserts") before considering any transport intervention.

Like the OSM query process, the creation of the aforementioned maps, graphs, and the treatment of associated data can be consulted and executed via the Script `4_1_Amenities.R`.

3.4. Modelling and Simulation Tools

The evaluation of the proposed scenarios requires the capacity to calculate multimodal travel times (walking + public transport) at a large scale. To this end, a computational workflow was implemented based on two main components: The generation of standardised transport specifications and the use of high-performance routing engines.

3.4.1. Generation of Synthetic Transport Networks (GTFS Generator)

To model the projected rail systems (First Metro Line and Rail Network), which are not yet in operation and thus lack official open data, a network generation tool was developed which can be consulted and executed via the Script `5_0_GTFS_Generator.R`, available on GitHub.

This algorithm allows for the conversion of geometric alignments (kml files) of the project tracks and stations into files under the GTFS (General Transit Feed Specification) standard. The process implies:

- **Infrastructure Digitalisation:** Definition of the spatial location of stations and track alignment.
- **Operational Parameterisation:** Assignment of estimated commercial speeds, dispatch frequencies as a function of day and time, and dwell times. The foregoing are based on technical studies or estimates for each project.
- **Data Structuring:** Compilation and subsequent compression of the files `agency.txt`, `calendar.txt`, `feed_info.txt`, `routes.txt`, `shapes.txt`, `stops.txt`, `stop_times.txt` and `trips.txt`.

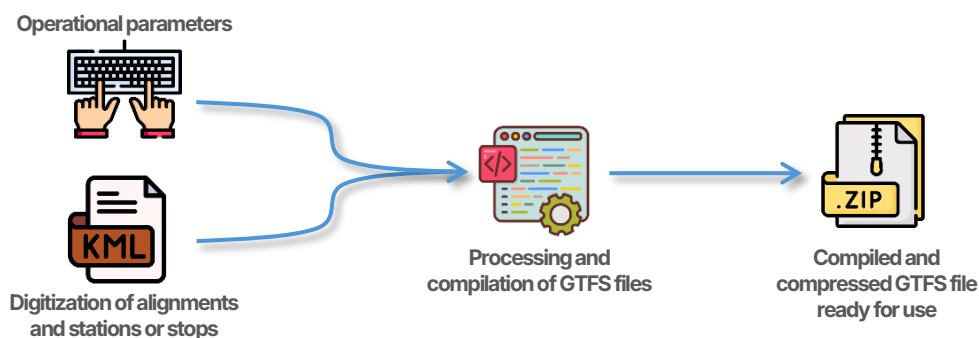


Figure 21. Simplified process of generating a GTFS file via Script `5_0_GTFS_Generator.R`. Source: Author's own elaboration.

The result is a "Synthetic GTFS" that allows the routing engine to interpret future trains and metros with the same logic with which it interprets the current system (SITP and Transmilenio), allowing for a virtual operational integration in the model.

With the aim of verifying the integrity of all GTFS files generated via this script, the validation tool recommended by gtfs.org was used: Canonical GTFS Schedule Validator, which can be found at <https://gtfs-validator.mobilitydata.org/>, showing that the generated files do not contain any type of integrity error—something that does not occur with the official GTFS of the Bogotá transport network.

Given the good performance of the GTFS file generation script developed within the framework of this research, its future implementation is projected as an open-access web tool, allowing any user to easily build custom transport networks and generate the corresponding GTFS files. Nevertheless, at the time of this document's publication, said tool is not yet available.

3.4.2. Routing Engine (r5r)

For the calculation of travel time matrices and accessibility indicators, the **r5r** package (Pereira & Herszenhut, 2024), was used, an interface in the R language for the R5 (Rapid Realistic Routing on Real-world and Reimagined networks) routing engine developed by Conveyal, which runs on Java.

The choice of this tool is founded on specialised literature (Pereira & Herszenhut, 2023), which highlights R5 for its computational efficiency in processing large urban networks and its capacity to model complex multimodal trips with realism.

The model for the development of this work integrates three layers of spatial information:

- **Pedestrian Road Network:** Obtained from OpenStreetMap (OSM), it allows for the modelling of trips and segments via walking, which also functions for access to and egress from transport stations, transfers, among others.
- **Public Transport Network:** Integrates the official GTFS of Bogotá (for the existing component), and for the proposed scenarios with rail lines, a set of synthetic GTFS.

The r5r engine also admits TIFF files to incorporate topography into mobility calculations. This Digital Elevation Model (DEM) allows for the penalisation of displacements as a function of terrain slope. In the present work, the elevation variable was not incorporated, given that the majority of the urban area, as

well as the zones of influence of the analysed rail systems, do not present slopes pronounced enough to justify the use of the DEM.

3.4.3. Transport Model Calibration

Before running the accessibility algorithm, it is necessary to calibrate the model. To this end, the input parameters of the engine (r5r) are systematically adjusted so that its results resemble as closely as possible the real data observed in the Mobility Survey.

The general methodology involves using an optimization algorithm that iteratively tests different combinations of parameters, calculates travel times with r5r for each combination, compares them with the survey data, and seeks the combination that minimises the error between the two. Naturally, to perform the travel calculations with r5r, a city scenario had to be established that was similar to the conditions under which the Mobility Survey was conducted. The most similar scenario is the current scenario of the city, and the official Bogotá GTFS was used for this purpose.

The calibration process was divided into the following steps:

a. Identification of parameters to calibrate

First, the parameters of the calculation function for obtaining the Travel Time Matrix (or O-D Matrix) that could be adjusted were selected. The most important parameters are:

- **walk_speed:** The walking speed (in km/h). This is one of the most influential parameters, as it affects the access and egress legs to public transport. The default value in r5r is 3.6 km/h. For this exercise, it will be varied between 1.8 and 7.2 km/h.
- **max_walk_time:** The maximum walking time (in minutes) that a user is willing to spend to reach or leave a stop. For this exercise, it will be varied between 15 and 60 minutes.
- **departure_datetime:** The departure date and time. For this exercise, only weekdays will be taken into account, not weekends. However, because the city's GTFS does not change between weekdays, the variation between days would not generate changes, unlike the change in departure time, which has been established to vary between 06:00 and 23:00.

b. Definition of the error function

A metric is required to quantify how different the travel times produced by the model (r5r) are from those observed in the Mobility Survey. This function will take a set of parameters, run the model, and return a single value—the error. The objective of the optimizer is to minimize this value through iterative computations while varying the parameters.

The metric used for this work is the Root Mean Square Error (RMSE), which penalises large errors more heavily. The formula for estimating this error is:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (T_{Model_i} - T_{Survey_i})^2}{n}}$$

Where:

- T_{Model_i} is the travel time calculated by r5r for the origin-destination pair i .
- T_{Survey_i} is the average survey travel time for the same pair i .
- n is the total number of origin-destination pairs being compared.

c. Choice of optimization algorithm

There are several algorithms to find the parameters that minimise the error function. For this type of problem, where each evaluation of the error function can be computationally expensive (because it implies running r5r hundreds of times), robust methods are best. Some of the most important ones are shown below:

- **Grid Search:** Only viable if there are 1 or 2 parameters with few values to test. It becomes computationally impossible very quickly.
- **Gradient-based Algorithms:** They are fast, but can get stuck in local minima if the error surface is very complex.
- **Genetic Algorithms:** They explore the solution space very efficiently and are less prone to getting stuck in local minima.
- **Bayesian Optimization:** A very powerful technique designed specifically to optimise "expensive-to-evaluate" functions. Very similar to Genetic Algorithms regarding not getting stuck in local minima.

Of the above optimization algorithms, Genetic Algorithms were selected due to the advantages they possess over the others. In the event that the optimization is not favourable, Bayesian Optimization, which is the most similar algorithm, will be implemented.

d. Implementation of the iterative process in R

For the implementation of the process in R, it was necessary to prepare the datasets with the Origin-Destination times from the Mobility Survey, the Origin-Destination points to be evaluated, and of course, the intervals of the parameters named previously (3.4.3.a.).

The complete implementation of the algorithm for model calibration can be consulted in the Script `6_0_Model_Calibration.R`.

Creation of the error function in R. This function is the heart of the process, as it receives a vector of parameters and their intervals, executes the `r5r` engine, and returns the RMSE:

```
Calcular_RMSE <- function(Parametros) {  
  # 1. Extract the parameters being tested in this iteration  
  Velocidad_Caminata <- Parametros[1] # The walking speed parameter  
  Tiempo_Max_Caminata <- Parametros[2] # The max walking time  
  Hora_De_Salida <- floor(Parametros[3]) # The hour parameter (Integer part)  
  # Pass the remainder to minutes:  
  Minuto_De_Salida <- round((Parametros[3]-floor(Parametros[3]))*60, digits = 0)  
  Tiempo_De_Partida <- as.POSIXct(paste("2024-07-10 ", Hora_De_Salida, ":", Minuto_De_Salida, ":00", sep =  
  ""))  
  
  # 2. Execute the travel time matrix calculation with r5r  
  Matriz_Tiempos_Modelo <- r5r::travel_time_matrix(  
    r5r_core = r5r_core_calibration,  
    origins = Origenes_Calibracion_sf,  
    destinations = Destinos_Calibracion_sf,  
    mode = c("WALK", "TRANSIT"),  
    departure_datetime = Tiempo_De_Partida,  
    walk_speed = Velocidad_Caminata,  
    max_walk_time = Tiempo_Max_Caminata,  
    percentiles = 50 # Using the median (50th percentile) is best practice  
  )  
  
  # 3. Join model results with survey results  
  Comparacion <- merge(Matriz_Tiempos_Modelo, Tiempos_Encuesta_df,  
    by.x = c("from_id", "to_id"),  
    by.y = c("from_id", "to_id"))  
  
  # 4. Calculate the error (RMSE)  
  RMSE <- sqrt(mean((Comparacion$travel_time_p50 - Comparacion$Tiempo_EM2023)^2, na.rm = TRUE))  
  
  # Print progress:  
  cat("Testing model with:",  
    "\n Walk_speed: ", Velocidad_Caminata,  
    "\n | Max_walk_time: ", Tiempo_Max_Caminata,  
    "\n | Departure time: ", Hora_De_Salida, ":", Minuto_De_Salida,  
    "\n -> RMSE: ", RMSE, "\n", sep = "")  
  
  return(RMSE)  
}
```

R Code Snippet 4. Function for RMSE calculation. Source: Author's own elaboration.

Execution of the optimizer. Using the GA package in R for Genetic Algorithms, the optimizer was run by establishing limits for each of the parameters between the intervals mentioned in point **3.4.3.a**. The code for execution is shown below:

```
library(GA)

# Define limits for parameters
# Lower and upper limit for walk_speed (in km/h)
Limite_Inferior <- c(1.8, 15, 6) # walk_speed > 1.8 km/h, max_walk > 10 min, Min hour: 6:00
Limite_Superior <- c(7.2, 60, 23) # walk_speed < 7.2 km/h, max_walk < 60 min, Max hour: 23:00

# Execute the genetic algorithm
# GA maximizes, so we must minimize the negative of our error function
Resultado_GA <- ga(
  type = "real-valued",
  fitness = function(params) -Calcular_RMSE(params), # Minimizing RMSE is maximizing -RMSE
  lower = Limite_Inferior,
  upper = Limite_Superior,
  popSize = 20, # Number of "individuals" (solutions) per generation
  maxiter = 100, # Maximum number of generations
  run = 50, # Stopping criterion if there is no improvement
  monitor = TRUE # Shows progress
)
```

R Code Snippet 5. Code for the Genetic Algorithms optimizer in R. Source: Author's own elaboration.

Once the optimal data for the model calibration were obtained, the integrity and sense of these values within the context of Bogotá's mobility were verified, specifically regarding walking speeds and times, which were compared with those of the Mobility Survey O-D Matrix.

```
> summary(Resultado_GA)
——— Genetic Algorithm ———

GA settings:
Type           = real-valued
Population size = 50
Number of generations = 100
Elitism         = 2
Crossover probability = 0.8
Mutation probability = 0.1

Search domain =
  x1 x2 x3
lower 1.8 15 6
upper 7.2 60 23

GA results:
Iterations           = 100x50
Fitness function value = 32.19441
Solutions =
      x1      x2      x3
[1,] 2.818053 10.27883 7.99333

# The best parameters found
Parámetros óptimos encontrados
Walk_speed: 2.818053
Max_walk time: 10.27883
Time: 2024-07-10 07:56:00
Mejor RMSE: 32.19441
```

R Code Snippet 6. Optimization results for model calibration. Source: Author's own elaboration.

The comparison matrix was used not only to compare real times with synthetic ones, but to calculate means and observe data behaviour via a scatter plot and histogram.

```
# Comparacion_df <- merge(Matriz_Tiempos_Viaje, Tiempos_Encuesta_df,
                        by.x = c("from_id", "to_id"),
                        by.y = c("from_id", "to_id")) |>
mutate(
  # Absolute error: the direct difference in minutes:
  Error_Absoluto = travel_time_p50 - Tiempo_EM2023,
  # Percentage error: useful for comparing errors in short vs long trips:
  Error_Porcentual = (Error_Absoluto / Tiempo_EM2023) * 100
)

Comparacion_df
Key: <from_id, to_id>
  from_id to_id travel_time_p50 Tiempo_EM2023 Error_Absoluto Error_Porcentual
  <char> <char>      <int>      <num>      <num>      <num>
1:   1003   176         103        180.0        -77.0       -42.777778
2:   1003   217         115        105.0         10.0        9.523810
3:   1003   230          97        100.0          3.0       -3.000000
4:   1003   254         102         61.0         41.0       67.213115
5:   1003   275         100         90.0         10.0       11.111111
---
166:   558   546          47         62.5        -15.5       -24.800000
167:   558   560          20         60.0        -40.0       -66.666667
168:   558   567          44         60.0        -16.0       -26.666667
169:   558   990          68         90.0        -22.0       -24.444444
170:   558   996          92         95.0         3.0       -3.157895
```

R Code Snippet 7. Comparative table of O-D times between model data and EM2023. Source: Author's own elaboration.

Upon reviewing the data, it was observed that the results were not only reasonable but revealed very useful information about the model.

- **Optimal walking speed parameter:** The optimal parameter found was 2.82 km/h, which is relatively close to the default value of 3.6 km/h. This is a walking speed that falls within the range of speeds accepted in transport studies (generally between 2.5 and 5.0 km/h). It is also widely recognised that Latin Americans have a lower walking speed—possibly due to physiognomy—compared to standard models from North American and European countries, where models are generally calibrated and default values established.
- **Optimal maximum walking time parameter:** The optimal parameter found was 10.28 minutes. This time is only 28 seconds away from the 9.59 minutes average time of the Mobility Survey data, which is also a good indicator that the parameter is well calculated.
- **Departure time parameter:** The optimal value found is 07:56:00, which coincided with the morning peak hour.
- **Similarity of means:** The difference between both means was only 2.16 minutes (Survey mean: 87.74 min vs Model mean: 85.58 min). This indicates that the model does not have a systematic bias and did not deviate from the survey data. In other words, on average, the model was not calculating trips systematically much faster or slower than reality. At an aggregated level, it captured the central tendency of travel times very precisely.

The fact that the optimization algorithm converged on such realistic values indicates that the calibration process functioned correctly and did not force parameters to strange values to minimise error.

The RMSE of 32.19 at first glance may seem like a high error, but in the context where the means are nearly identical, it says something very specific: Although on average the model was accurate, the dispersion of the error was high. That is, for some Origin-Destination (O-D) pairs it predicted times much shorter than reality, and for others, times much longer. To see this more clearly, the dispersion was plotted with the survey time on the X-axis and the model time on the Y-axis. If the model were perfect, all points would lie on the diagonal line ($Y=X$), which is impossible to achieve. From the results, one can expect:

- **Similar means:** Means that the cloud of points is centred around that diagonal line.
- **High RMSE:** Means that, although the cloud is centred, the points are very dispersed and far from the line.

Obtaining a high RMSE with similar means could be due to:

- **Unmodeled complexity:** r5r is an ideal model. It does not capture real-world variability such as peak hour traffic congestion, unforeseen delays in public transport, or the time people actually take to find or access it.
- **Traveller heterogeneity:** The walking speed of 2.82 km/h is only an average. In reality, there are people who walk much faster and others much slower. The model cannot capture this variability on its own. Even the walkability of the city can be very variable, depending on location, infrastructure conditions, and safety. Inconsistencies may also present themselves in the synthetic pedestrian network used by r5r to determine times.
- **Specific O-D pair problems:** It is very likely that the error is not distributed uniformly. For example, it could be that the model is very imprecise for very long trips, or for trips requiring many transfers, whilst being very precise for short and direct trips, or vice-versa.
- **Limitation in the number of parameters to evaluate:** Unfortunately, r5r does not allow for the adjustment of many parameters; arriving at a more advanced level of calibration—extracting and simulating different scenarios separately and in detail to manually attempt to adjust the model to reality—could be too cumbersome and time-consuming, as well as being outside the scope of a work such as the present one.

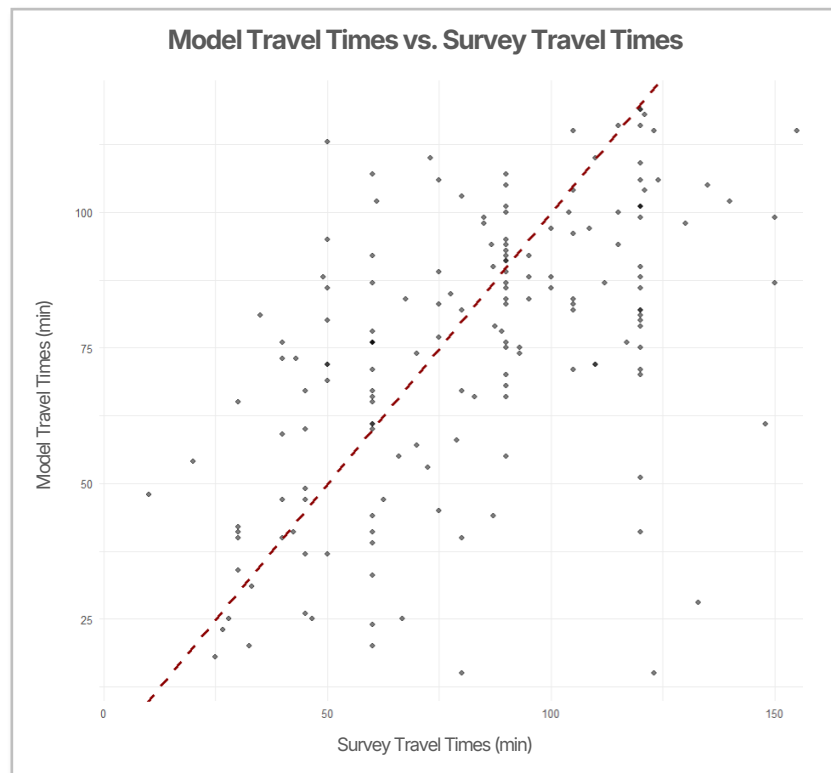


Figure 22. Scatter plot of average model travel times versus the survey. Source: Author's own elaboration.

As observed in **Figure 22**, there is great dispersion, which is in concordance with having obtained a high RMSE. Points above the red line are trips the model overestimates (predicts as slower). Points below are trips it underestimates (predicts as faster).

In the corresponding histogram (**Figure 23**), it is expected that the error distribution is centred near zero (the red line). The large dispersion on both sides of this value is what generates a high RMSE.

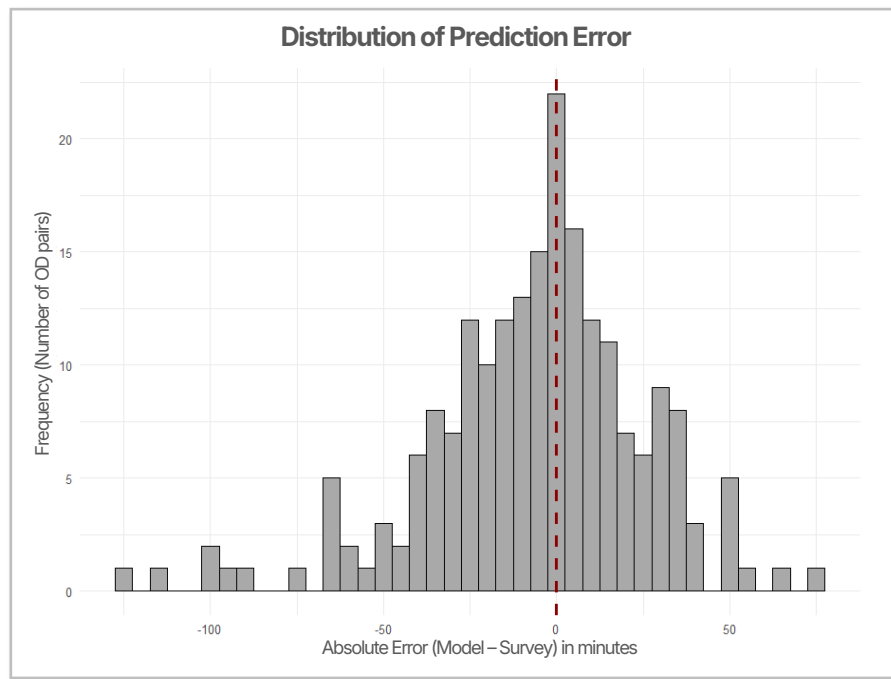


Figure 23. Histogram of prediction error distribution. Source: Author's own elaboration.

Having validated the integrity of the optimal parameters obtained as a result of the model calibration, these could be used for all scenarios where a synthetic model for Bogotá running with the r5r engine was involved.

3.4.4. Accessibility Calculation

For the calculation of accessibility, the algorithm generates the number of isochrones established as time parameters, which, for the case of this work, are 15 and 30 minutes. The origin of the isochrones was established as the centroid of each of the more than 40,000 blocks (manzanas) in the city.

Once the isochrones are generated, the algorithm filters the points of interest (facilities) falling within them, and calculates a Travel Time Matrix (Origin-Destination Matrix or O-D Matrix) from the block centroids towards the destination facilities, considering waiting times, transfers, and walking. As a result, the algorithm returns a matrix with the number of facilities accessible to each block identifier, for each of the established time units (15 and 30 minutes).

An example of said matrix is shown below. The id column represents the block identifier; opportunity is the classification of the facility, whether by category, social function, or composite indicator. Note that in this example, "Learning" and "Enjoying" are two of the social functions; "Well-being" and "Sociability" are composite indicators; and "Education and Culture" and "Health and Care" are facility categories. The **percentile** column indicates that the accessibility level was calculated using the mean of travel times from

the O-D Matrix—recalling that the 50th percentile is the mean. The `cutoff` column represents the cut-off times established at the moment of running the algorithm (in our case: 15 and 30 minutes), and the `accessibility` column is the number of facilities within that time to which the block has access.

	id	opportunity	percentile	cutoff	accessibility
45	1100110000000011010102	Learning	50	15	16
46	1100110000000011010102	Learning	50	30	167
47	1100110000000011010102	Enjoying	50	15	137
48	1100110000000011010102	Enjoying	50	30	1125
49	1100110000000011010102	Well_being	50	15	708
50	1100110000000011010102	Well_being	50	30	5399
51	1100110000000011010102	Sociability	50	15	105
52	1100110000000011010102	Sociability	50	30	898
53	1100110000000011010102	Education_and_Culture	50	15	16
54	1100110000000011010102	Education_and_Culture	50	30	203
55	1100110000000011010102	Health_and_Care	50	15	9
56	1100110000000011010102	Health_and_Care	50	30	129

R Code Snippet 8. Example of the matrix resulting from the accessibility calculation using r5r. Source: Author's own elaboration.

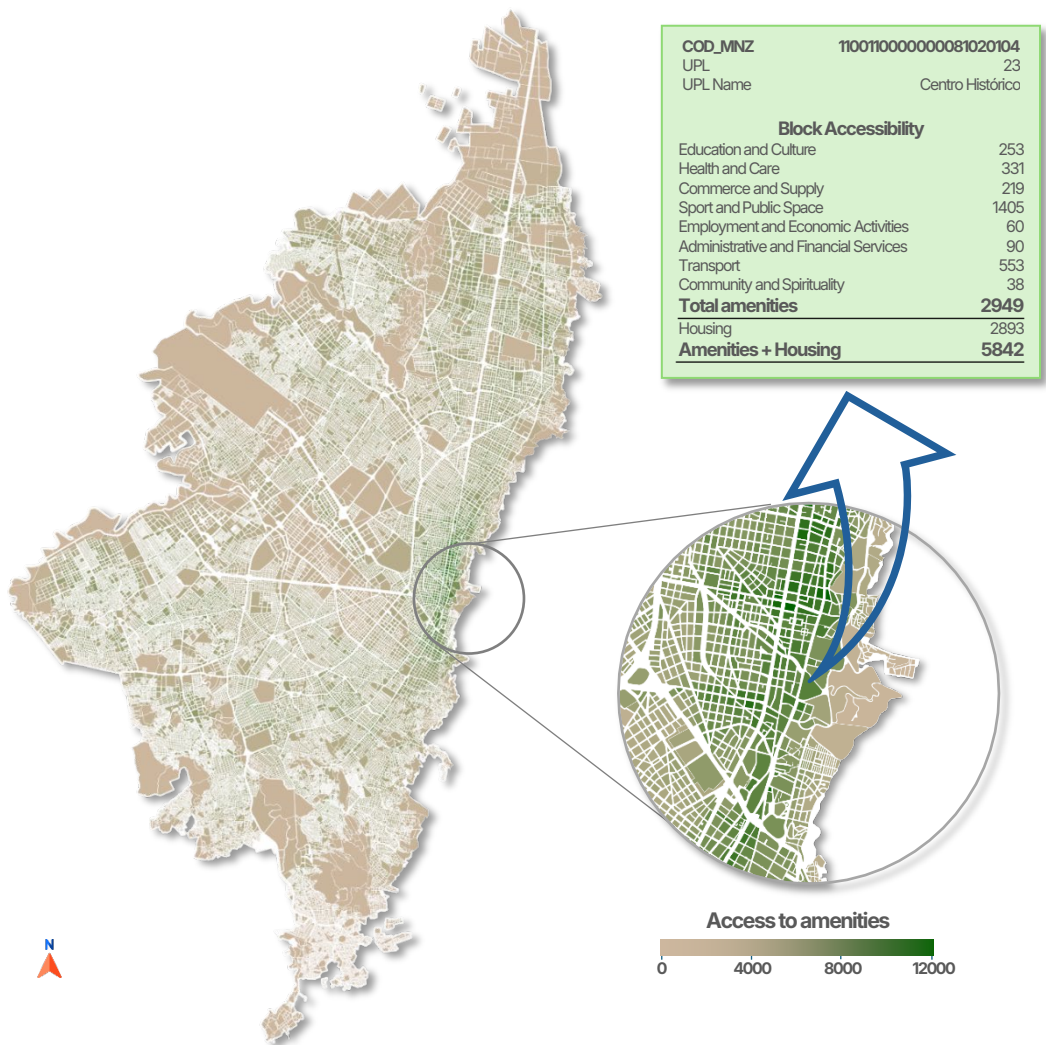


Figure 24. Representation of the accessibility level at the block scale. Source: Author's own elaboration.

Figure 24 presents an example of accessibility at the block scale. Based on the geoprocessing of the accessibility calculation results, each vector layer—corresponding to a block—incorporates information on the facilities accessible from it. Although the example shows information disaggregated by category, the source data table also contains information related to social functions and composite indicators, which are useful for the HQSL calculation.

The accessibility calculation for the different scenarios comprising this thesis can be consulted in the GitHub Scripts `6_x_Accessibility_x.R`, where `x` represents the number of the scenario of interest.

3.4.5. Calculation of the High Quality of Societal Life (HQSL) Index

The representation at the block level allows for the identification of concentrations of blocks with high levels of accessibility to facilities, as well as "islands" with significant deficits. Nevertheless, to estimate the HQSL index at the Local Planning Unit (UPL) scale, it was necessary to aggregate the results to said territorial level using data science and geoprocessing techniques implemented in R.

To transfer the datasets resulting from the accessibility calculation from the block scale to the UPL scale, performing a simple grouping of information was insufficient. It was also necessary to apply a data normalisation process, taking as a reference the highest accessibility level recorded for each social function and composite indicator within each UPL. This procedure allows for consistent comparisons between different areas of the city and, subsequently, for the calculation of the HQSL index and the integration of the resulting information into a Proximity ScoreCard for each Local Planning Unit.

The normalisation applied to each value of the vector layer is expressed as follows:

$$x'_{ij} = \frac{x_{ij}}{\max(x_{i1}, x_{i2}, \dots, x_{ik})} \times 100$$

Where:

- x'_{ij} : Is the normalised value of UPL i , for the social function or composite indicator j .
- x_{ij} : Is the original accessibility value calculated for UPL i , for the social function or composite indicator j .
- $\max(x_{i1}, x_{i2}, \dots, x_{ik})$: Is the maximum value found among all social functions, or all composite indicators, for the same UPL i .

It is important to note that the normalisation process is independent for social functions and composite indicators, and these are not mixed, even though they are stored in the same tabular data structure for processing. In Script `6_x_Accessibility_x.R` (where `x` represents the number of the scenario of interest), the normalisation process can also be observed. The function for this process is declared in Script `0_Initialization.R` as `Normalizacion()`.

With the information on normalised accessibility, the construction of the HQSL Index proceeded. There are several methodologies for constructing the index (Moreno, 2023), but for the realisation of this work, the urban social functions methodology was chosen, as follows:

$$HQSL = iH + jT + kS + lE + mA + nD$$

Where:

- *H*: Living
- *T*: Working
- *S*: Supplying
- *E*: Caring
- *A*: Learning
- *D*: Enjoying
- *i, j, k, l, m, n* are weighting coefficients.

Due to the synthetic nature of this work, it was decided not to use weighting coefficients, as this would require a much more detailed study of the territory, which is outside the scope of this study. Thus, the value for coefficients *i, j, k, l, m, n* was set equal to 1.

Proximity ScoreCard

As seen in **Figure 9**, the ScoreCard is a card that integrates a set of indicators allowing for the comparison of study areas regarding proximity, providing a panoramic view of the situation of each one with respect to the rest of the city. The ScoreCard takes information from the proximity table and the scores obtained for each of the Social Functions and Composite Indicators. The representation of data in each of the elements comprising the ScoreCard is explained below:

- **Header:** The ScoreCard header provides information about the study UPL, the scenario to which the accessibility calculation belongs, and the temporal limit for which the model was run (15 or 30 minutes).
- **Basic UPL Information:** Shows basic information of the UPL, such as population, area, and density. All of these are compared with the same indicators at the city level.
- **HQSL Index Score:** The score obtained for the High Quality of Societal Life (HQSL) Index of the UPL is visibly displayed. This score is calculated as indicated on the previous page.
- **Radar Chart for Social Functions:** The radar chart takes the score information for social functions and represents each one on an axis (Living, Working, Supplying, Caring, Learning, and Enjoying). The chart also shows the mean values obtained at the city level for each of these Social Functions. The charts are created with the R function `radarchart(df)`, where `df` (data.frame) is the accessibility data table.
- **Comparative Bars of Composite Indicators:** The comparative bars of composite indicators take the information of Well-being, Sociability, and Sustainability, and represent it via horizontal bars. The bars also show the average value of these indicators at the city level, so that a reference of the current state of the UPL in question can be obtained. The bars and their components were created using the `ggplot()` function in R, with data from the accessibility calculation result table.
- **UPL Map:** The map of the UPL is shown with visible blocks, where each one represents, via a colour level, the number of facilities to which it is accessible for the indicated scenario. The map is generated with the `ggplot()` function in R.
- **UPL Location Map:** The UPL location map is a map showing the relative location of the UPL with respect to the city. The map is generated with the `ggplot()` function in R.

4

Diagnosis of the Current State

This chapter establishes the research baseline, which henceforth will be denominated **Scenario 1**. A spatial and statistical analysis of the demographic configuration, mobility patterns, and current distribution of facilities in Bogotá is presented, identifying the structural gaps that justify the need for the proposed interventions (Scenarios 2 to 7, which will be studied in subsequent chapters of this work).

4.1. Sociodemographic Characterisation and Population Distribution

Understanding mobility begins with understanding the residential structure. Based on the processing of census data (DANE, 2018), it is evident that Bogotá presents a territorial occupation model characterised by high peripheral densities.

The spatial analysis reveals a marked population concentration in the western and southern arc of the city, encompassing the zones corresponding to the localities of *Suba*, *Engativá*, *Kennedy*, *Bosa*, and *Ciudad Bolívar*. These zones function predominantly as "dormitories", housing the city's workforce. In contrast, the central zones (*La Candelaria*, *Chapinero*, or *Teusaquillo*) present comparatively lower residential densities but concentrate economic activity, as will be seen later.

The figures below show the number of inhabitants and the population density per locality. In each one, a 3D map is included, which allows for the easier observation of the marked difference between the localities of the western arc and the central-eastern zone of the city. It is important to note that what we call the "centre" in a city, referring to the place where the greatest economic activity takes place, is found towards the eastern edge in Bogotá.

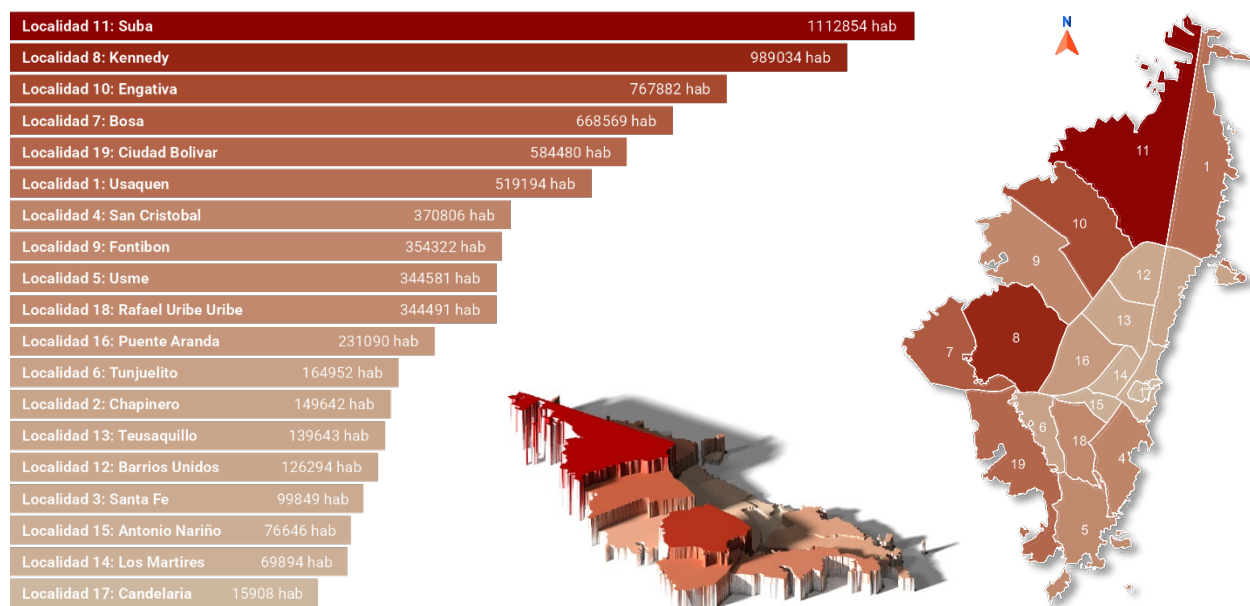


Figure 25. Population by localities. Source: Author's own elaboration with data from (DANE, 2018).

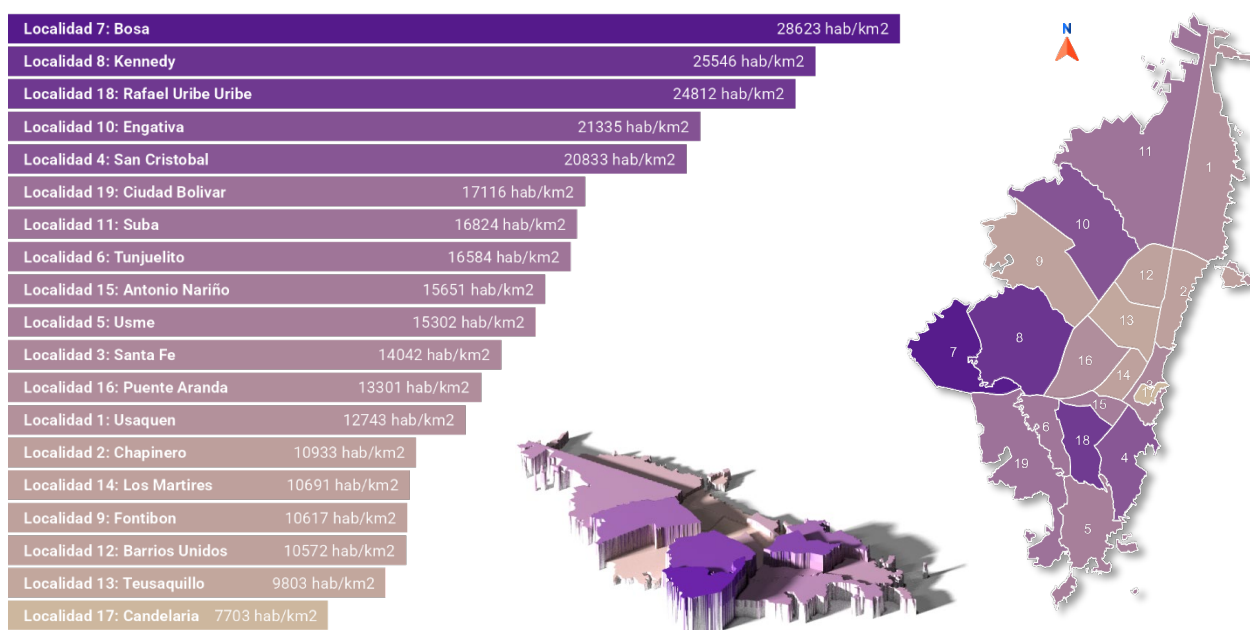


Figure 26. Population density by localities. Source: Author's own elaboration with data from (DANE, 2018).

As mentioned in Chapter 3.3.1, at the date of the National Population and Housing Census, the implementation of Local Planning Units (UPL) was not yet in force. Through the treatment of the vector layers described in that chapter, it was possible to bring the population level to the UPL scale.

Figure 27 presents the population distribution at the scale of Local Planning Units (UPL), allowing for a more detailed analysis of population concentration patterns. In this sense, the case of the Suba locality is illustrative: Although in **Figure 25** it is identified as the locality with the highest absolute number of inhabitants, said population is distributed across six UPLs, which dilutes concentration levels at this scale of analysis. In contrast, UPLs 11 and 29, corresponding to locality 10 of *Engativá*, evidence a considerably higher population concentration, something that was not as evident even in **Figure 26** of population density by localities.

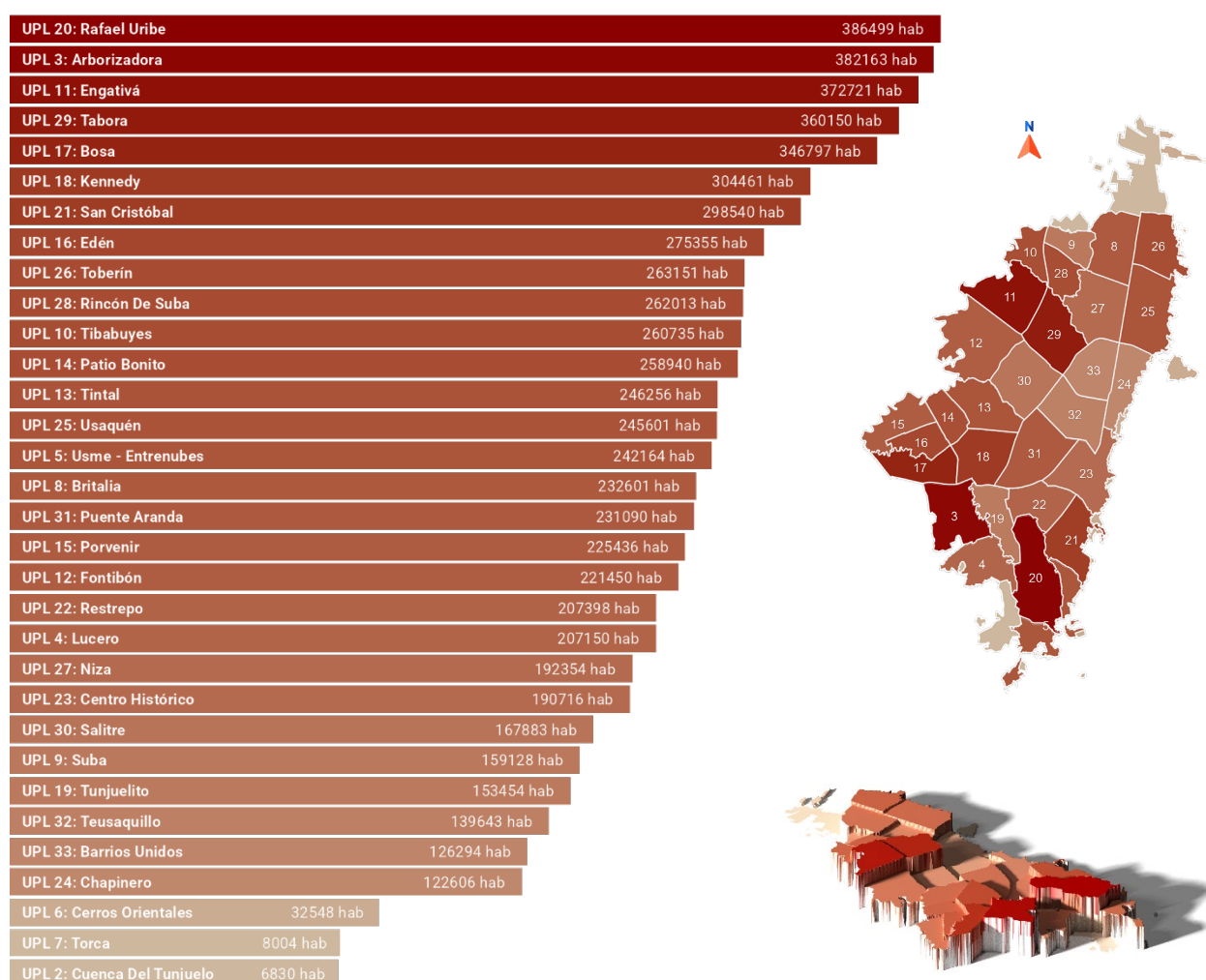


Figure 27. Population by Local Planning Units. Source: Author's own elaboration with data from (DANE, 2018).

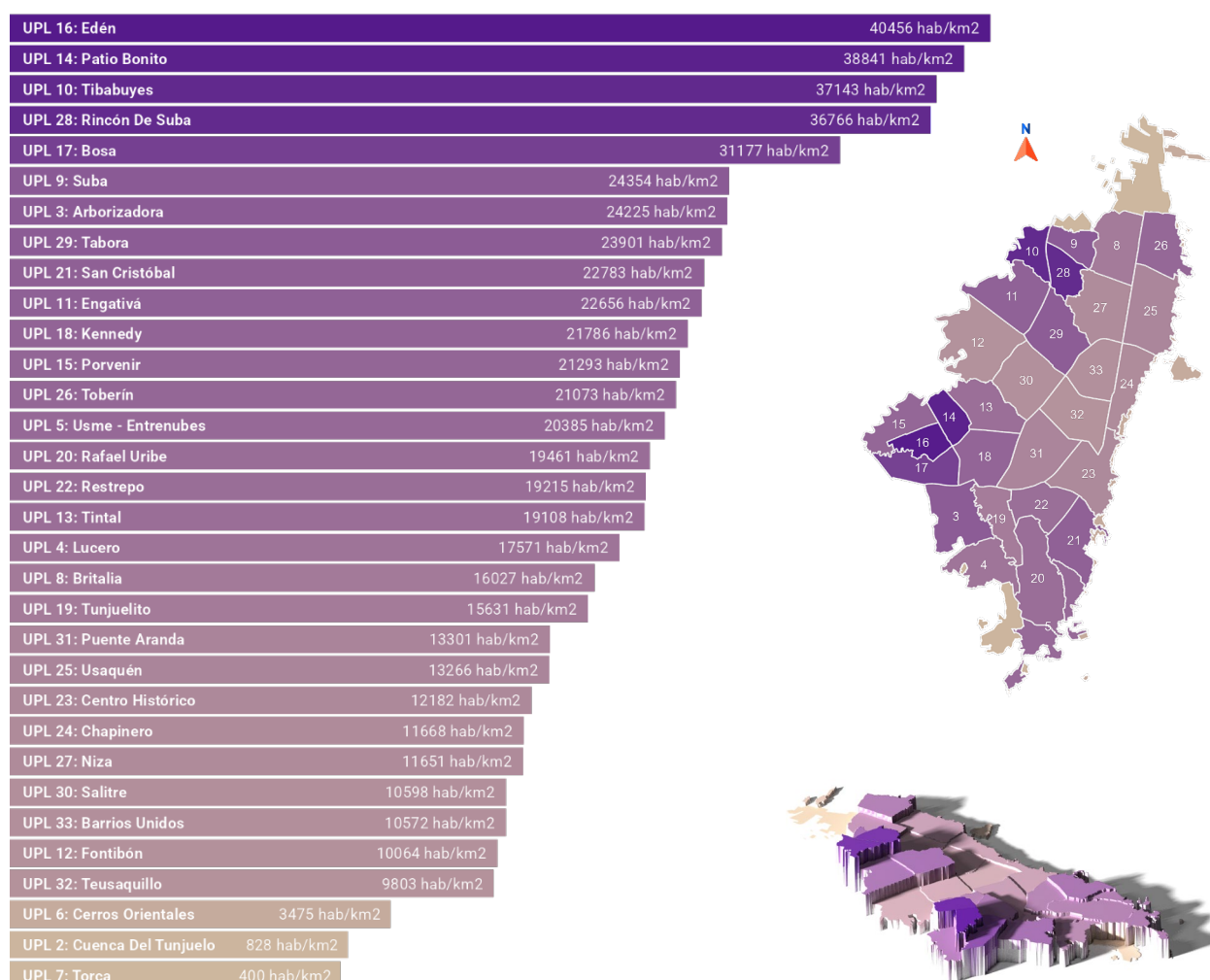


Figure 28. Population density by Local Planning Units. Source: Author's own elaboration with data from (DANE, 2018).

Demographic Profile

The disaggregation of information shows a distribution by gender and age that influences transport demand (Figure 29, Figure 30, and Figure 31). The prevalence of the population of productive and student age in zones of high peripheral density prefigures the massive pendular flows (Figure 32) analysed in the following section.

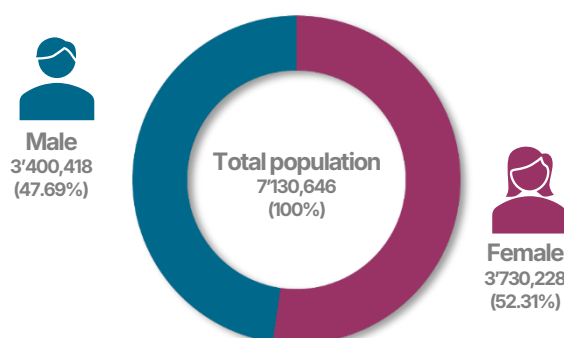


Figure 29. Distribution of inhabitants by gender. Source: Author's own elaboration with data from (DANE, 2018).

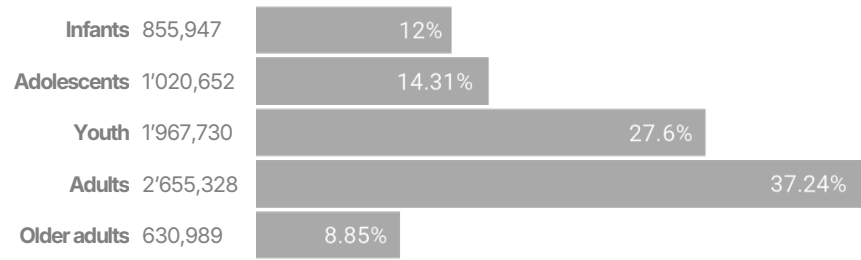


Figure 30. Population distribution by age groups. Source: Author's own elaboration with data from (DANE, 2018).

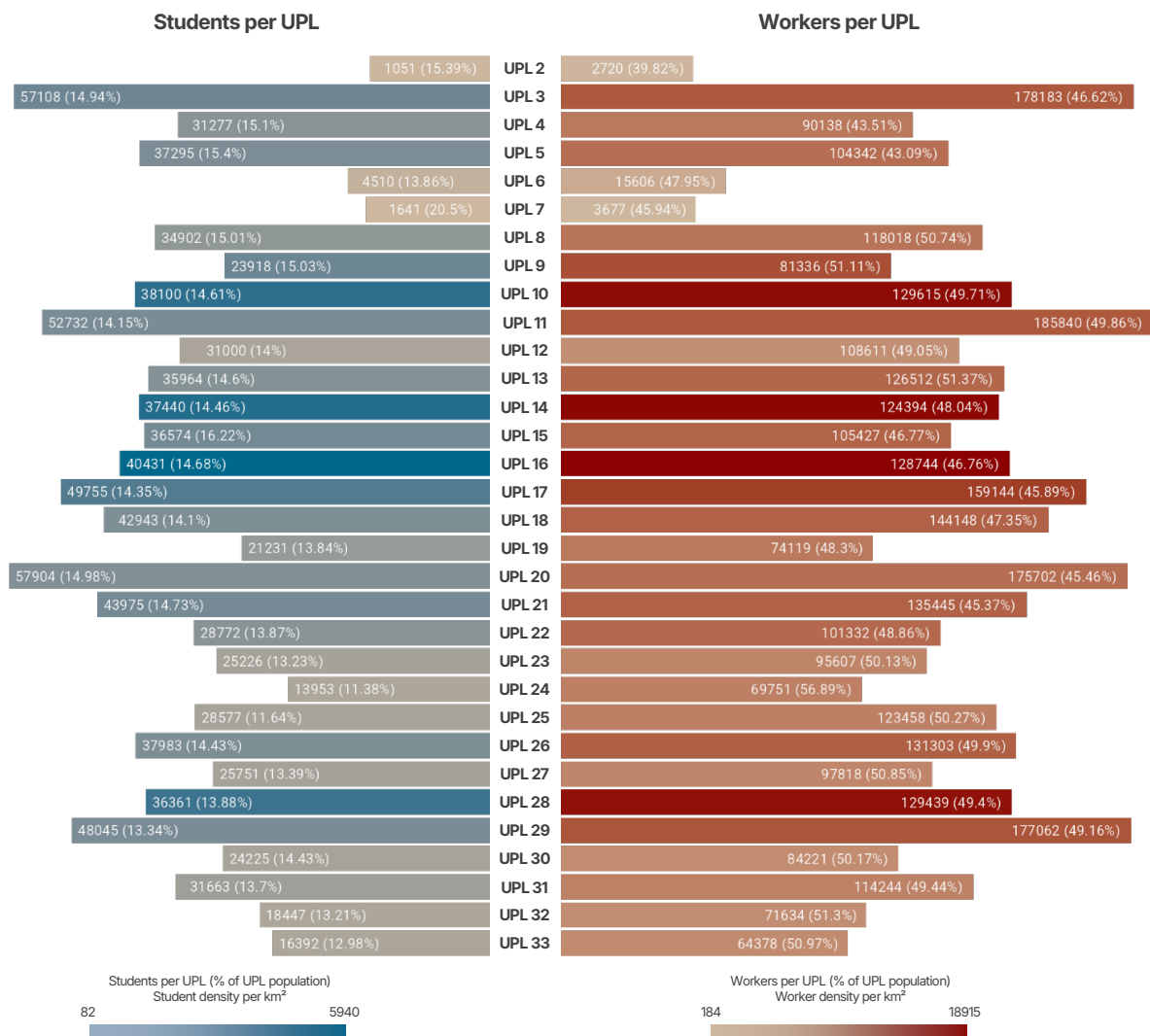


Figure 31. Student and worker population by UPL. Source: Author's own elaboration with data from (DANE, 2018).

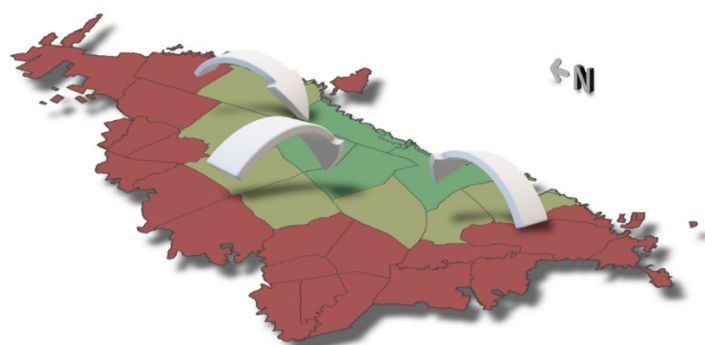


Figure 32. Expected trip flow in morning hours, from the most populated western arc towards the centre of Bogotá. Source: Author's own elaboration.

4.2. Mobility Patterns and Travel Dynamics

The 2023 Mobility Survey allowed for the characterisation of how the inhabitants of Bogotá move and at what temporal costs. The analysis of the survey can be consulted and replicated via Script `3_EM2023.R` in the GitHub repository.

4.2.1. Modal Split and Sustainable Mobility

Bogotá maintains a mobility profile with a high participation of sustainable modes, with public transport usage standing out predominantly.

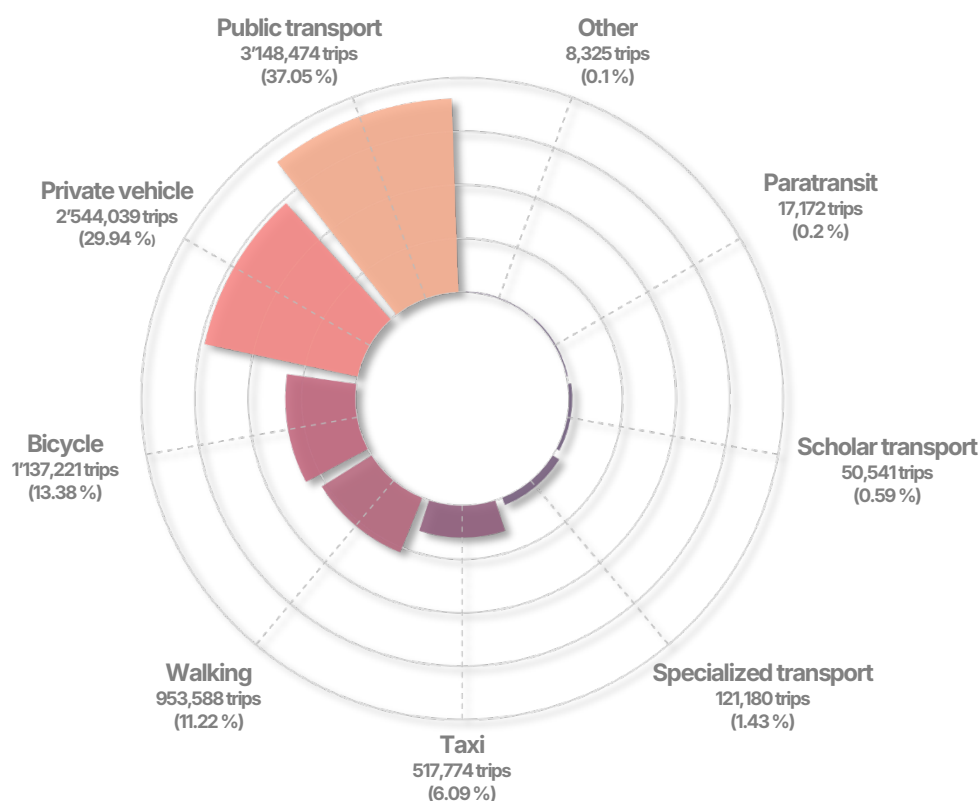


Figure 33. Modal split of Bogotá. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023)

However, private vehicle usage has a high participation. Compared to the 2019 Mobility Survey (Secretaría Distrital de Movilidad, Bogotá, DC., 2023) the increase in motorcycle usage was 11%. Of that percentage, it is estimated that 9% migrated from sustainable modes (3% from public transport, 1% from walking, and 5% from cycling).

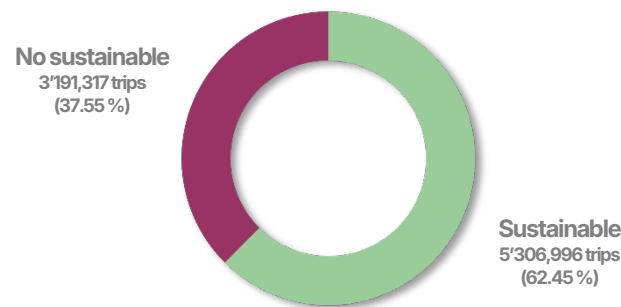


Figure 34. Distribution of sustainable and non-sustainable modes. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

Figure 34 reveals the paradox that, although a large proportion of trips are made by public transport, bicycle, and on foot, the quality of these displacements (in terms of time, comfort, and safety) is low. Dependence on public transport (SITP and Transmilenio) is critical in middle and lower strata, whilst private vehicle usage tends to increase with income level.

4.2.2. Spatial Structure of Trips

The visualisation of Origin-Destination flows represented in the thread chart in Figure 35 confirms the city's monocentric model.

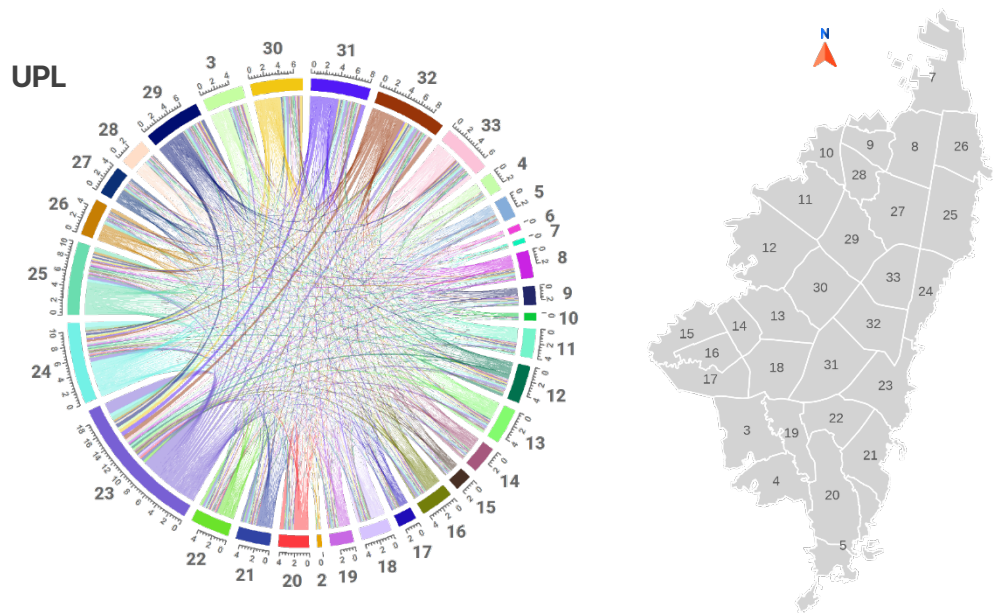


Figure 35. Thread chart for the representation of O-D flows. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

As observed in **Figure 35**, UPLs 23, 24, and 32 are major trip attractors, whilst the opposite is true for UPLs with higher population density such as 10, 28, 14, 15, 16, or 17, which are precisely the furthest from the former.

Because the smallest geographic unit of analysis used by the Mobility Survey is the Traffic Analysis Zone (ZAT), a trip flow map was produced with a random sample of 10% of these trips, with the objective of observing how trips are distributed and concentrated in different corridors. **Figure 36** below shows how trip flows are concentrated mainly on the eastern edge, which, in addition to confirming that the UPLs located there are major trip attractors, also shows they are protagonists of these flows, although the totality of these trips do not necessarily end there.

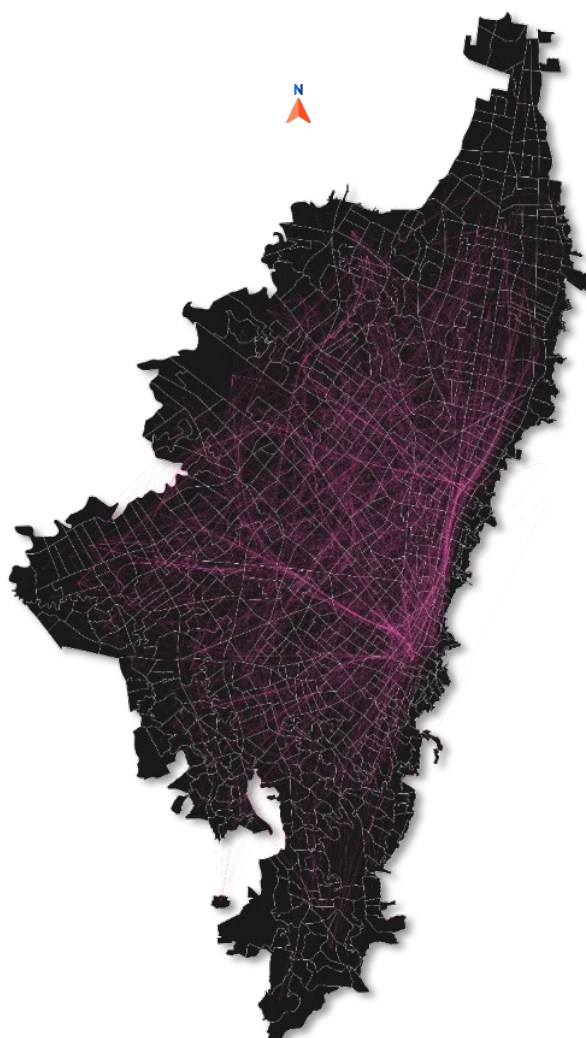


Figure 36. Representation of the O-D trip flow of Bogotá. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

In summary, these two figures (**Figure 35** and **Figure 36**) evidence a strong radial tension; that is, large volumes of trips originate in the mornings from the west and south (high residential density zones identified in section 4.1) destined for the "Eastern Edge"—or "Expanded Centre" as it is commonly known

in Bogotá—where jobs and higher education services are largely concentrated. This structure generates bidirectional pressure on the road network and the mass transit system, collapsing its capacity during peak hours.

4.2.3. Travel Times and Inequality Gaps

As stated in section 1.1, travel time is a critical indicator of the mobility diagnosis in Bogotá. **Figure 37** presents the average travel times according to socioeconomic stratum, alongside their spatial distribution and georeferencing within the city territory.

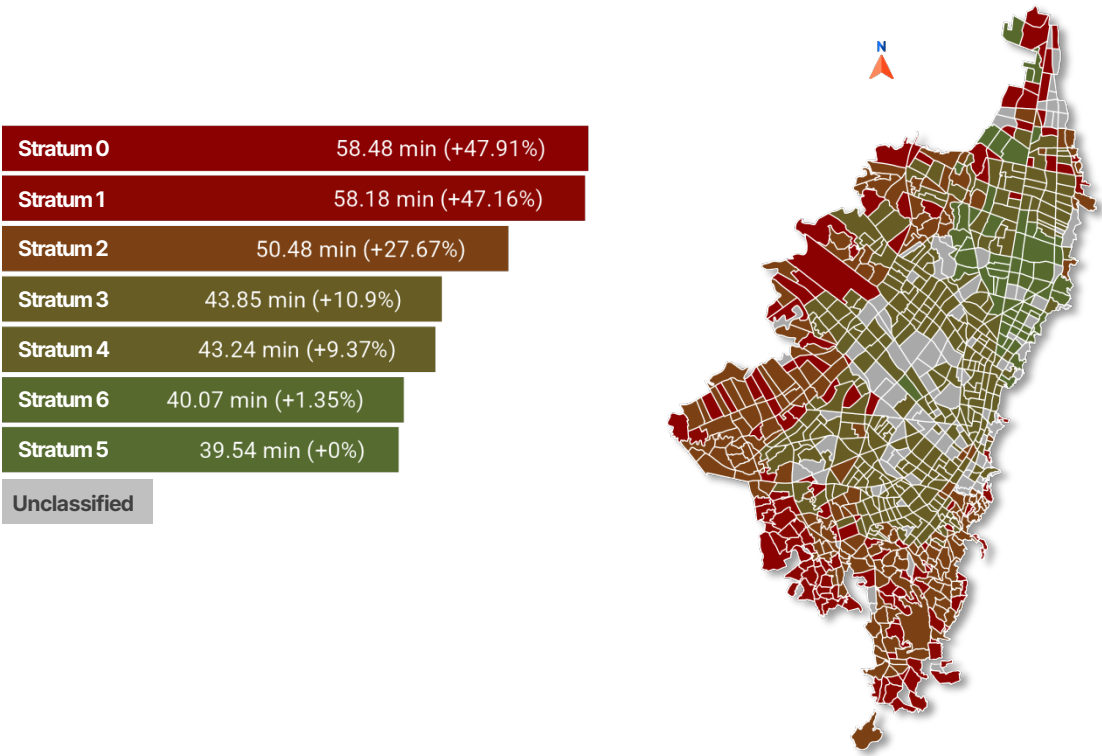


Figure 37. Average travel times by socioeconomic stratum. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023)

The processed data show a worrying inverse correlation: **The lower the socioeconomic stratum, the higher the average travel time.** Inhabitants of peripheral UPLs of strata 0, 1, and 2 invest significantly more time in their daily displacements than residents of strata 5 and 6, located closer to the centralities. This confirms that mobility in Bogotá acts as a mechanism for the reproduction of inequality.

Regarding travel times by gender, the survey shows that men invest the most time in displacements, followed very closely by women.



Figure 38. Travel times by gender. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

In Bogotá, nearly 675,216 care trips are made daily. Women represent more than 90% of the caregiving population, mostly without receiving remuneration according to the same survey (Secretaría Distrital de Movilidad, Bogotá, DC., 2023). Nearly 330,000 daily care trips are made on foot in Bogotá.

The survey shows that the main trip purpose is returning home, which is logical considering that, for example, regardless of the purpose of the morning trip, one must return home in the afternoon or evening. Thus, with the objective of determining the main trip purposes of Bogotanos, trips to the home are discarded and Work, Study, and Errands/Procedures (Trámites) are taken into account.

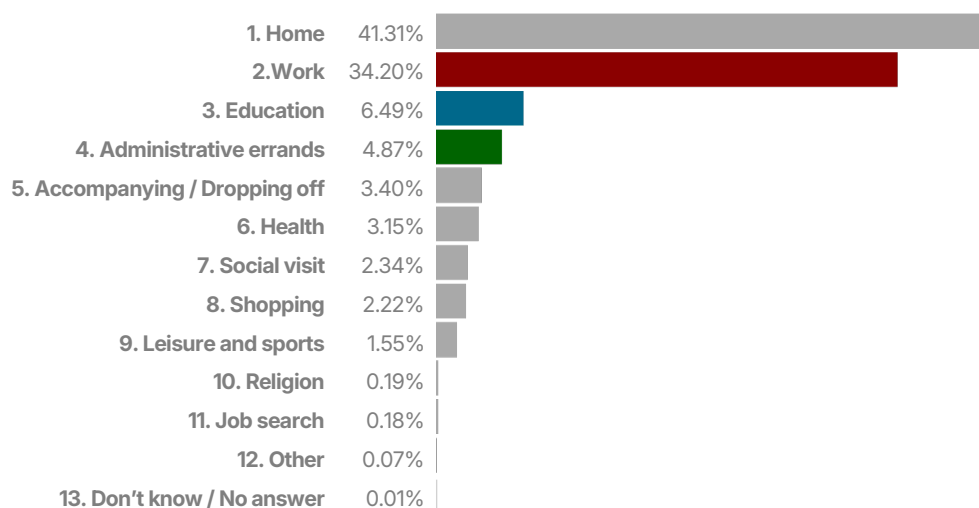


Figure 39. Main trip purposes in Bogotá. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023)

Note that in **Figure 39**, the three main trip purposes (Work, Study, and Errands) have been formatted with a distinctive colour each. This is to avoid confusion for the readers of this document when viewing the detail of displacement times shown in the upcoming figures, especially **Figure 42**, **Figure 43** and **Figure 44**.

When analysing travel times by purpose, it is observed that trips for work, study, and errands also consume the greatest amount of time, subtracting available hours for leisure or care, directly affecting the quality of life index that will be addressed subsequently.



Figure 40. Average displacement times for the three main trip purposes. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

Given that Bogotá has a wide supply of kindergartens and primary and secondary education establishments, distributed across different zones of the city as a result of significant investments oriented towards strengthening this sector, travel times associated with these educational levels do not present values as high as those corresponding to higher education. Upon filtering the information considering solely trips for higher education purposes, a significant increase in displacement times is observed, as shown in **Figure 41**, placing this type of trip in first place in terms of longest travel times, after previously occupying the third.



Figure 41. Displacement times for the three main trip purposes, when study considers only higher education trips. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

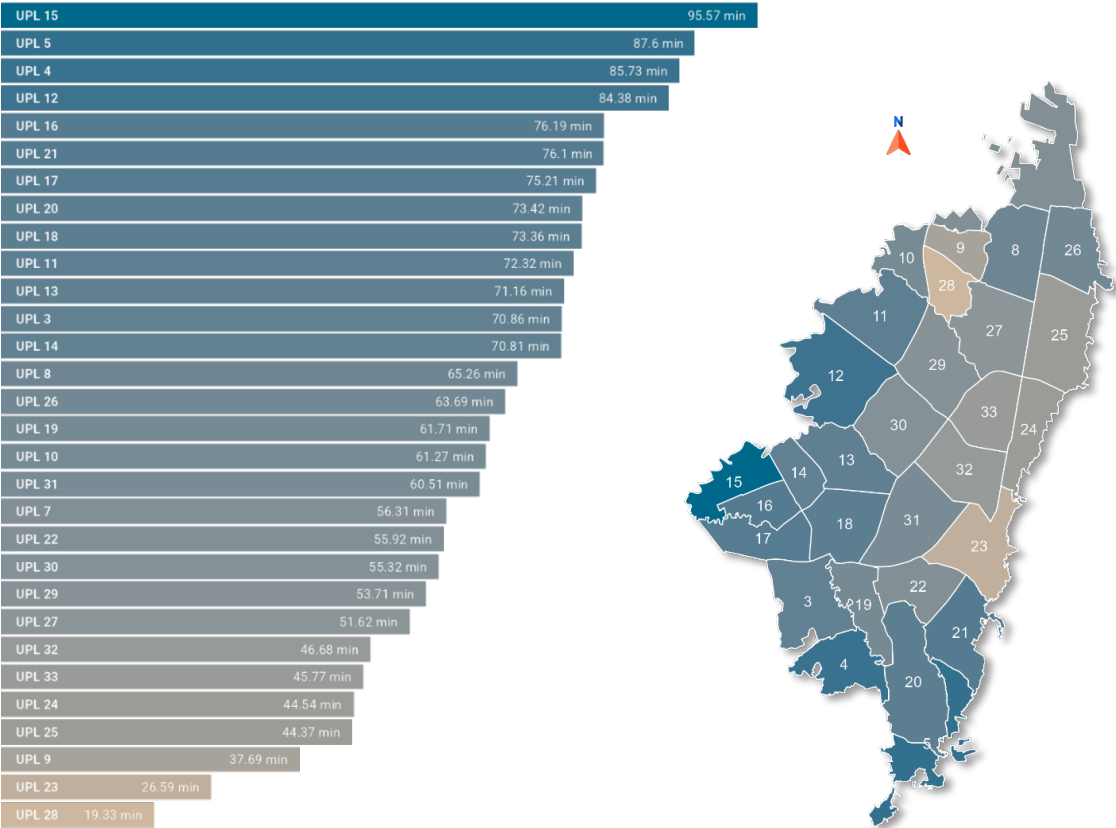


Figure 42. Average displacement times per UPL for higher education study purposes. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

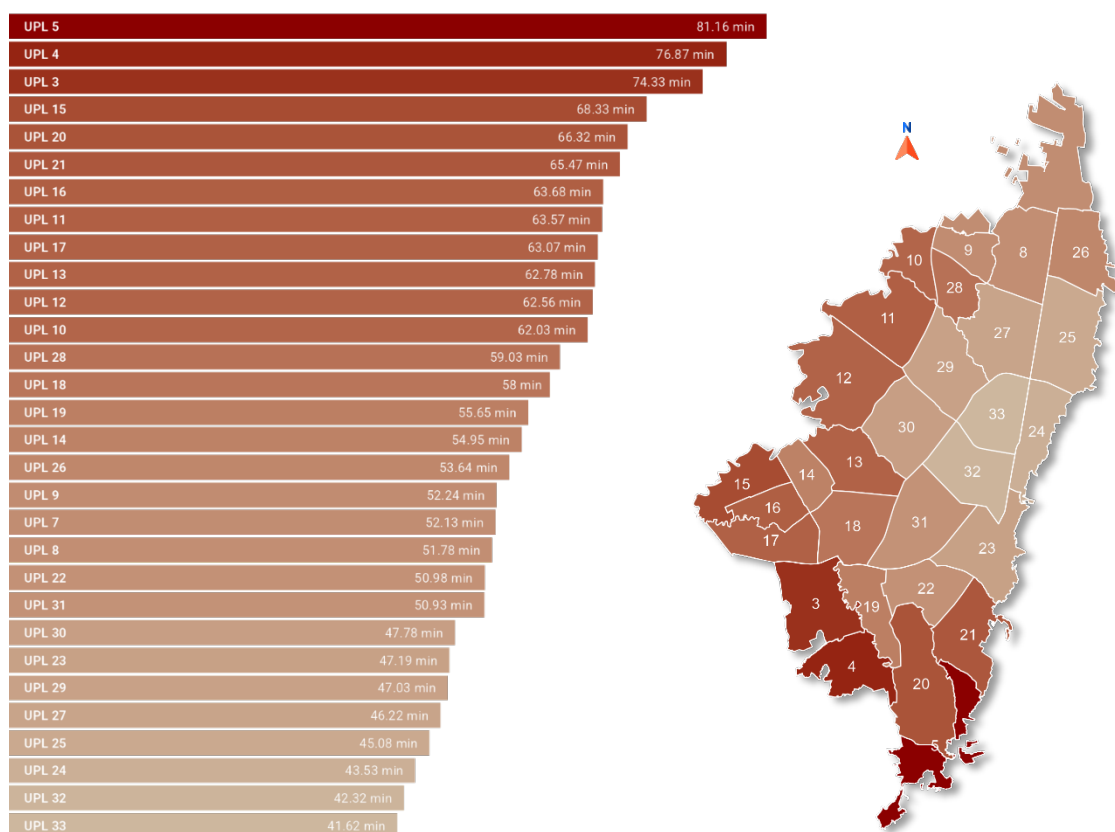


Figure 43. Average displacement times per UPL for work purposes. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

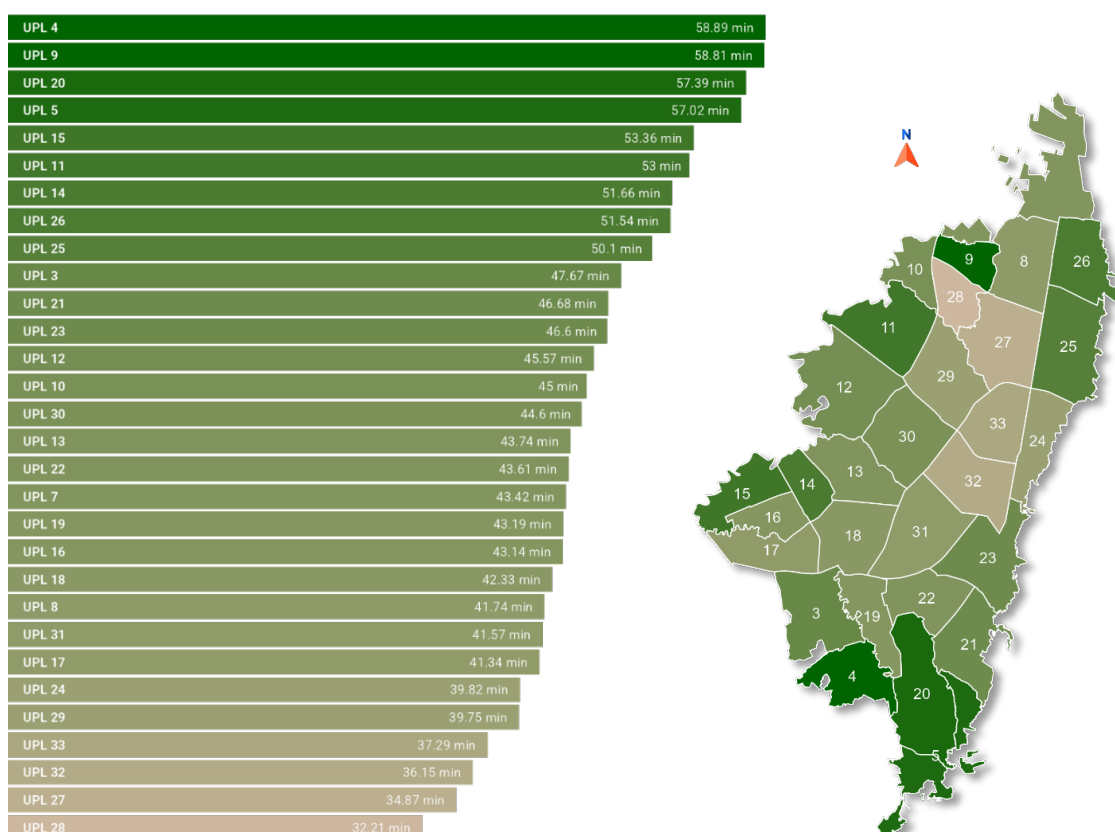


Figure 44. Average displacement times per UPL for errands/procedures purposes. Source: Author's own elaboration with data from (Secretaría Distrital de Movilidad, Bogotá, DC., 2023).

4.3. Diagnosis of Facilities

Once demand was analysed—understood as the residential location of the population and their mobility patterns—it was necessary to evaluate the supply of opportunities. Based on the georeferenced inventory of facilities obtained through the OpenStreetMap (OSM) query for the current state, corresponding to Scenario 1 (Script 4_1_Amenities_Scn1.R), the diagnosis of the "City of Proximity" was constructed under the High Quality of Societal Life (HQSL) Index approach.

The OSM query and subsequent processing and cross-referencing with the National Population and Housing Census (DANE, 2018), oriented towards verifying housing supply, yielded a total of 591,267 facilities. This volume of information, significant in analytical terms, required the application of data science (Big Data) and geoprocessing techniques for its adequate handling. Whilst this quantity might not seem high at a first approximation, the fact that each facility is georeferenced and classified by categories, social functions, and composite indicators substantially increases the complexity and size of the dataset.

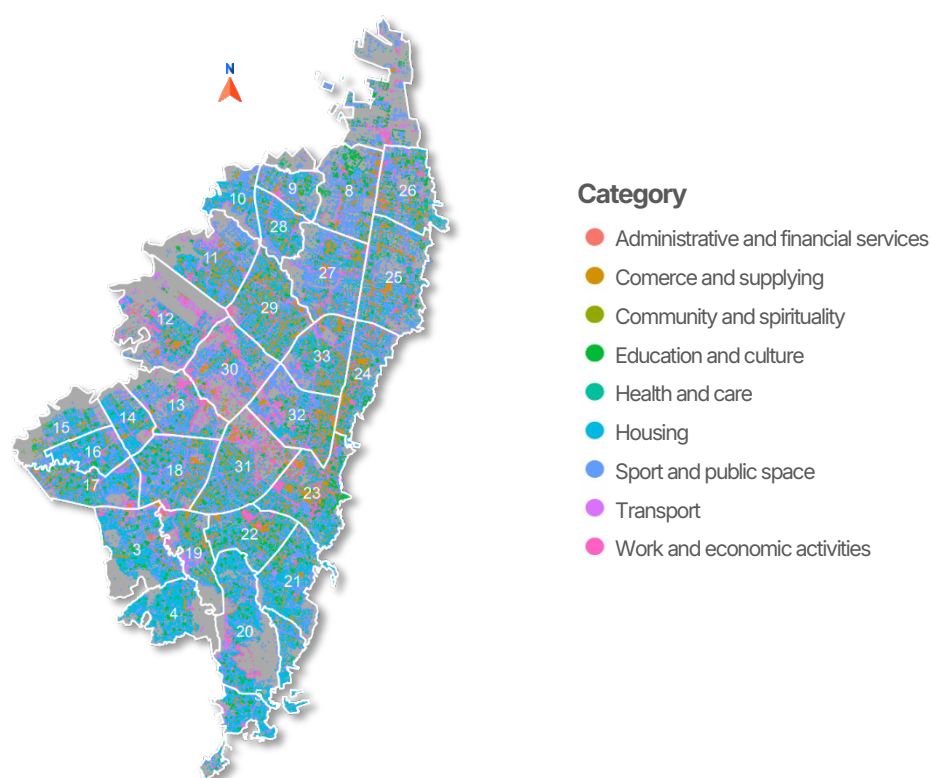


Figure 45. Location and categorisation of urban facilities in Bogotá.
Source: Author's own elaboration with data from (OpenStreetMap Contributors, 2025) and (DANE, 2018).

4.3.1. Spatial Distribution of Supply

Whilst **Figure 45** represents the totality of queried facilities, identifying concentration patterns proves complex due to the high number of points visualised. In order to facilitate spatial reading and analyse facility density, these were aggregated using the hexagon (HEX) vector layer, which, being a homogeneous spatial unit of reduced area, allows for better identification of concentrations. Mapping facilities onto the hexagonal grid evidences a marked spatial segregation, as service supply is concentrated mainly on the eastern edge of the city, as observed in **Figure 46**.

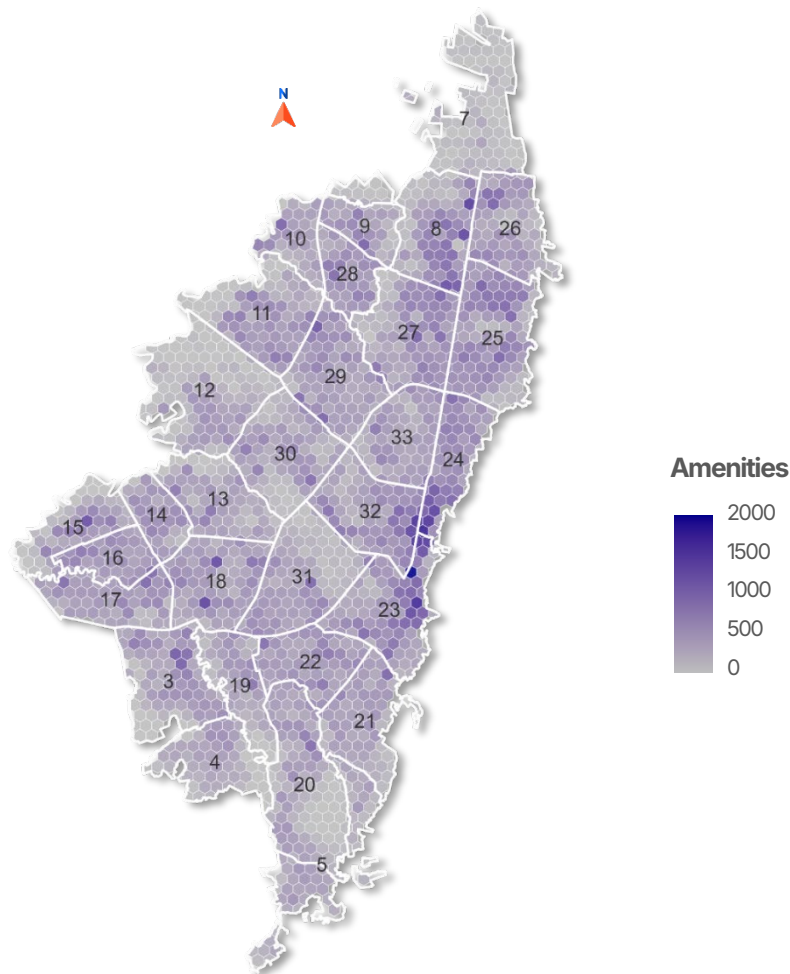


Figure 46. Spatial concentration of urban facilities on a hexagonal grid (HEX).
Source: Author's own elaboration with data from (OpenStreetMap Contributors, 2025) and (DANE, 2018).

This concentration is corroborated by performing the same exercise and transferring the facility supply to the Local Planning Unit vector layer, as observed in **Figure 47** and **Figure 48** below.

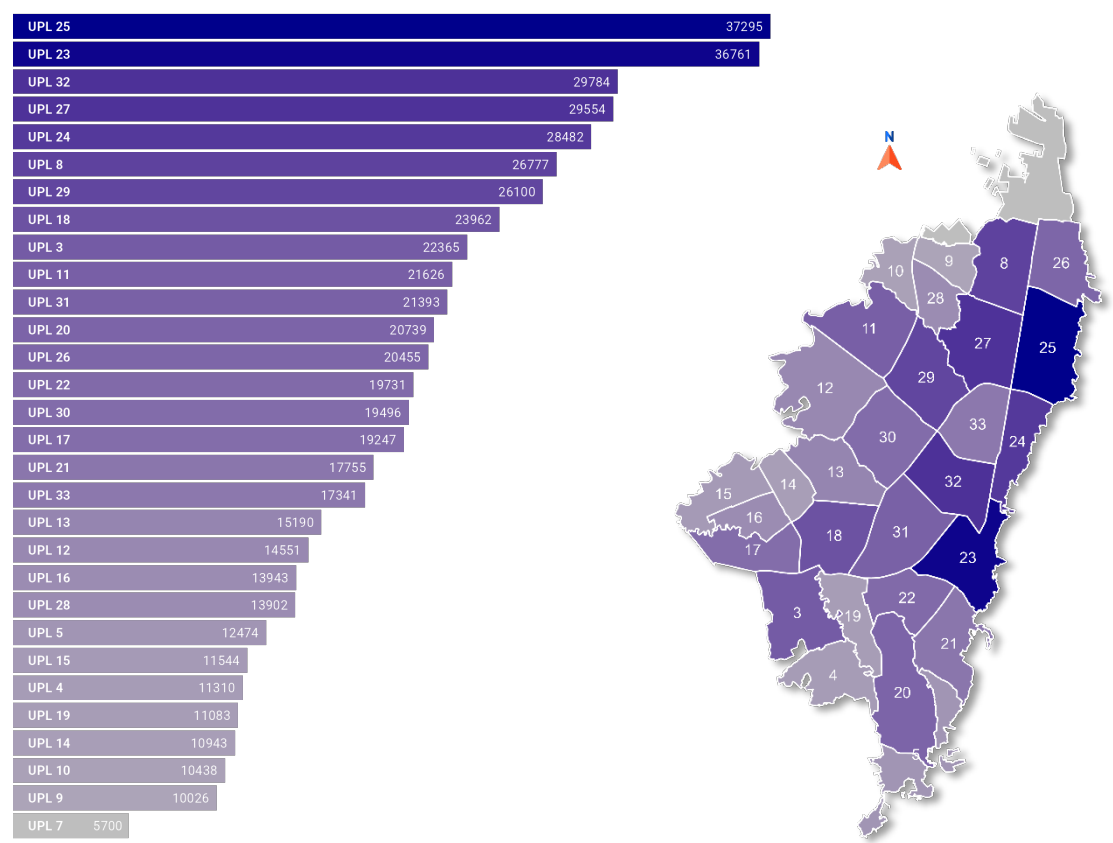


Figure 47. Concentration of facilities by UPL. Source: Author's own elaboration with data from (OpenStreetMap Contributors, 2025) and (DANE, 2018).

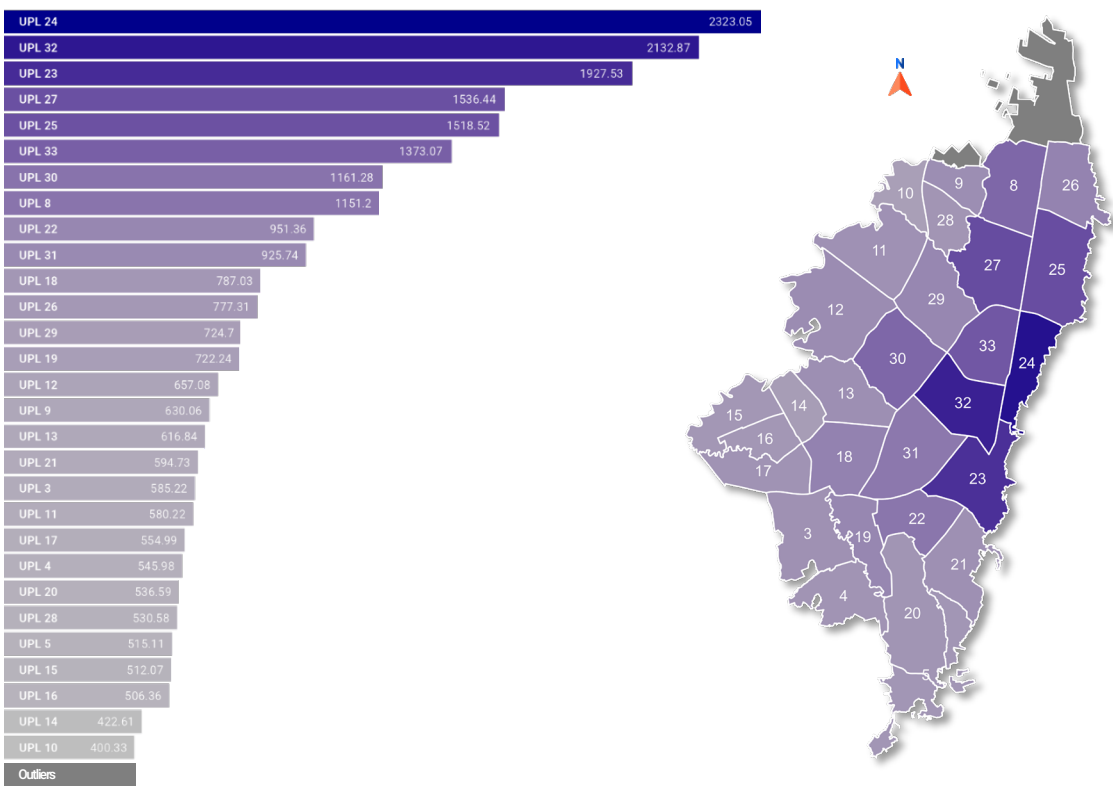


Figure 48. Facilities per UPL per every 10,000 inhabitants. Source: Author's own elaboration with data from (OpenStreetMap Contributors, 2025) and (DANE, 2018).

In section 4.2.3, it was evidenced that travel times to access higher education present, on average, the highest values for the inhabitants of Bogotá. Upon filtering facilities and considering solely those corresponding to higher education supply, a marked hyper-concentration is observed in Local Planning Units 23, 32, and 24, located on the eastern edge of the city and corresponding to the *Centro Histórico*, *Teusaquillo*, and *Chapinero*. This spatial pattern supports the hypothesis whereby inhabitants of other UPLs are forced to travel considerable distances to access this type of service..

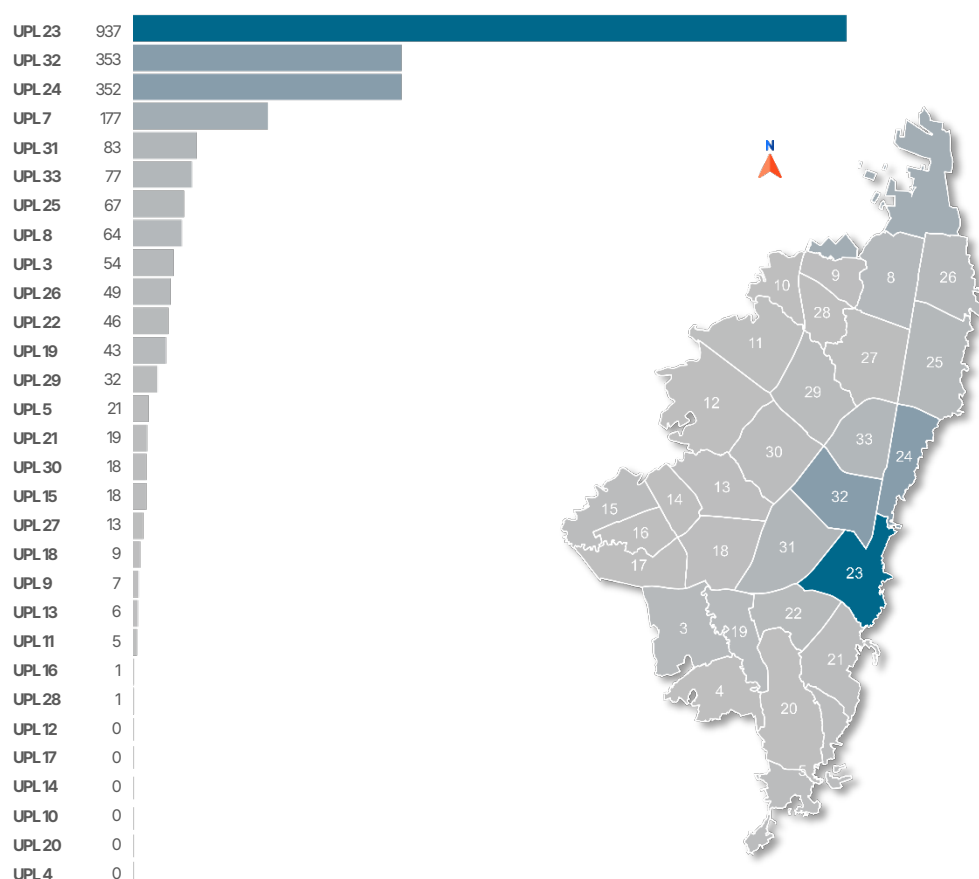


Figure 49. Supply of higher education facilities by UPL. Source: Author's own elaboration with data from (OpenStreetMap Contributors, 2025).

This hyper-concentration of higher education facility supply is evidenced more clearly by performing the extrusion of the aggregated density at the hexagon level, in order to represent said concentration in a third dimension. The result of this visualisation exercise is presented in **Figure 50** below.

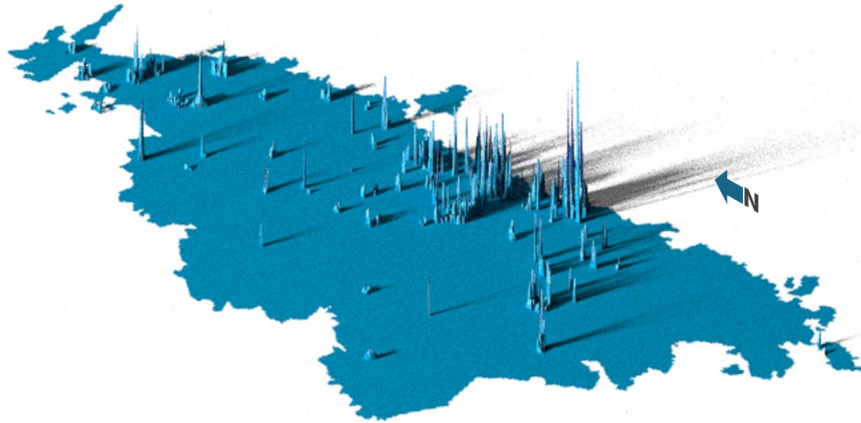


Figure 50. Three-dimensional visualisation of the concentration of higher education facilities. Source: Author's own elaboration with data from (OpenStreetMap Contributors, 2025).

The spatial analysis of facility supply yielded three structural findings:

- **Hyper-concentration of supply on the eastern edge:** The UPLs located along the eastern hills (*cerros orientales*) —especially *Centro Histórico*, *Chapinero*, and *Teusaquillo*— concentrate the vast majority of metropolitan-scale facilities. The 3D map of universities dramatically illustrates how higher education supply is stacked in a narrow area, forcing students from all over the city to converge on the same point daily.
- **Deficit in the periphery:** Upon normalising supply via the "Facilities per every 10,000 inhabitants" indicator, it is evident that the southern and western UPLs (coinciding with the current localities of *Bosa*, *Kennedy*, *Ciudad Bolívar*, and *Engativá*), despite being the most population-dense, present the lowest indices of services per capita (**Figure 48**). There are vast extensions of residential fabric lacking high-level health, cultural, or higher education facility supply, or formal employment.
- **The "Dormitory City":** The superposition of population maps (section 4.1) with facility maps reveals a functional disconnection. That is, where people live, services are not found, and where services are found, the majority of people do not live.

4.4. Proximidad Urbana y HQSL

Prior to the construction of quality of life indicators, the accessibility model (Script [6_1_Accessibility_Scn1.R](#)) configured for Scenario 1 was executed. This calculation was performed via the *r5r* engine, the technical operation of which was detailed in section 3.4.2. For this purpose, the current public transport network (official Transmilenio and SITP GTFS) (Transmilenio S.A., 2025), the current road and pedestrian network (OpenStreetMap Contributors, 2025), and naturally, the optimal parameters resulting from the model calibration for the current scenario (explained in detail in chapter 3.4.3) were utilised.

4.4.1. Cumulative Accessibility of Scenario 1

The model generated travel matrices reflecting cumulative accessibility, understood as the quantity of opportunities (facilities) reachable from each block (manzana) within specific time thresholds (15 and 30 minutes), as shown in **R Code Snippet 8** in section 3.4.4. A fraction of the accessibility matrix for Scenario 1 is shown below:

```
> Acceso_a_Equipamientos_Scn1
```

	id	opportunity	percentile	cutoff	accessibility
1	1100110000000011010101	Living	50	15	807
2	1100110000000011010101	Living	50	30	8628
3	1100110000000011010101	Working	50	15	0
4	1100110000000011010101	Working	50	30	275
5	1100110000000011010101	Supplying	50	15	21
6	1100110000000011010101	Supplying	50	30	597
7	1100110000000011010101	Caring	50	15	41
8	1100110000000011010101	Caring	50	30	347
9	1100110000000011010101	Learning	50	15	19
10	1100110000000011010101	Learning	50	30	450
11	1100110000000011010101	Enjoying	50	15	214
12	1100110000000011010101	Enjoying	50	30	2921
13	1100110000000011010101	Well_being	50	15	972
14	1100110000000011010101	Well_being	50	30	10197
15	1100110000000011010101	Sociability	50	15	139
16	1100110000000011010101	Sociability	50	30	2564
17	1100110000000011010101	Environmental_Impact	50	15	155
18	1100110000000011010101	Environmental_Impact	50	30	1934

R Code Snippet 9. First 18 rows of the matrix resulting from the accessibility calculation for Scenario 1. Source: Author's own elaboration.

Based on a process of information organisation using data science techniques and its subsequent geoprocessing in R, accessibility maps were generated at the block and UPL scale for each of the time thresholds established in the model execution (15 and 30 minutes).

In **Figure 51**, it is observed that the eastern edge of the city concentrates the highest levels of accessibility to facilities, for both the 15-minute and 30-minute thresholds, which again confirms the hyper-concentration of supply in this zone of Bogotá. In the 30-minute scenario, a significant proportion of the urban territory presents improvements in facility access levels, an expected result as the temporal threshold is expanded. This increase is particularly evident in areas close to the main transport corridors. In quantitative terms, the improvement in accessibility is notable, given that by doubling the travel time it is possible to access up to six times more facilities.

When representing accessibility at the UPL scale (**Figure 52**), the contrast between the 15 and 30-minute scenarios is less visually evident. Nevertheless, the extreme values of the legend allow for the identification of a significant improvement in general accessibility levels at the city scale..

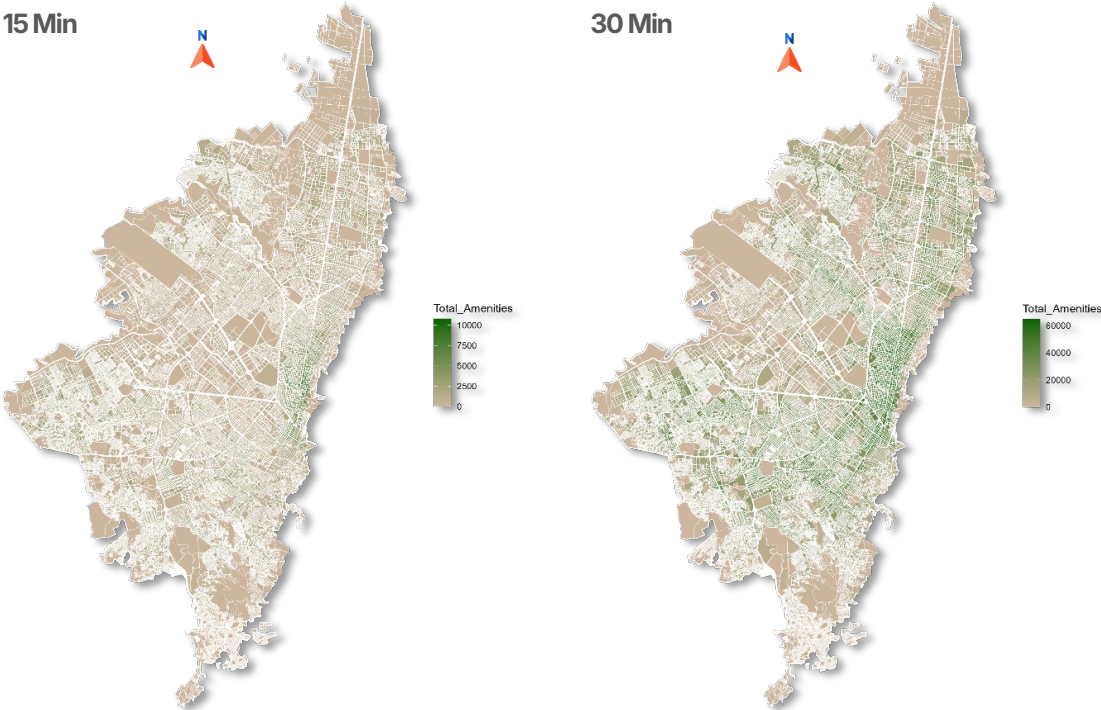


Figure 51. Cumulative access to facilities for Scenario 1 at the block level, for 15 and 30 minutes. Source: Author's own elaboration.

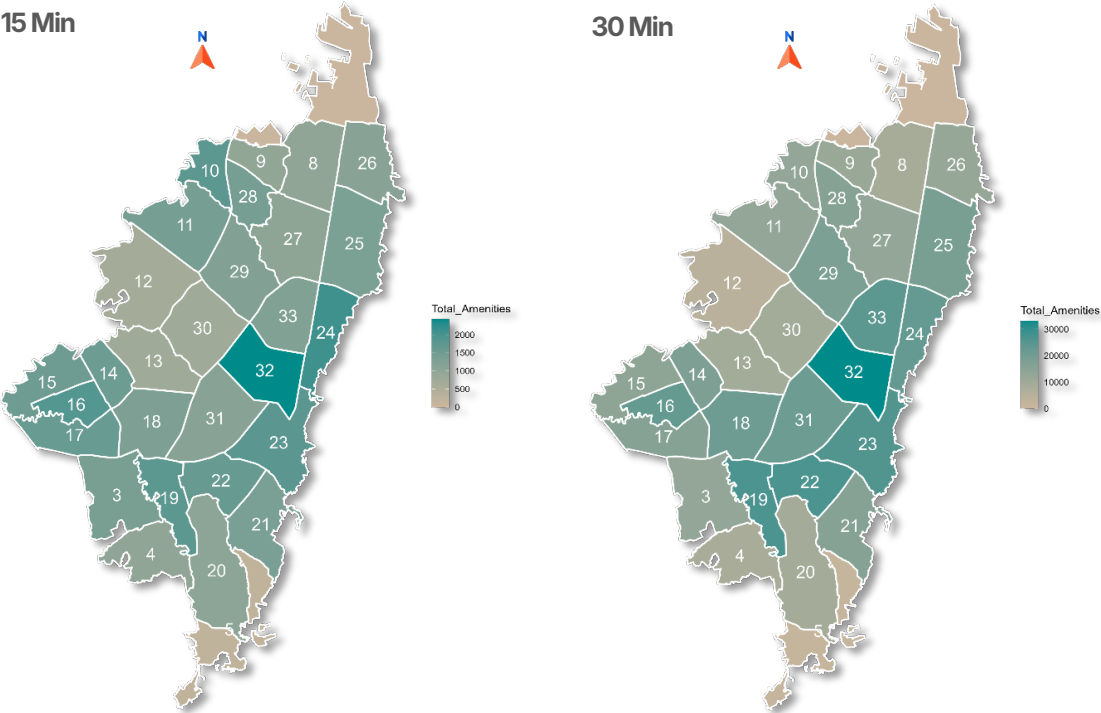


Figure 52. Cumulative access to facilities for Scenario 1 at the UPL level, for 15 and 30 minutes. Source: Author's own elaboration.

In general terms, it is at the block scale where a strong spatial inequality is most evidenced.

- **15-minute threshold:** In the periphery (western arc), access is limited almost exclusively to basic supply (local commerce) and basic education. Health and cultural services are practically inaccessible within this time range.
- **30-minute threshold:** Even expanding the travel time, the edge zones (such as UPLs 14, 15, and 16, corresponding to the Bosa locality) access less than 10% of the supply that a resident of the expanded centre reaches within the same time threshold.

4.4.2. HQSL Index and ScoreCards for Scenario 1

Based on accessibility data, the High Quality of Societal Life (HQSL) Index was calculated following the methodology described in section 3.4.5, which translates the quantity of accessible facilities into an indicator of vital need fulfillment. The index was estimated at both the block and Local Planning Unit (UPL) scales via Script 6_1_Accessibility_Scn1.R. However, for practical purposes, the present work focuses on the analysis of the HQSL Index at the UPL scale, which is employed for the elaboration of the ScoreCards and the comparison between different areas at this same territorial level.

```
Simple feature collection with 85434 features and 11 fields
Geometry type: MULTIPOLYGON
Dimension: XY
Bounding box: xmin: -74.22353 ymin: 4.464889 xmax: -74.01135 ymax: 4.833006
Geodetic CRS: MAGNA-SIRGAS
# A tibble: 85,434 × 11
  COD_MNZ cutoff Living Working Supplying Caring Learning Enjoying Well_being Sociability Environmental_Impact
  <chr>    <int> <int> <int> <int> <int> <int> <int> <int> <int> <int>
1 1100110... 15 563 0 81 7 10 261 772 150 175
2 1100110... 30 10732 59 509 160 175 2765 12841 1762 2056
3 1100110... 15 524 0 80 9 10 237 715 134 160
4 1100110... 30 10229 59 507 158 167 2704 12287 1722 2008
5 1100110... 15 500 0 80 13 8 206 678 121 135
6 1100110... 30 9661 56 489 154 148 2530 11572 1610 1860
# i 85,429 more rows
```

R Code Snippet 10. Information at the block level, resulting from the calculation, processing, and aggregation process of accessibility indicators.
Source: Author's own elaboration.

```
Simple feature collection with 60 features and 24 fields
Geometry type: GEOMETRY
Dimension: XY
Bounding box: xmin: -74.22358 ymin: 4.464759 xmax: -74.01206 ymax: 4.833003
Geodetic CRS: MAGNA-SIRGAS
# A tibble: 60 × 25
  COD_UPL cutoff Living Working Supplying Caring Learning Enjoying Well_being Sociability Environmental_Impact
  <chr>    <int> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 10      15 1038 0 11 2 10 137 1173 95 93
2 10      30 6348. 15 228. 59 104 1474. 7554. 958. 1029
3 11      15 668 3 69 21 9 265 926 200 178
4 11      30 5367 65 582 203 121 2279 7285 1588 1574
5 12      15 465 12 11 1 14 108 578 93 59
# i 55 more rows
```

R Code Snippet 11. Information aggregated at the UPL level, resulting from the calculation, processing, and aggregation process of accessibility indicators.
Source: Author's own elaboration.

The calculation of Social Functions at the UPL level was not performed cumulatively via the sum of values corresponding to each block belonging to a UPL. Instead, the median of the values per block within each UPL was used as a representative measure. This methodological decision allows for the reduction of the influence of outliers present in particular blocks, preventing these from artificially inflating the UPL's aggregated accessibility value. It is for this reason that the range of values for the scales in **R Code Snippet 11**, in **Figure 53**, and **Figure 54** for the UPL level, may in some cases appear lower than that of the block level.

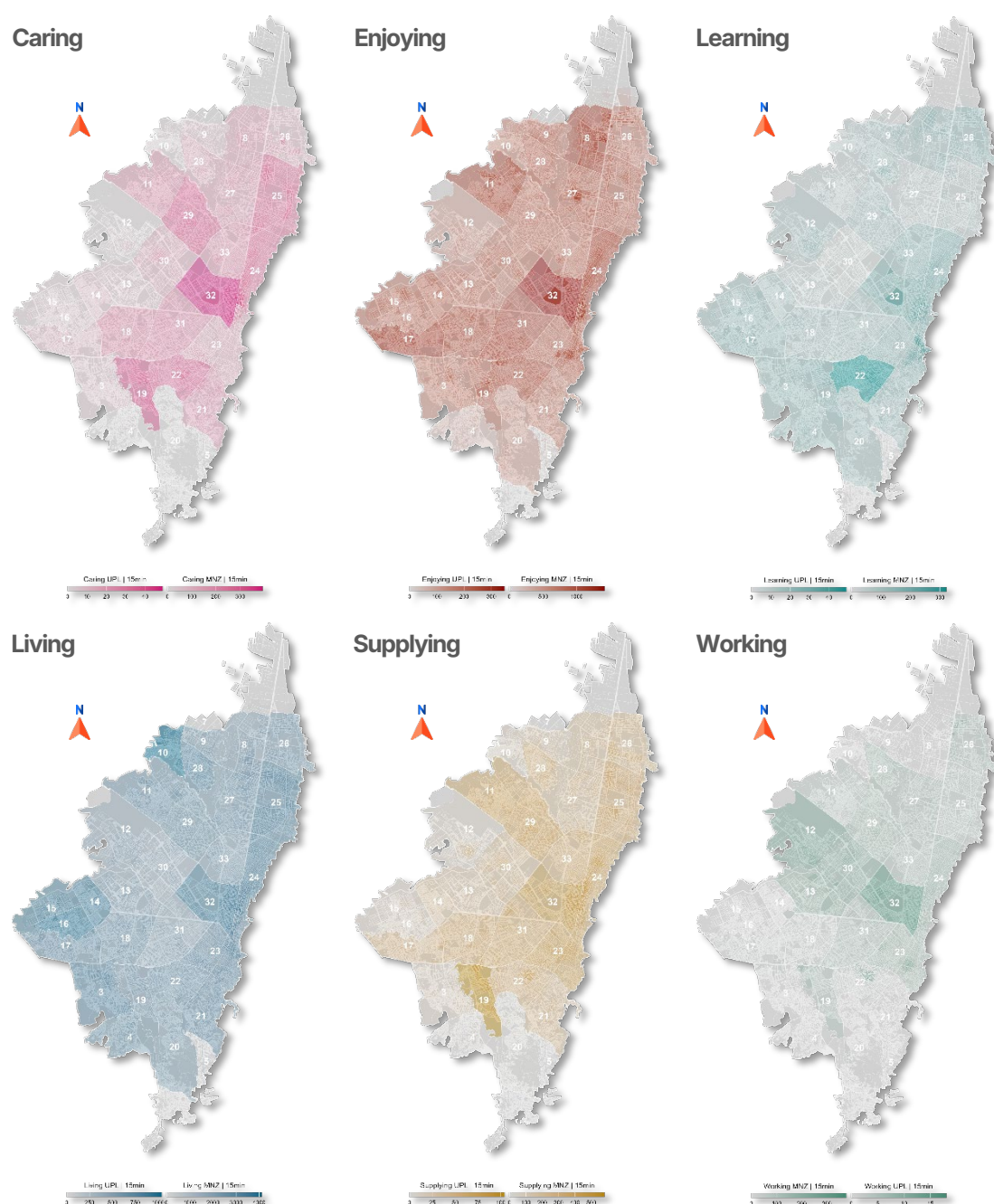


Figure 53. Map of Social Functions at the UPL and Block level for Scenario 1. Source: Author's own elaboration.

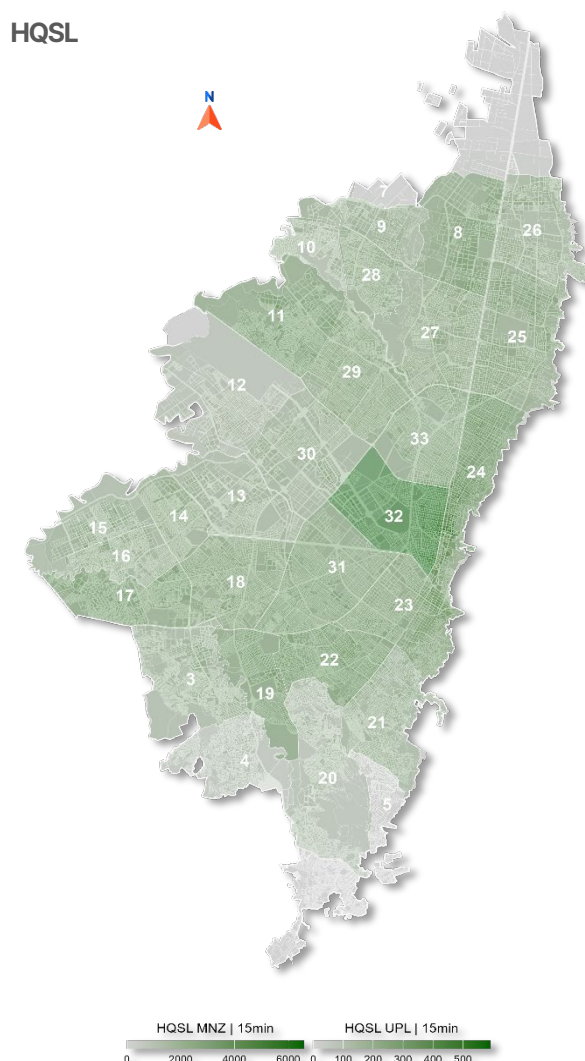
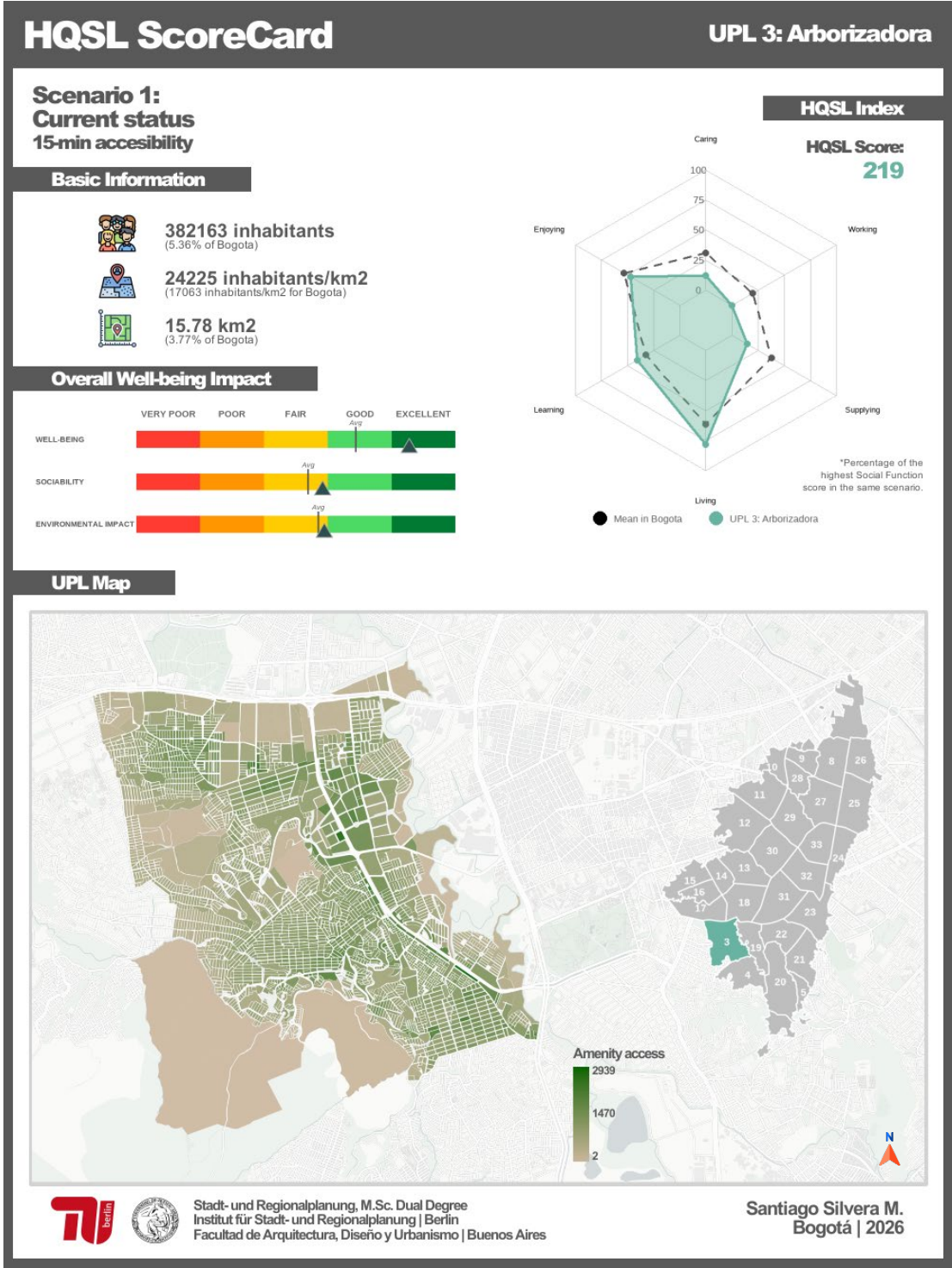


Figure 54. HQSL Map at the UPL and Block level for Scenario 1. Source: Author's own elaboration.

The maps by social functions and of the HQSL Index in **Figure 53** and **Figure 54** respectively, show the functional segregation of Bogotá. The areas with high scores coincide with the Expanded Centre, whilst vast residential zones—where the largest amount of the population is found—are classified in low ranges, indicating that their inhabitants are forced to rely on long journeys to satisfy many of their basic needs, thus reducing their urban quality of life.

With the purpose of evaluating and showing in greater detail the High Quality of Societal Life situation for each UPL, the ScoreCard for Scenario 1 (Current State) was elaborated for each of them. In each of these "cards", statistical data of the UPL are shown and compared with the rest of the city. This tool is especially useful for individual diagnosis by UPL. **Figure 55** below shows an example ScoreCard for a UPL, for Scenario 1, at the 15-minute temporal threshold.



5

Intervention Scenarios

This chapter presents, firstly, the operational characteristics and the description of the alignments of the rail infrastructure projects—corresponding to the Bogotá Metro Line 1 and the proposed Rail Network—as well as the land-use strategies associated with the implementation of new urban facilities. Based on these two components, the configuration of the modelled future scenarios is subsequently developed, with the aim of evaluating their impact on accessibility and urban quality of life.

5.1. Proposed Transport Infrastructure

The main modification to the transport supply consists of the integration of high-capacity rail-based modes, modelled through the generation of synthetic GTFS files (as described methodologically in section 3.4).



Note: In Bogotá, streets ("*Calles*"), avenues ("*Avenidas*"), and north–south corridors ("*Carreras*") follow a grid-based and hierarchical naming system; original Spanish names are retained alongside their English equivalents for clarity.

5.1.1. Bogotá Metro Line 1 (PLMB – Primera Línea del Metro de Bogotá)

The PLMB was modelled following the alignment and technical specifications defined in the project's technical and legal structuring (Metro de Bogotá, 2018):

- **Trace and Alignment:** The corridor begins at the depot (*patio taller*) in the *Bosa* locality, traverses the *Avenida Villavicencio*, *Avenida Primero de Mayo*, *Avenida NQS*, *Avenida Calle 8 Sur*, and *Avenida Calle 1*, connecting the south-west with the centre, and subsequently takes the avenue Caracas corridor northwards up to *Calle 72 Station*.
- Stations: 16 elevated stations were georeferenced.
- Operational Parameters: For the simulation in r5r, an average commercial speed of 43 km/h and a peak hour dispatch frequency (headway) of 3 minutes were configured, representing a significant increase in speed and capacity regarding current road-based transport..

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_M1-Actual.zip` is available on GitHub

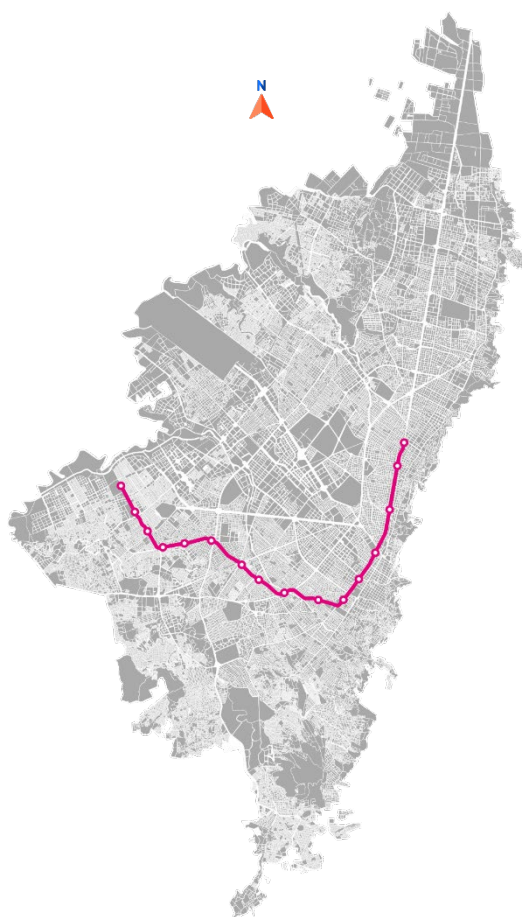


Figure 56. Alignment of the Bogotá Metro Line 1. Source: Author's own elaboration with data from (Alcaldía de Bogotá, 2019).

At the date of this work's realisation, the PLMB project is in the construction phase. For the scenarios in which was implemented the PLMB, the infrastructure was simulated exactly as structured and described in this section. For the implementation of the Bogotá Metro Network discussed in the following section, which forms part of Scenarios 5, 6, and 7, two extensions to the PLMB were performed; one of them to

the *Suba* locality as contemplated in the 1996 Mobility Master Plan (Agencia de Cooperación Internacional de Japón - JICA, 1996).

5.1.2. Implementation of the Bogotá Metro Network

Based on transport demand, the deficit thereof, density, and trip attraction zones, the alignments and station locations were defined for the implementation of the Bogotá Metro Network, which would accompany the PLMB.

Extension and branch for the PLMB.

As pointed out in section 5.1.1, for the full implementation of the Bogotá Metro Network, the extension of the Bogotá Metro Line 1 (PLMB) towards the north and west of the city was structured, with the purpose of improving metropolitan accessibility, expanding the system's territorial coverage, and balancing the distribution of travel demand.

From the perspective of accessibility to employment and urban activities, the north and west extension responds to three planning criteria:

- Serving an area of high trip generation for work purposes in the vicinity of the *El Chicó* neighbourhood and *Calle 100*.
- Correcting transport coverage deficits in sectors of north-eastern *Suba* and incorporating UPL 10 *Tibabuyes* and UPL 28 *Rincón de Suba*—the third and fourth densest UPLs in Bogotá (see **Figure 28**)— into the Mass Transit network; these are located in *Suba*, the locality with the largest population in the city (see **Figure 25**). This decision is consistent with the provisions of the 1996 Mobility Master Plan (Agencia de Cooperación Internacional de Japón - JICA, 1996), which already anticipated the need to connect *Suba* to the PLMB.

From the perspective of territorial coverage and spatial equity, a southern branch was proposed that extends service using the same PLMB infrastructure towards the localities of *Rafael Uribe Uribe*, *Tunjuelito*, and *Usme*. In this proposal, the branch terminates at the *Portal Usme*, the current terminal station of the Transmilenio BRT system. This seeks to expand metro coverage towards the south-east, incorporating territories that, although served by Transmilenio, present a peak hour supply that overflows the system's capacity and generates recurrent saturation conditions.

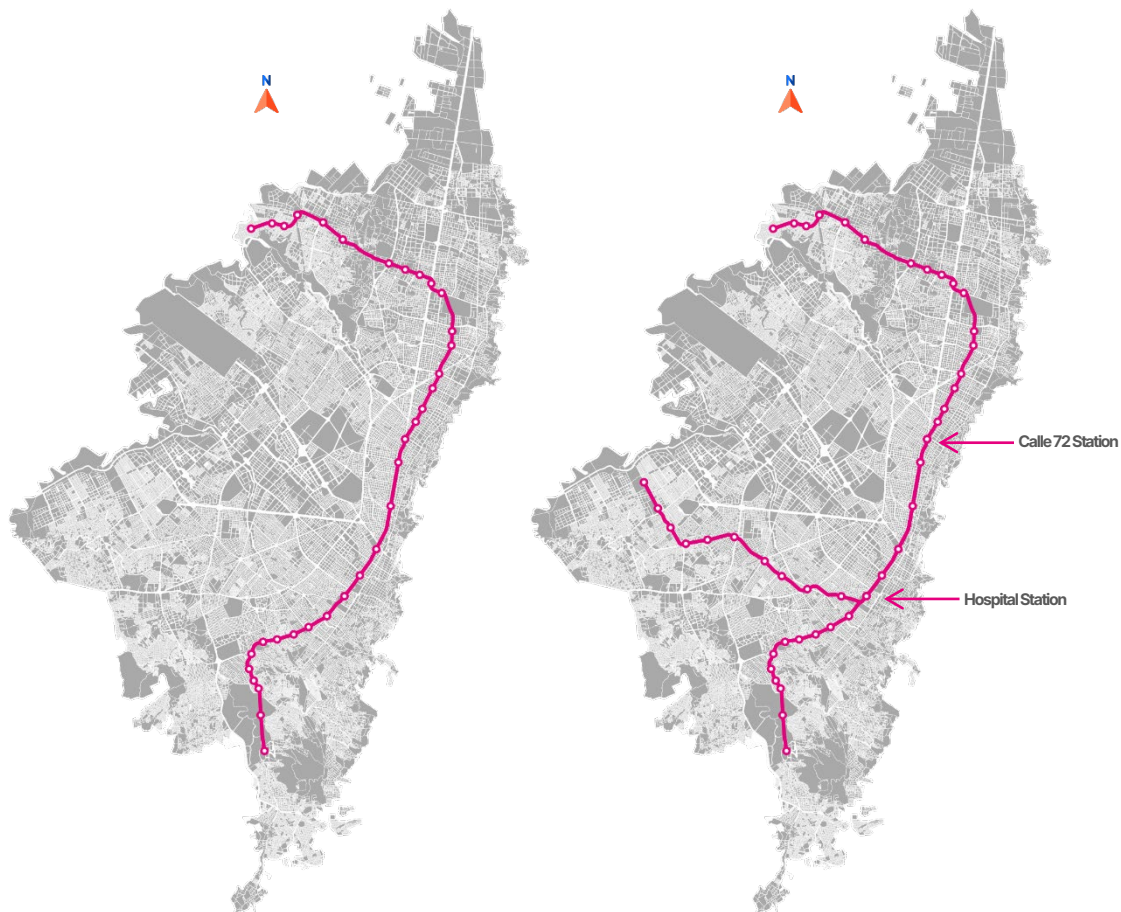


Figure 57. Left: Alignment of the extension and branch proposed for the PLMB.
Right: Complete alignment PLMB + Extension and proposed branch. Source: Author's own elaboration.

In operational terms, the proposed configuration implies that the PLMB must function with two overlapping services that will share infrastructure between the *Hospital* and *Calle 72* stations. This scheme increases supply in the heart of the Expanded Centre—where the greatest demand is concentrated—and improves system resilience against load peaks. Operation with multiple services over a single infrastructure is an internationally consolidated practice; as a reference, the central section of the Berlin S-Bahn operates with services S3, S5, S7, and S9, which, upon leaving the Ring (functionally analogous to the Expanded Centre of Bogotá), separate into distinct routes to maximise regional coverage..

- **Trace and Alignment:** In the north-west, the corridor begins in the Lisboa neighbourhood, in UPL 10 (*Tibabuyes*), and advances along *Avenida Suba* to guarantee accessibility to the centre of the locality, then connects underground to the neighbourhoods of *Colina Campestre* and subsequently *Prado Veraniego*. Later, the alignment is oriented towards the employment and service hubs of *Unicentro* and *El Chicó* via *Carrera 15*, prioritising accessibility to jobs and metropolitan facilities. From the intersection of *Calle 100* and *Carrera 15*—where an intermodal station is proposed, connecting the PLMB, the future TransMilenio trunk line along *Avenida 68*, and the Northern Regional Train described later in section 5.1.3—it continues south to *Avenida*

Caracas, where it connects with the original PLMB alignment and the SLMB (Bogotá Metro Line 2, described below). In the south, starting from the *Hospital* station, the PLMB extension follows a branch describing the same trace as *Avenida Caracas*, until reaching *Portal Usme*, where the route terminates.

- **Stations:** 28 new stations were georeferenced, located strategically to integrate with existing Transmilenio trunk lines (*Avenida Suba*, *Autopista Norte* (Northern Highway), *Avenida Caracas*, and the future *Avenida 68/Calle 100*), and one of the suburban/regional rail lines proposed in this work, in order to increase transport coverage. In addition to the connection with other transport lines, stations were conceived to be located in both high residential density zones and trip attractors, in accordance with the O-D matrices.
- **Operational Parameters:** Like the original PLMB, for the simulation in r5r, the PLMB extension and its southern branch were configured at an average commercial speed of 43 km/h and a peak hour dispatch frequency of 3 minutes.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_M1-Extensiones.zip` is available on GitHub.

Bogotá Metro Line 2 (SLMB – Segunda Línea del Metro de Bogotá)

The Second Line of the Bogotá Metro is in technical, legal, and financial structuring at the date of this study's realisation. However, the alignment and its stations are already defined, and the SLMB proposal for modelling was established based on them.

- **Trace and Alignment:** The corridor begins at the depot (*patio taller*) in the *Suba* locality, travels along the road reserve of the *Avenida Longitudinal de Occidente* (ALO), incorporating onto *Avenida Ciudad de Cali* in the vicinity of *Calle 127*. It continues its route along *Avenida Ciudad de Cali* southwards, up to *Calle 72*, where it takes that trace eastwards and reaches the end of its line at *Avenida Caracas*, where it connects with the PLMB.
- **Stations:** 11 stations were georeferenced (2 elevated and 9 underground). A slight modification was made to the *Calle 72* station of the SLMB so that it connects directly with the station of the same name serving the PLMB, given that in the original designs, the two stations are not georeferenced or connected to each other; this implies performing transfers leaving the system, which is considered a major design flaw.
- **Operational Parameters:** Like the PLMB, for the simulation in r5r, the SLMB was configured with an average commercial speed of 43 km/h and a peak hour dispatch frequency of 3 minutes.

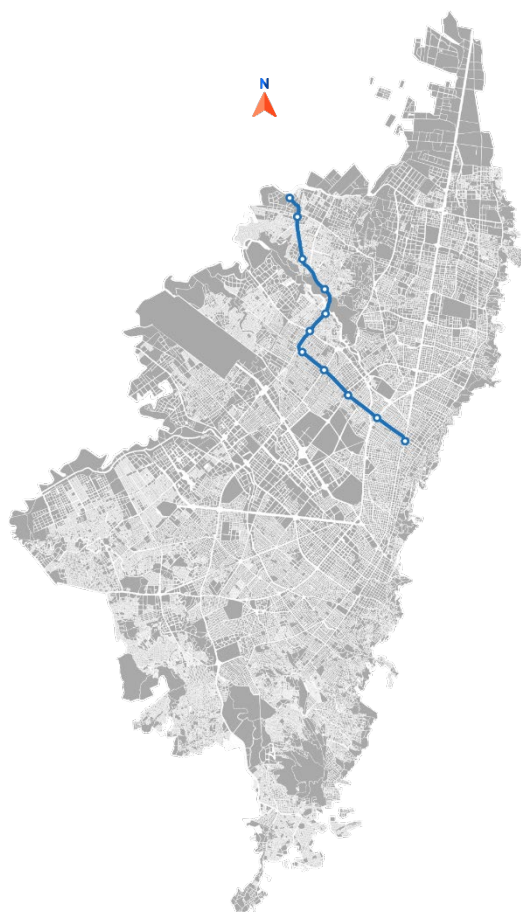


Figure 58. Alignment of the Bogotá Metro Line 2. Source: Author's own elaboration with data from (Alcaldía de Bogotá, 2019).

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_M2-Modificada.zip` is available in the GitHub repository.

Extension and branch for the SLMB.

Considering that the SLMB has not been contracted for construction and operation, an optimisation thereof was proceeded with, aiming to improve its accessibility levels and territorial coverage, as was done for the PLMB.

The first part of this optimisation was based on the line's extension towards the centre of Bogotá and, subsequently, towards the south-east of the city, allowing for the incorporation of areas with high concentrations of urban activities and population, expanding system coverage towards the southern zone of *Chapinero* and the localities of *La Candelaria* and *San Cristóbal*. Additionally, the proposed extension contributes to a more balanced distribution of demand and transfer loads at the *Calle 72* station—avoiding transfers to Transmilenio or the PLMB—which is the only SLMB station located in the Expanded Centre and which, foreseeably, would present the highest congestion levels.

The second part of the optimisation is related to increasing line coverage towards UPL 11 *Engativá* and UPL 29 *Tabora*, the third and fourth UPLs with the largest populations in the city (**Figure 27**).

Analogously to the configuration proposed for the PLMB, the SLMB would require the operation of two overlapping services, coinciding between the *Avenida Ciudad de Cali* and *Calle 72* stations. Said overlap could be extended up to the *Museo Nacional* station—characterised by high demand volumes—or, eventually, up to the terminus of the extension in *San Cristóbal*, should operational loads justify it.

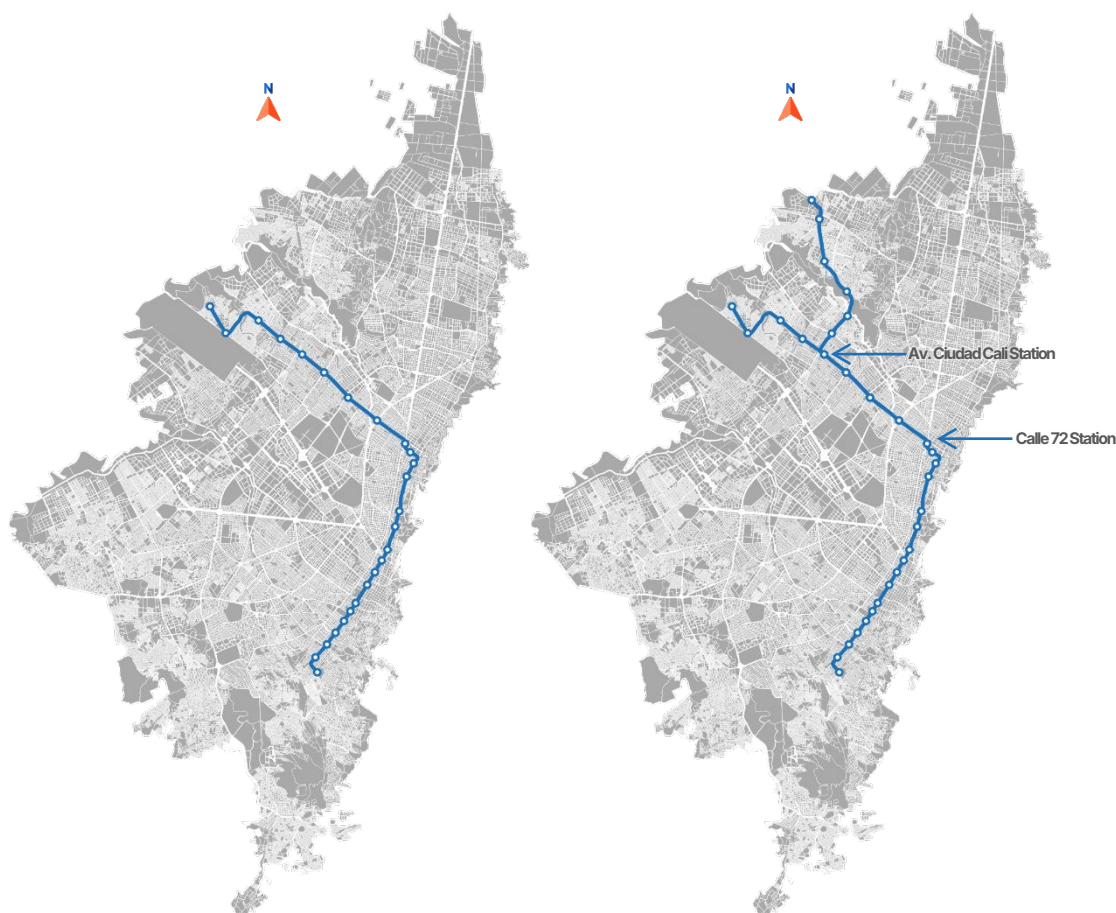


Figure 59. Left: Alignment of the extension and branch proposed for the SLMB. Right: Complete alignment SLMB + Extension and proposed branch. Source: Author's own elaboration.

- **Trace and Alignment:** In the north-west, the corridor begins at the *Engativá Pueblo* square, in the *Engativá* locality, passing through high-density neighbourhoods such as *Garcés Navas*, and subsequently advancing along *Calle 72* eastwards to the homonymous station where it connects with the PLMB. From there, it begins the turn towards the south of the city passing under the traditional neighbourhood of *Quinta Camacho*, to take *Carrera 7* southwards until connecting with the current *Museo Nacional* station. From there, it connects with *Avenida Carrera 10* southwards, following its trace until reaching the *20 de Julio Portal* at *Calle 31 Sur*, where its route will terminate. The extension and branch proposal for the SLMB is 100% underground.

- **Stations:** 20 new stations were georeferenced, located strategically to integrate with existing Transmilenio trunk lines (*Avenida NQS, Avenida Caracas, Avenida Carrera 10*, and the *Avenida 68*), three of the suburban/regional train lines, and two Tram (LRT) lines proposed in this work. In addition to the connection with other transport lines, stations were conceived to be located in both high residential density zones and trip attractors, in accordance with the O-D matrices.
- **Operational Parameters:** Like the original PLMB, for the simulation in r5r, the SLMB extension and its branch to Engativá were configured at an average commercial speed of 43 km/h and a peak hour dispatch frequency of 3 minutes.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_M2-Extensiones.zip` is available on GitHub.

Bogotá Metro Line 3 (TLMB – Tercera Línea del Metro de Bogotá)

Although the Bogotá Metro Line 3 (TLMB) is currently in the technical structuring phase and is planned with an origin in the Centre of Bogotá and a terminus in the *Bosa* locality—similar to the PLMB—its alignment partially utilises the Southern Rail Reserve, through which the railways connecting Bogotá with the southern municipalities historically circulated. For this reason, in this work, it was decided to allocate said rail reserve to the structuring of a suburban or commuter rail project, designed to be compatible with the circulation of medium- and long-distance trains, necessary for the rail articulation of Bogotá with other cities in the region. This project will be explained in section **5.1.3**.

In this context, the TLMB proposed in this work was designed along one of the corridors with the highest transport demand in the city, where multiple mass transit interventions have historically been announced but, to the date of this work's elaboration, have not materialised, and for which no current high-capacity system proposals exist.

Whilst metro projects in Bogotá predominantly follow a radial logic —connecting the western arc with the Expanded Centre— the TLMB adopts a distinct configuration, proposed as a north-south structuring axis through the west of the city that avoids the Expanded Centre, following the trace of *Avenida Boyacá*, the most extensive road corridor in Bogotá. This approach seeks to balance the metropolitan network by strengthening transversal connections and redistributing demand without overloading the central core.

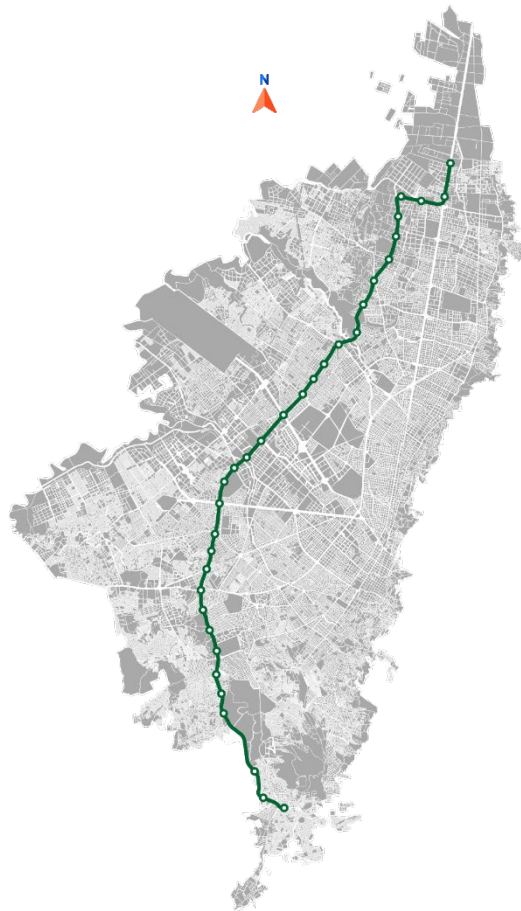


Figure 60. Alignment of the Bogotá Metro Line 3. Source: Author's own elaboration.

- **Trace and Alignment:** The TLMB begins at the *Terminal Norte* (Northern Bus Terminal) , on *Autopista Norte* and *Calle 191*, coinciding with the Transmilenio "*Terminal Norte*" station. From there, it begins its route southwards along *Autopista Nort* up to *Calle 170*, where it turns west to *Avenida Boyacá* and then takes the alignment of this avenue southwards, covering the entire western arc crossing the localities of *Suba*, *Engativá*, *Fontibón*, *Kennedy*, *Bosa*, and *Ciudad Bolívar*, finally terminating at the *Yomasa* junction, in the *Usme* locality.
- **Stations:** 33 stations were georeferenced, located strategically to integrate with existing Transmilenio trunk lines (*Autopista Norte*, *Calle 80*, *Calle 26*, *Avenida de Las Américas*, *Autopista Sur*, and the future *Calle 13*), the PLMB, and those proposed in this work: PLMB Extension, SLMB, two Regional Trains, and a Tram line discussed later in section 5.1.4.
- **Operational Parameters:** Like the lines explained previously, for the simulation in r5r, the TLMB was configured at an average commercial speed of 43 km/h and a peak hour dispatch frequency of 3 minutes.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_M3-Boyaca.zip` is available in the GitHub repository.

Bogotá Metro Line 4 (CLMB – Cuarta Línea del Metro de Bogotá)

The Bogotá Metro Line 4 (CLMB), like the TLMB, fulfills the role of strengthening transversal connections between the city's most populated localities, articulating zones that currently present deficits in direct connectivity and avoiding the need to make unnecessary trips through the Expanded Centre. In this sense, a displacement between *Suba* (to the north) and *Bosa* (to the south) could be made in approximately 30 minutes, providing a fast and high-capacity mass transit alternative for the UPLs with the highest population density (see **Figure 28**).

With the purpose of optimising the use of existing infrastructure, the CLMB is articulated operationally and physically with the PLMB and the SLMB, sharing track sections and transfer nodes. In particular, the line integrates with two PLMB stations and at least one SLMB station, infrastructure onto which it could channel part of the passenger demand to its origin in the *Suba* locality. Additionally, the CLMB functions as a direct technical link between the city's two main depots (*patios-talleres*), acting as a connection branch for the transfer of rolling stock, maintenance tasks, or eventual operational support between lines. For this functionality to be viable, it is indispensable that the technologies and rolling stock of the different lines be technically compatible in terms of signalling systems, electrification, loading gauge, and operational characteristics.

- **Trace and Alignment:** The CLMB begins in *Bosa*, at the *Carrera 96* station shared with the PLMB; from there, it moves east to take *Avenida Ciudad de Cali* northwards and continue along the alignment of that avenue up to the "Av. Cali-Calle 80" station of the SLMB. Just before that station, the line will have physically incorporated into the SLMB infrastructure, allowing it to operate until the end of the latter. The line's coverage passes through the localities of *Bosa*, *Kennedy*, *Fontibón*, *Engativá*, and *Suba*—four of them being the most populated in the city.
- **Stations:** 17 stations were georeferenced, of which 2 are shared with the PLMB and 6 with the SLMB. Stations were located strategically to integrate with existing Transmilenio trunk lines (*Avenida de Las Américas*, *Calle 26*, and *Calle 80*), as well as with the Regional Train of the West proposed in this work, presented in section **5.1.3**.
- **Operational Parameters:** Like the lines explained previously, for the simulation in *r5r*, the CLMB was configured at an average commercial speed of 43 km/h and a peak hour dispatch frequency of 3 minutes.

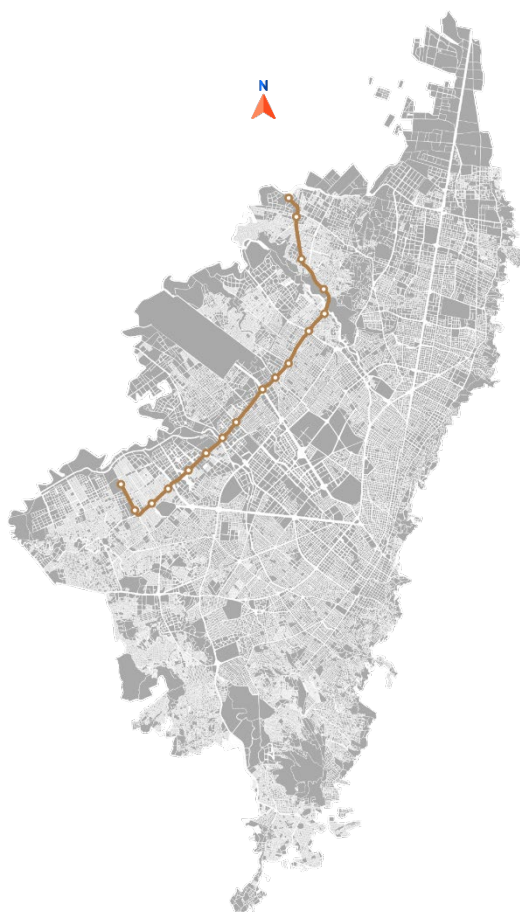


Figure 61. Alignment of the Bogotá Metro Line 4. Source: Author's own elaboration..

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_M4-Cali.zip` is available on GitHub.

5.1.3. Regional Rail Network

This component integrates commuter rail projects utilising the old national rail corridors to connect the municipalities of the region with the heart of the city. Although some projects are already contracted (*Regiotram Occidente*) or in the structuring process (*Regiotram Norte*) under a regional tramway scheme with Tram-Train type rolling stock, international evidence suggests that, for a city of Bogotá's size and suburban demand level, robust regional or commuter trains—capable of transporting high passenger volumes at competitive speeds and with comfort standards typical of interurban services—are more adequate (Kursawe, et al., 2019).

Whilst cities like Paris have an extensive commuter rail network—not counting the RER regional express network—they operate some Tram-Train lines with suburban functions, serving distances and demand levels considerably lower than those characterising the Bogotá Metropolitan Region. The same occurs in medium-sized cities in Germany, where this hybrid model originated. In contrast, Bogotá would become

the first large-scale capital to adopt a Tram-Train system for suburban trips, despite warnings regarding possible limitations in service quality in relation to the magnitude of demand. For this reason, for the purposes of this work, it was decided to model the operation with commuter trains specialised in regional and suburban transport, prioritising rolling stock designed specifically for these services, and not Tram-Train type vehicles.

The three considered lines employ the alignments of the old national railway tracks that historically converged in the centre of Bogotá. Although originally these trains were destined for the *La Sabana* Station, in this work it is proposed to conduct the corridors underground to the heart of the Expanded Centre, so that they can functionally integrate with current transport systems and proposed rail projects. This connection materialises at the *Museo Nacional* and *Estación Central* stations, conceived as a strategic intermodal node between commuter trains, the metro network, and Transmilenio.

For all three projects, operation exclusively within the urban perimeter was taken into account, so that the last station considered for each line is the last one inside the urban zone of Bogotá.

Regional Train of the North

- **Trace and Alignment:** The Regional Train of the North project begins at the underground *Museo Nacional* station, heading west to connect with *Estación Central*, located on *Avenida Caracas* and connecting with the PLMB. From there, it continues underground until connecting with the old rail reserve. At the height of the *Ciudad Salitre* neighbourhood, the line takes the northern branch of the line to continue along that alignment until leaving the city to the north, destined for the northern municipalities. It is expected that in a first stage, the service will cover the transport needs of the municipalities of *Chía*, *Cajicá*, and *Zipaquirá*. As mentioned before, for the purposes of this study, only the urban section of the project in Bogotá was considered.
- **Stations:** 17 stations were georeferenced within the urban perimeter of Bogotá, located to provide connection with existing Transmilenio trunk lines (*Avenida Carrera 10*, *Avenida Caracas*, *Avenida NQS*, *Calle 26*, and the future *Avenida 68*), as well as with the PLMB, SLMB projects, and a Tram line. Stations were also located at major trip attractors, significantly increasing accessibility levels to zones not covered by the Transmilenio system or the Metro Network. The intermodal connection at *Calle 100* and *Carrera 15* stands out, where the Regional Train of the North, the PLMB extension, and the future *Avenida 68* Transmilenio trunk line converge.

- **Operational Parameters:** The line was prepared for simulation in r5r with a commercial speed configuration of 43 km/h and a peak hour dispatch frequency of 5 minutes—a lower frequency than the metro network, but similar to suburban systems in cities like Paris, Berlin, Madrid, or London.

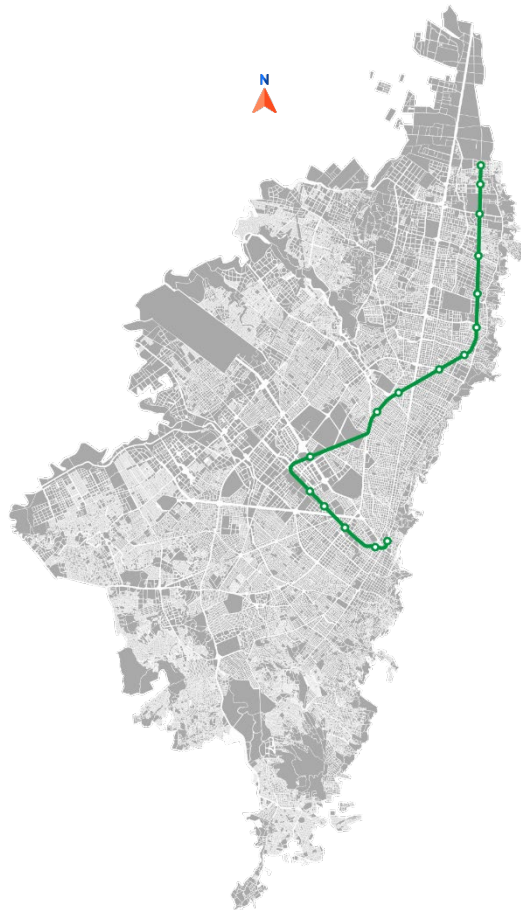


Figure 62. Urban alignment of the Regional Train of the North. Source: Author's own elaboration.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_RE-Norte.zip` is available on GitHub.

Regional Train of the South

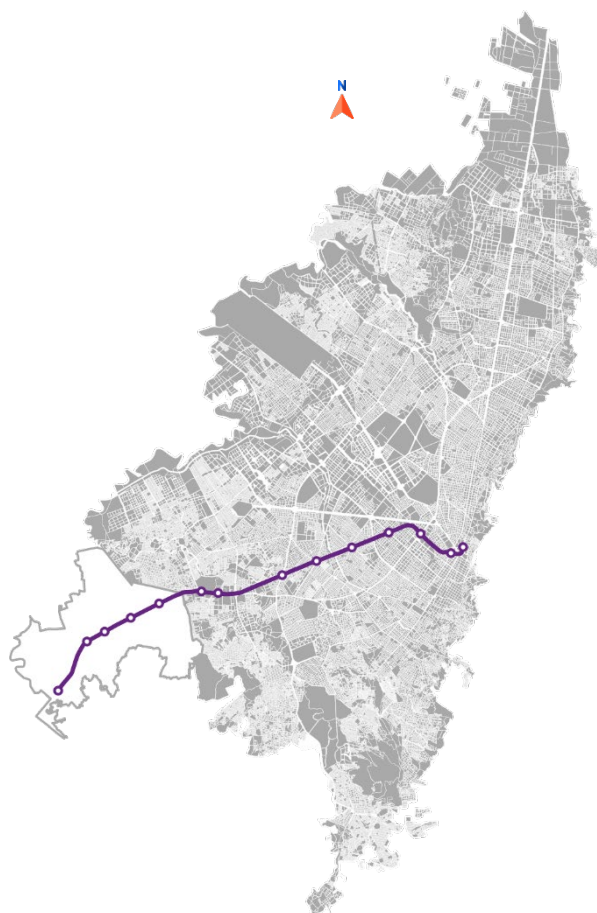


Figure 63. Urban alignment of the Regional Train of the South. The white area at the southern end of the line corresponds to the conurbated municipality of Soacha. Source: Author's own elaboration.

- **Trace and Alignment:** The Regional Train of the South project, like the northern one, begins at the underground *Museo Nacional* station, heading west to connect with *Estación Central* station, located on *Avenida Caracas* and connecting with the PLMB. From there, it continues underground until connecting with the old rail reserve. Once it crosses *Avenida NQS*, the line takes the southern branch and continues along that branch up to the intersection with *Avenida Villavicencio*, where the rail reserve disappears as a consequence of the construction of Transmilenio Phase II, which used part of the reserve for the alignment and construction of the BRT system's *Portal Sur*. Thus, the Regional Train of the South line is forced to continue underground following the alignment of *Autopista Sur* (Southern Highway), up to the *El Vínculo Portal* (last Transmilenio stop), to the south-west of the municipality of *Soacha*.
- **Stations:** 9 stations were georeferenced within the urban perimeter of Bogotá, and 5 in the municipality of *Soacha*, which is conurbated with Bogotá. As in previous projects, stations were located to provide connection with existing Transmilenio trunk lines (*Avenida Carrera 10*, *Avenida*

Caracas, Avenida NQS, Avenida de Las Américas, the future Avenida 68, and Autopista Sur), as well as with the PLMB, SLMB, TLMB projects, and a Tram line. Stations were also located at the main trip attractors of the alignment, significantly increasing accessibility levels to zones not covered by the transport network.

- **Operational Parameters:** The line was prepared for simulation in r5r with a commercial speed configuration of 43 km/h and a peak hour dispatch frequency of 5 minutes.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_RE-Sur.zip` is available on GitHub.

Regional Train of the West



Figure 64. Urban alignment of the Regional Train of the West. Source: Author's own elaboration.

- **Trace and Alignment:** The Regional Train of the West project, like the northern and southern regional trains, begins at the underground *Museo Nacional* station, heading west to connect with *Estación Central* station, located on *Avenida Caracas* and connecting with the PLMB. From there, it continues underground until connecting with the old rail reserve that directs the service directly to the western end of the city destined for the municipalities of *Mosquera, Funza, Madrid*, and

Facatativá. As mentioned before, for the purposes of this study, only the urban section of the project in Bogotá was considered.

- **Stations:** 10 stations were georeferenced within the urban perimeter of Bogotá, located to provide connection with existing Transmilenio trunk lines (*Avenida Carrera 10*, *Avenida Caracas*, *Avenida NQS*, and the future *Avenida 68*), as well as with the PLMB, SLMB, TLMB, CLMB projects, and two Tram lines. As in previous projects, stations were located at the main trip attractors of the alignment, highlighting the *CATAM* station at *Carrera 116*, where a Tram connection destined for Eldorado International Airport is expected.
- **Operational Parameters:** Like the other two regional rail projects, the line was prepared for simulation in r5r with a commercial speed configuration of 43 km/h and a peak hour dispatch frequency of 5 minutes.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_RE-Occidente.zip` is available in the GitHub repository.

5.1.4. Tram Lines

Complementarily, two Tram lines were proposed to cover specific needs in two zones that, at the date of this work's realisation, lack quality transport coverage, and where demand and investment costs do not merit a metro line.

Carrera 7 Tram (T1)

The Carrera 7 Tram is a project in Bogotá that has been stalled for years and has not reached structuring or contracting, especially due to strong pressure for BRT to be constructed in said corridor (Instituto de Desarrollo Urbano, 2021). In this study, the implantation of the line is proposed, covering the needs of the city's eastern edge, crossing the heart of the city's economic activity.

- **Trace and Alignment:** The T1 project begins its route from the *20 de Julio Portal*, where it connects with the PLMB extension, the Transmilenio *Avenida Carrera 10* trunk line, and the *San Cristóbal* Cable Car. From there, it takes *Carrera 7* northwards crossing the Expanded Centre of the city until reaching *Calle 191*, where it turns west to connect with the *Terminal del Norte* station, linking with the TLMB and the Transmilenio *Autopista Norte* trunk line.
- **Stations:** 46 at-grade stations were georeferenced within the urban perimeter of Bogotá, located to provide connection with Transmilenio trunk lines along the route, as well as with the PLMB, SLMB, TLMB projects, and the three Regional Train projects at the height of *Museo Nacional*.

Given their nature, inter-station distances are much shorter than those of the metro network or regional trains, which resulted in a much more significant number of stations.

- **Operational Parameters:** Given passenger demand in the Expanded Centre (Instituto de Desarrollo Urbano, 2021), for the simulation in r5r, the headway (interval between trains) was established at a minimum of 2 minutes in peak hours and 5 minutes in off-peak hours, with a commercial speed of 25 km/h.

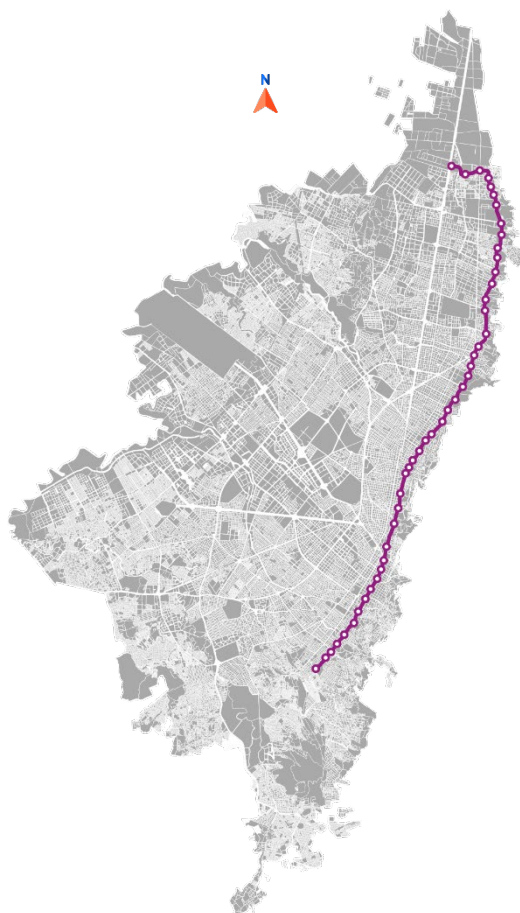


Figure 65. Alignment of the Carrera 7 Tram. Source: Author's own elaboration.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_T1-K7.zip` is available on GitHub.

El Dorado International Airport Tram (T2)

The El Dorado International Airport Tram connects the localities of *Fontibón* and *Engativá* with the Airport of which they are neighbours, providing a direct connection between the neighbourhoods of these localities and the main air terminal in Latin America (BBC Mundo, 2025). Currently, it is absolutely necessary to travel east to reach the airport via *Calle 26*. It is expected that, with this direct connection,

citizens of these localities can save significant time connecting not only with the airport but for the *Fontibón-Engativá* connection.

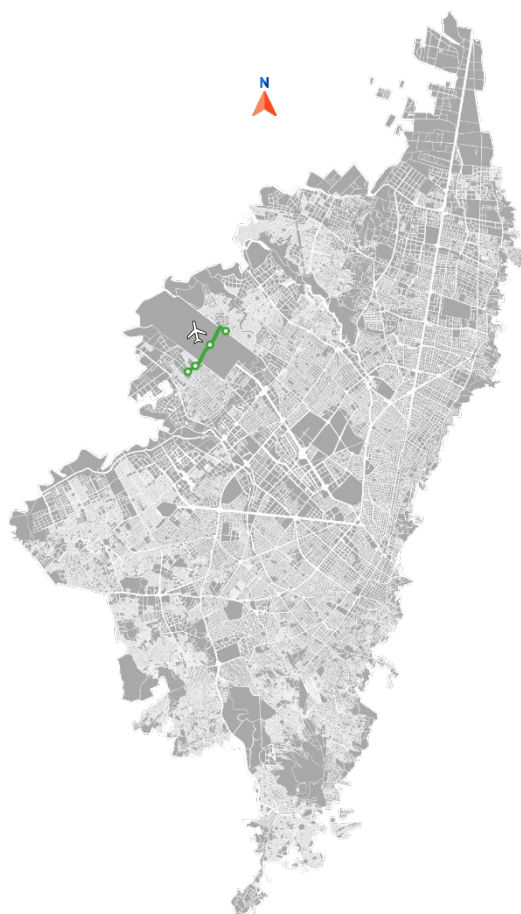


Figure 66. Alignment of the El Dorado Airport Tram. Source: Author's own elaboration.

- **Trace and Alignment:** The T2 project begins its route at the *CATAM-Calle 116* station of the Regional Train of the West project, heading north to the *Atahualpa* neighbourhood, where it has its first stop and begins its descent underground to enter the airport premises and connect directly with Terminal 1, where it has its second stop. From there, it continues its route northwards, to surface in the *Villa Gladys* neighbourhood of the *Engativá* locality, where it has its last stop at the station bearing the same name as the neighbourhood, and connects with the SLMB extension.
- **Stations:** 4 stations were georeferenced (3 at-grade and 1 underground). The stations connect the Regional Train of the West projects and the SLMB extension. The most important stop of the project regarding trip attraction is the El Dorado Airport station, a connection that justifies the project.
- **Operational Parameters:** For the simulation in r5r, a headway between 2 and 5 minutes was established as a function of time and demand, as well as a commercial speed of 25 km/h.

For reference and consultation regarding the trace, stations, and operation of the line and its services, the file `GTFS_T2-ElDorado.zip` is available on GitHub.

Figure 67 below shows the map of the proposed rail network with all its components.

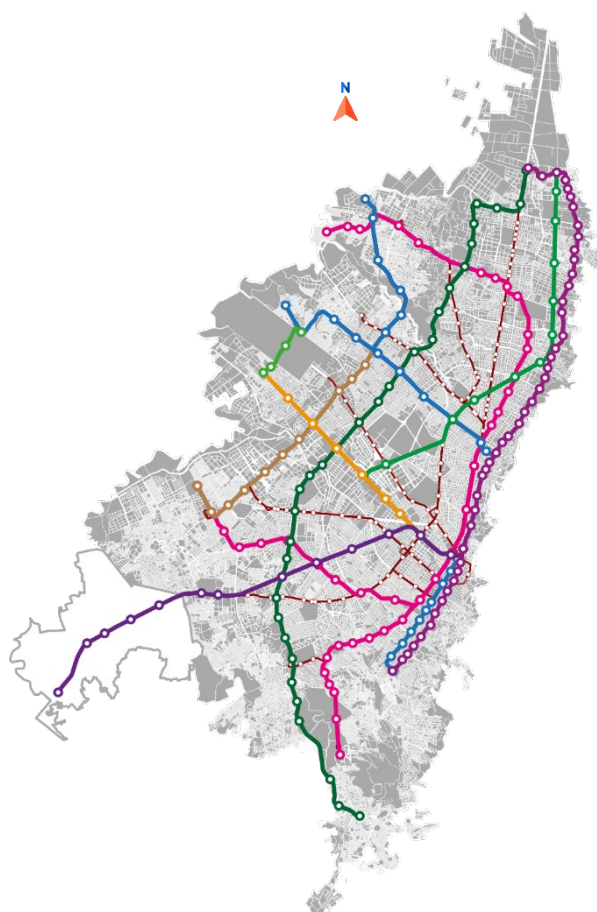


Figure 67. Alignment of the proposed Rail Network, overlaid on existing Transmilenio trunk lines. Source of proposed Rail Network: Author's own elaboration. Source of Transmilenio network: Author's own elaboration with data from (Transmilenio S.A., 2025)

Appendix C presents the complete schematic map of the Bogotá mass public transport system, including the previously described projects. Said map was elaborated by the author of this study within the framework of a mobility consultancy developed during the year 2023 and subsequently adapted for this work, for illustrative purposes, in order to represent the rail network and its stations in integration with the TransMilenio BRT system.

5.2. Facility Implantation Strategies

Recognising that accessibility depends on both transport speed and the proximity of destinations, Scenarios 3, 4, 6, and 7 incorporate modifications in service supply.

Unlike transport scenarios 2 and 5, in which only new Metro lines and the Rail Network are implemented respectively, scenarios 3 and 6 simulate a Transit-Oriented Development (TOD) policy. For their part, scenarios 4 and 7 simulate the full implantation of facilities in zones where there is a deficit thereof, ignoring the TOD policy.

5.2.1. Implantation of Facilities in the Vicinity of the PLMB and Rail Network

The theoretical insertion of new facilities in deficit was modelled within the areas of influence of the metro system and rail network stations, utilising the 15-minute walking isochrone with an additional 50-metre buffer as the spatial limit. This decision is founded on two main criteria:

- A 15-minute isochrone, calculated with the optimal parameters derived from the model calibration process (see section 3.4.3), guarantees that the resulting maximum distance—upon adding the 50-metre buffer—does not exceed 800 metres, a threshold widely accepted in literature as the maximum walking distance for access to urban services, particularly public transport.
- The incorporation of the 50-metre buffer allows for the inclusion of blocks that, otherwise, would only be tangentially intersected by the isochrone. This ensures their consideration within the facility implementation area and reinforces the coherence of spatial coverage within the established temporal limit.

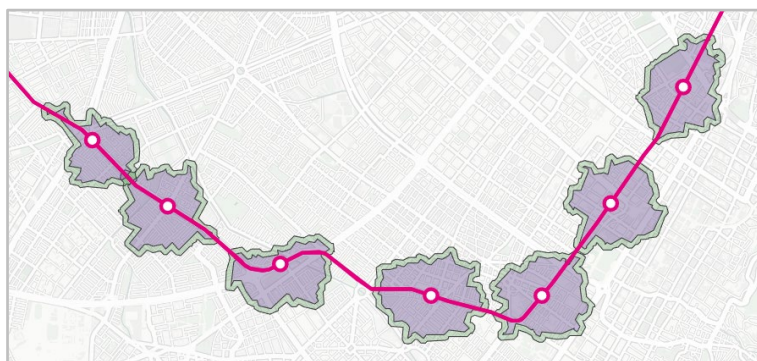


Figure 68. 15-minute isochrones and 50-metre buffers for the implantation of facilities in deficit in the immediate vicinity of rail system stations. Source: Author's own elaboration.

With these areas already determined, the facility deficit by category was sought, and those necessary to reach the city mean were implanted in the blocks found within the temporal threshold (15-minute isochrone + buffer).

This new configuration allows for the evaluation of the City of Proximity hypothesis, verifying whether bringing services closer to transport nodes is more efficient than simply moving people faster towards existing distant services.

The code for the implementation of facilities in deficit in each of the areas (Isochrone + buffer) can be consulted in Script `4_3_Amenities_Scn3.R` and `4_6_Amenities_Scn6.R` available on GitHub, for scenarios 3 and 6 respectively.

5.2.2. Full Implantation of Facilities at the UPL Level

The theoretical insertion of facilities was modelled in those UPLs presenting a supply deficit. The location of these facilities was not delimited by the 15-minute isochrone around stations nor by the associated 50-metre buffer, but was executed at the UPL scale as a territorial intervention unit. This change in implantation scale led to a more homogeneous spatial distribution of facilities between UPLs and, at the same time, to a greater intensity of intervention on the territory.

The code for the implementation of facilities in deficit in each of the UPLs can be consulted in Script `4_4_Amenities_Scn4.R` available on GitHub. Given that facility implantation is performed at the UPL level, this implantation is common between scenarios 4 and 7, meaning the previously mentioned Script is valid for these two scenarios.

5.3. Definition of Modelling Scenarios

To isolate the effects of each intervention and evaluate the synergies between transport and land use, six (6) progressive scenarios were structured (Scenario 2 to 7). These scenarios combine the base transport network (SITP/Transmilenio) with the new projected infrastructures in the rail domain, and the facility implantation and densification strategies in deficit zones. **Table 3** below presents the different scenarios and a short description of each.

Scenario	Description
Scenario 1	Current State (Base)
Scenario 2	Scenario 1 + PLMB
Scenario 3	Scenario 2 + Implantation of facilities in the vicinity of PLMB stations.
Scenario 4	Scenario 2 + Full implantation of facilities at the UPL level
Scenario 5	Scenario 1 + Rail Network
Scenario 6	Scenario 5 + Implantation of facilities in the vicinity of Rail Network stations.
Scenario 7	Scenario 5 + Full implantation of facilities at the UPL level

Table 2. Definition of modelling scenarios. Source: Author's own elaboration.

As explained in section **3.4 Modelling and Simulation Tools**, the intervention scenarios were modelled and run with the aid of computational tools based on R and the r5r library. All scripts necessary for the replication of this study can be consulted in the GitHub repository.

In the following chapter, the results and comparative analysis between the different interventions are presented, as well as their impact on accessibility and urban quality of life.

6

Results and Comparative Analysis

This chapter presents the results obtained from the simulation of mobility and land-use scenarios. The analysis is structured by systematically comparing future scenarios against the Baseline (Scenario 1), allowing for the quantification of benefits in terms of coverage (cumulative accessibility), territorial equity, and impact on urban quality of life.

6.1. Evolution of Cumulative Accessibility

Beyond travelling fast, the objective is to access opportunities. Cumulative accessibility was calculated to measure how many services are reachable within fixed time thresholds of 15 and 30 minutes for each of the scenarios, just as was done for Scenario 1 in section 4.4.

6.1.1. Changes in Accessibility for Scenario 2 with the Implementation of the PLMB

The implementation of the Bogotá Metro Line 1 (PLMB) introduces improvements in urban accessibility levels, particularly along its direct influence corridor. However, these effects present a differentiated spatial distribution according to the temporal threshold considered. For trips of up to 15 minutes, changes in accessibility are relatively contained or marginal, and are concentrated mainly in the immediate vicinity of stations, as observed on the left side of **Figure 69**. Maps of change in urban accessibility at the block level with the implementation of the PLMB, for trips of up to 15 and 30 minutes. Source: Author's own

elaboration., suggesting that the initial impact of the rail system manifests primarily in areas of immediate proximity.

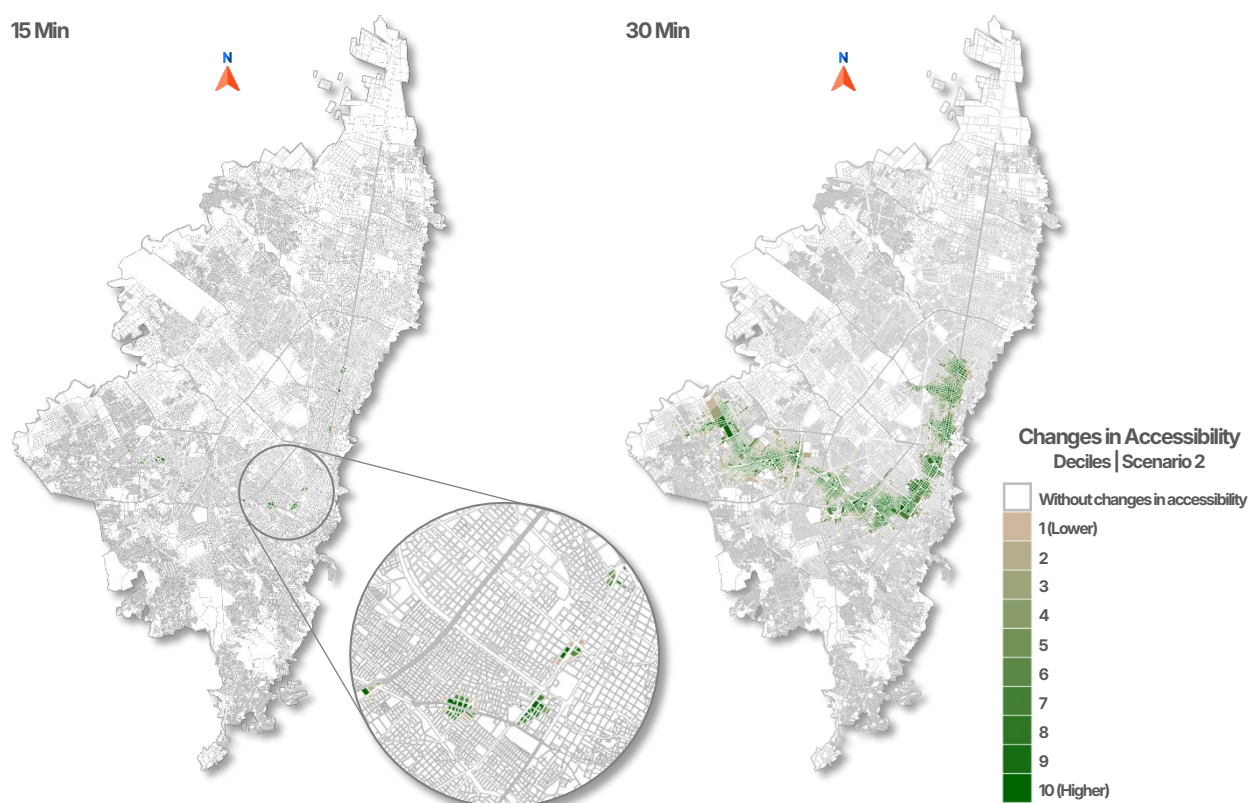


Figure 69. Maps of change in urban accessibility at the block level with the implementation of the PLMB, for trips of up to 15 and 30 minutes. Source: Author's own elaboration.

For trips of up to 30 minutes, conversely, a broader and more significant effect of the PLMB on cumulative accessibility is evidenced. In this case, improvements extend with greater intensity along the project corridor and in its area of influence, where exponential increases are recorded in the number of facilities reachable from multiple blocks.

The results show that **253** blocks (**0.58%** of the city's total) increase their accessibility by an average of **169** amenities per block for 15-minute trips, whereas for 30-minute trips, **5,543** blocks (**12.79%** of the city's total) increase their accessibility by an average of **1,543** amenities per block.

These figures evidence that the effect of the PLMB not only expands the spatial coverage of accessibility when considering larger temporal thresholds, but also substantially multiplies the volume of urban opportunities accessible from each benefited block, reinforcing the structuring role of the metro in the city's functional organisation.

6.1.2. Changes in Accessibility for Scenario 3 with the Implementation of the PLMB and the Implantation of Facilities in Deficit in the Vicinity of its Stations

The combination of the Bogotá Metro Line 1 with the implantation of facilities in deficit in the vicinity of its stations notably expands the accessibility benefits observed in Scenario 2. Unlike the latter, where improvements were concentrated mainly in the metro corridor, in Scenario 3 a more extensive diffusion of accessibility increases is appreciated, particularly for trips of up to 30 minutes as can be observed in **Figure 70**. This suggests that the strategic provision of facilities near stations not only potentiates the effect of the rail system but also generates positive externalities in adjacent areas.

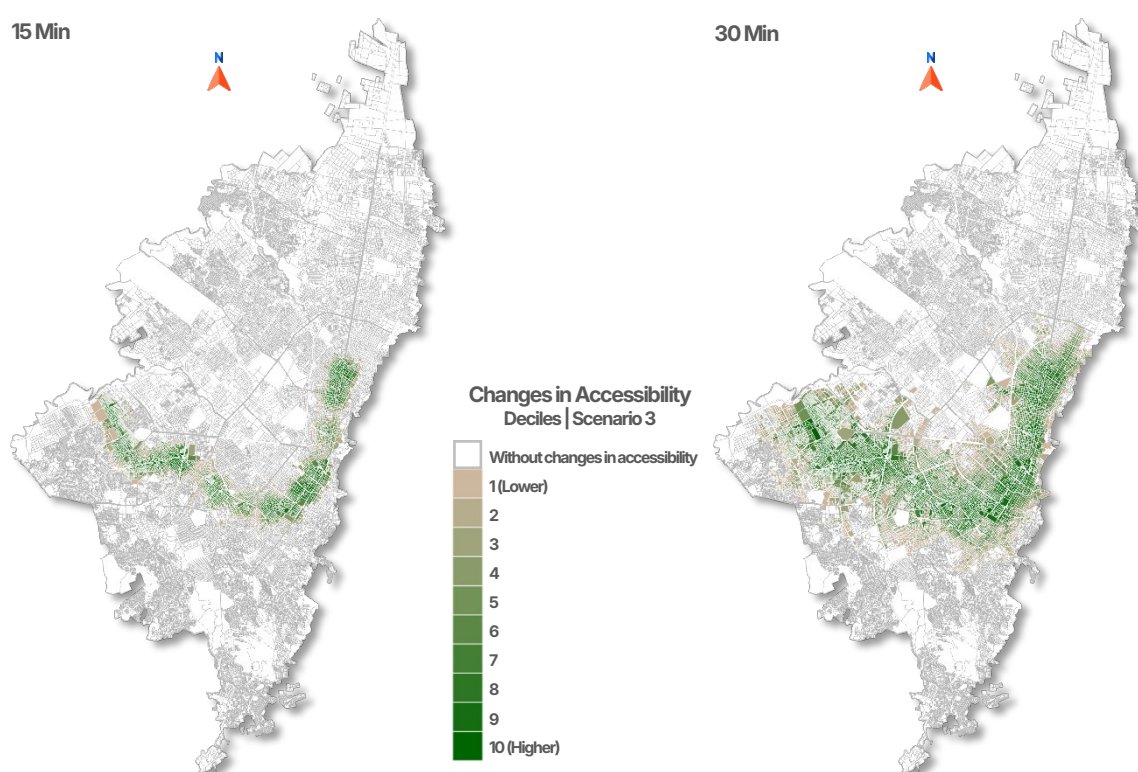


Figure 70. Maps of change in urban accessibility at the block level with the implementation of the PLMB and the implantation of facilities in deficit in the vicinity of its stations, for trips of up to 15 and 30 minutes. Source: Author's own elaboration.

In quantitative terms, the results show that **5,289 blocks (12.21% of the city total)** experience improvements in 15-minute trips, with an average increase of **108** facilities per block, whilst for 30-minute trips the impact extends to **14,031 (32.38% of the city)**, with an average of **1,084** additional accessible facilities. This evidences that the location of facilities near metro stations not only benefits immediately proximate areas, but also improves service access in medium-duration displacements, contributing to a more balanced territorial distribution of urban opportunities.

6.1.3. Changes in Accessibility for Scenario 4 with the Implementation of the PLMB and the Implantation of Facilities in Deficit at the UPL Level

The implantation of facilities in deficit at the UPL level, together with the PLMB, represents a structural change in the spatial distribution of urban accessibility. Unlike previous scenarios, where improvements were concentrated in specific areas or in the metro corridor, in this case practically the entirety of the city experiences significant increases in accessibility levels. This indicates that the strategy of facility provision at the UPL scale has a much broader redistributive effect than punctual interventions around stations.

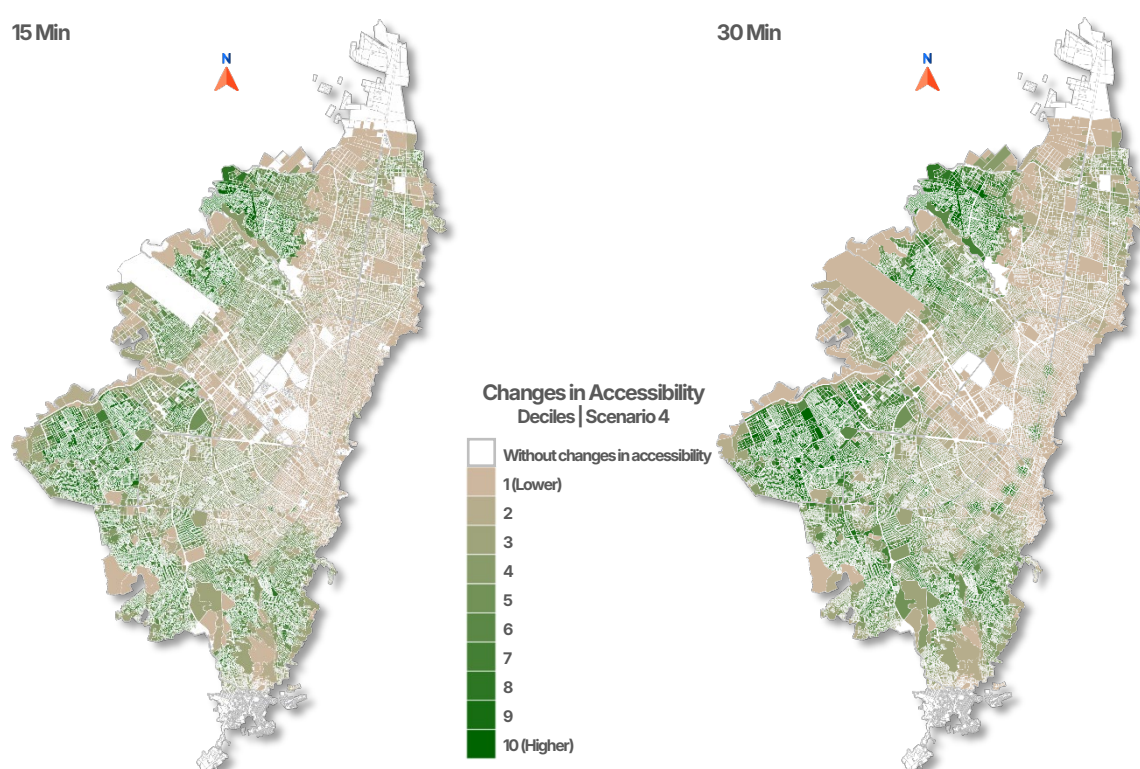


Figure 71. Maps of change in urban accessibility at the block level with the implementation of the PLMB and the implantation of facilities in deficit at the UPL level, for trips of up to 15 and 30 minutes. Source: Author's own elaboration.

Quantitative results confirm this trend: **41,240 blocks (95.17% of the city total)** increase their accessibility for 15-minute trips, with an average of **1,093** additional facilities per block, whilst for 30-minute trips the impact reaches **41,539 blocks (95.86% of the city)**, with an average increase of **8,723** facilities. These figures suggest that facility planning at the UPL scale is much more determinant than transport infrastructure for improving urban accessibility.

6.1.4. Changes in Accessibility for Scenario 5 with the Implementation of the Rail Network

The implementation of the urban Rail Network generates a substantive transformation in accessibility patterns, extending benefits beyond the PLMB corridor and better articulating the urban system at a metropolitan scale. Compared to Scenario 2, a much broader spatial coverage of improvements is observed, evidencing the structuring role of an integrated rail network in connecting areas that previously presented lower accessibility levels.

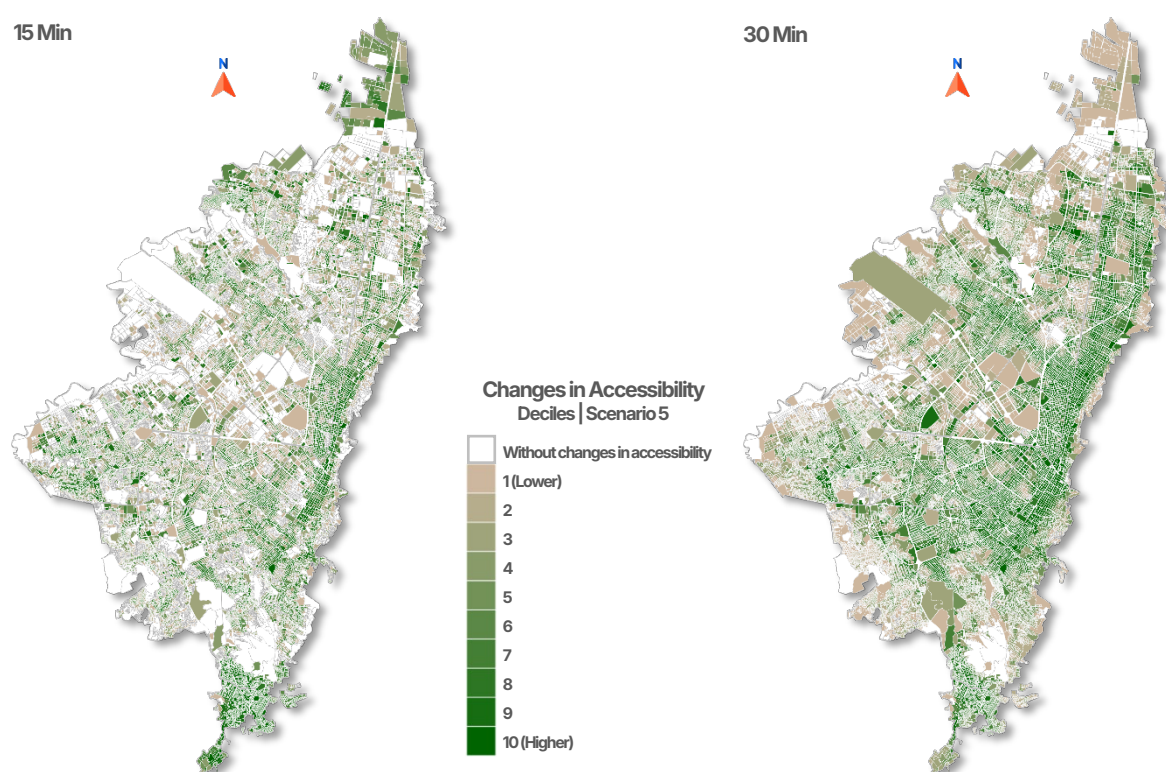


Figure 72. Maps of change in urban accessibility at the block level with the implementation of the Rail Network, for trips of up to 15 and 30 minutes. Source: Author's own elaboration.

Data show that **32,188** blocks (**74.28%** of the city total) record increases for 15-minute trips, with an average of **566** additional facilities per block, whilst for 30-minute trips the effect expands to **41,399** blocks (**95.54%** of the city), with an average increase of **11,928** accessible facilities. This suggests that the consolidation of a metropolitan rail network not only improves travel times but redefines functional relationships between distinct parts of the city by considerably expanding the set of reachable opportunities.

6.1.5. Changes in Accessibility for Scenario 6 with the Implementation of the Rail Network and the Implantation of Facilities in Deficit in the Vicinity of its Stations

The combination of the complete Rail Network with the implantation of facilities in deficit in the vicinity of its stations deepens the effects observed in Scenario 5. In this case, the strategy of locating facilities at transport nodes reinforces the network's capacity to structure urban accessibility, generating improvements in both spatial coverage and the intensity of increases per block.



Figure 73. Maps of change in urban accessibility at the block level with the implementation of the Rail Network and the implantation of facilities in deficit in the vicinity of its stations, for trips of up to 15 and 30 minutes. Source: Author's own elaboration.

Results indicate that **36,457 blocks (84.14% of the city)** experience improvements in accessibility for 15-minute trips, with an average of **667** additional facilities, whilst for 30-minute trips the impact reaches **41,962 blocks (96.84% of the city)**, with an average increase of **13,994** facilities. This evidences that the articulation between transport infrastructure and localised service provision is key to maximising accessibility benefits across distinct temporal thresholds.

6.1.6. Changes in Accessibility for Scenario 7 with the Implementation of the Rail Network and the Implantation of Facilities in Deficit at the UPL Level

Scenario 7 represents the most comprehensive level of intervention, combining the complete Rail Network with the implantation of facilities in deficit at the UPL level. This approach integrates both the metropolitan scale of transport and the territorial planning of services, generating the greatest accessibility increases among all analysed scenarios. In this case, practically the totality of city blocks experience substantive improvements, suggesting a profound reconfiguration of access to urban opportunities.

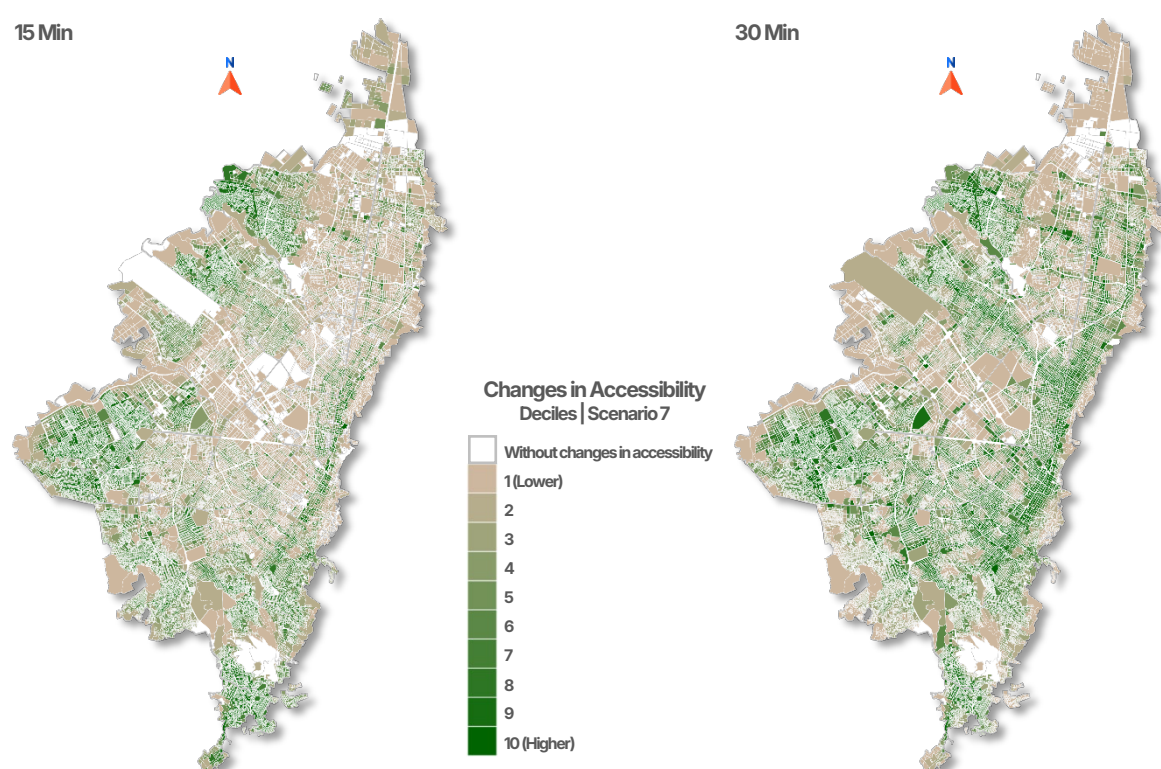


Figure 74. Maps of change in urban accessibility at the block level with the implementation of the Rail Network and the implantation of facilities in deficit at the UPL level, for trips of up to 15 and 30 minutes. Source: Author's own elaboration.

In quantitative terms, **42,060 blocks (97.07% of the city total)** increase their accessibility for 15-minute trips, with an average of **1,628** additional facilities per block, whilst for 30-minute trips the impact extends to **42,877 blocks (98.95% of the city)**, with an average increase of **22,363** facilities. These figures evidence that the combination of rail infrastructure and decentralised facility provision constitutes the most effective strategy for guaranteeing broad and equitable accessibility throughout the urban territory.

6.2. Spatial Equity Analysis

One of the central objectives of the research is to determine whether the interventions reduce the inequality gap. By cross-referencing the model results with socioeconomic stratification, a progressive correlation is evidenced: the greatest relative accessibility gains are concentrated in strata 0, 1, 2, and 3.

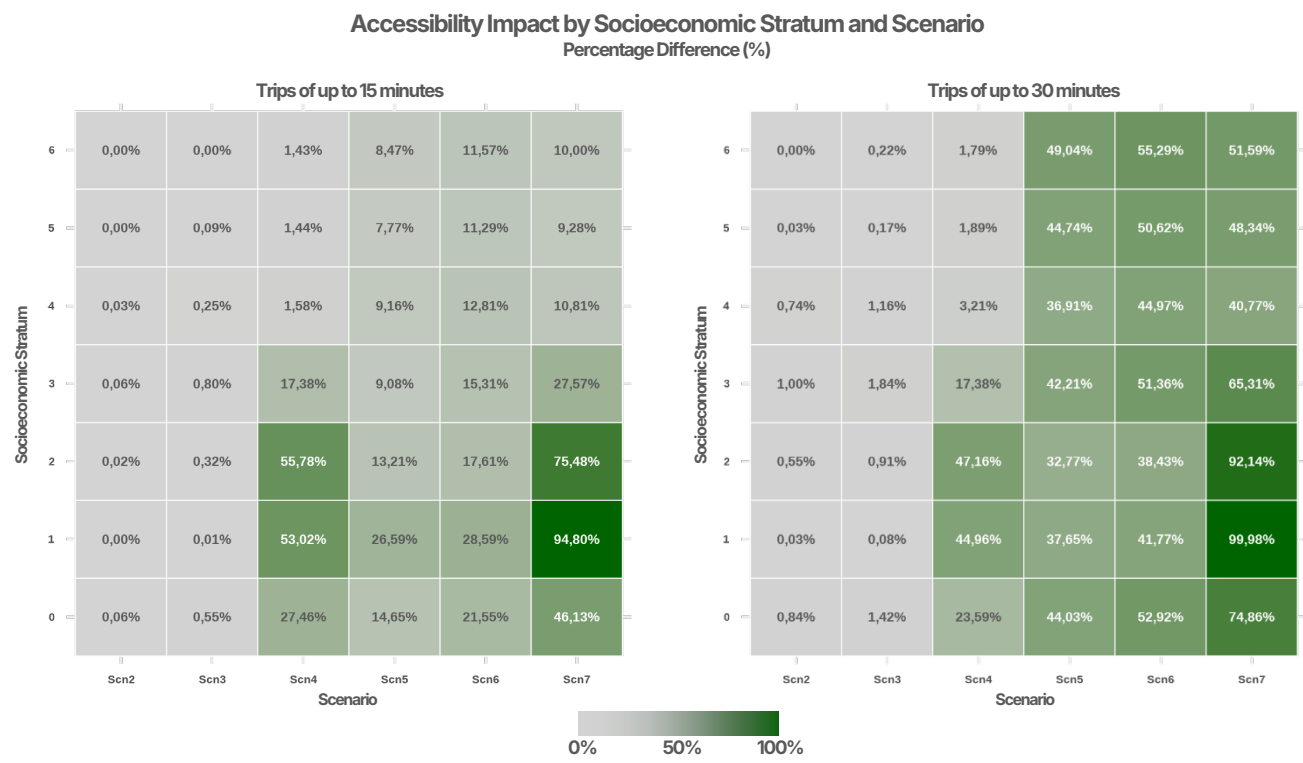


Figure 75. Heatmap of the percentage difference in improvement by socioeconomic stratum in accessibility for Scenarios 2 to 6, with respect to the baseline scenario. Source: Author's own elaboration.

In general, for trips of up to 15 minutes, the higher socioeconomic strata do not present improvements as pronounced as the lower socioeconomic strata, presumably because they already possessed relatively high levels of accessibility in the baseline scenario. In contrast, the lower socioeconomic strata — historically segregated and with structural deficits in access to services— exhibit the most significant increases in these indicators.

For trips of up to 30 minutes, the increase in accessibility becomes even more marked, even in the higher socioeconomic strata, particularly in Scenarios 5 to 7, where the rail network is incorporated. Nevertheless, the most substantive improvements are observed in Scenarios 4 and 7, in which facilities in deficit are implanted at the UPL level. This confirms that a policy of full proximity can be even more effective than the sole implementation of the urban rail network for advancing towards greater equity in access to services.

By cross-referencing the blocks recording improvements in accessibility with their resident population, it is possible to estimate the number of inhabitants benefited by said improvements. In line with what was exposed in section 6.1, the results show that, in the most advanced scenarios, practically the totality of the urban population is favoured by the increases in accessibility. **Figure 76** presents the distribution of the benefited population according to the different interventions analysed. In particular, a high concentration of benefited population is observed in strata two and three, which correspond precisely to the socioeconomic groups with the greatest demographic weight in the city.

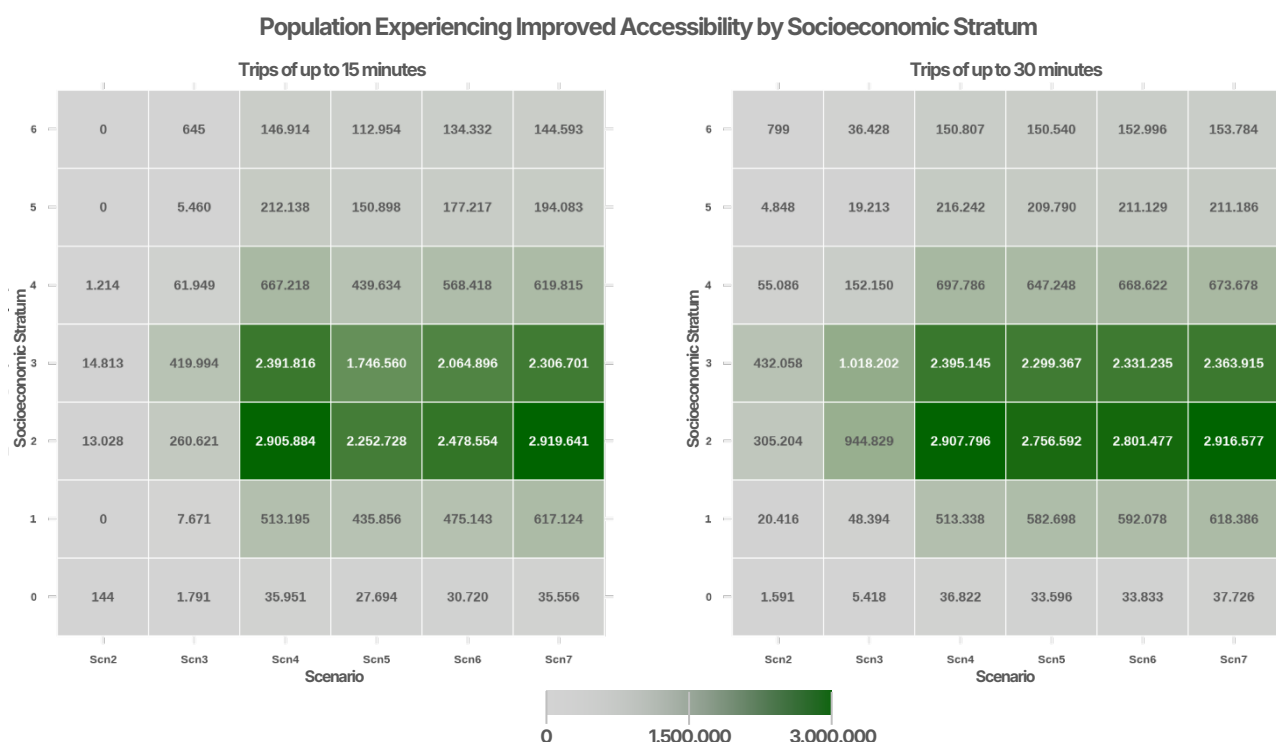


Figure 76. Heatmap for the population with improvement in accessibility by socioeconomic stratum. Source: Author's own elaboration.

6.3. Impact on the High Quality of Societal Life Index

Upon analysing the evolution of the High Quality of Societal Life (HQSL) Index throughout the seven scenarios, a clear correlation is observed between this indicator and improvements in accessibility, given that the components associated with Social Functions are directly linked to access to urban facilities and services. Consistent with the results presented in the percentage differences by socioeconomic stratum (**Figure 75**) and in the heatmap of the population benefited by accessibility improvements (**Figure 76**), **Figure 77** shows particularly significant increases in the HQSL Index in Scenarios 4 and 7, characterised by the implementation of a policy of full implantation of facilities at the UPL level.

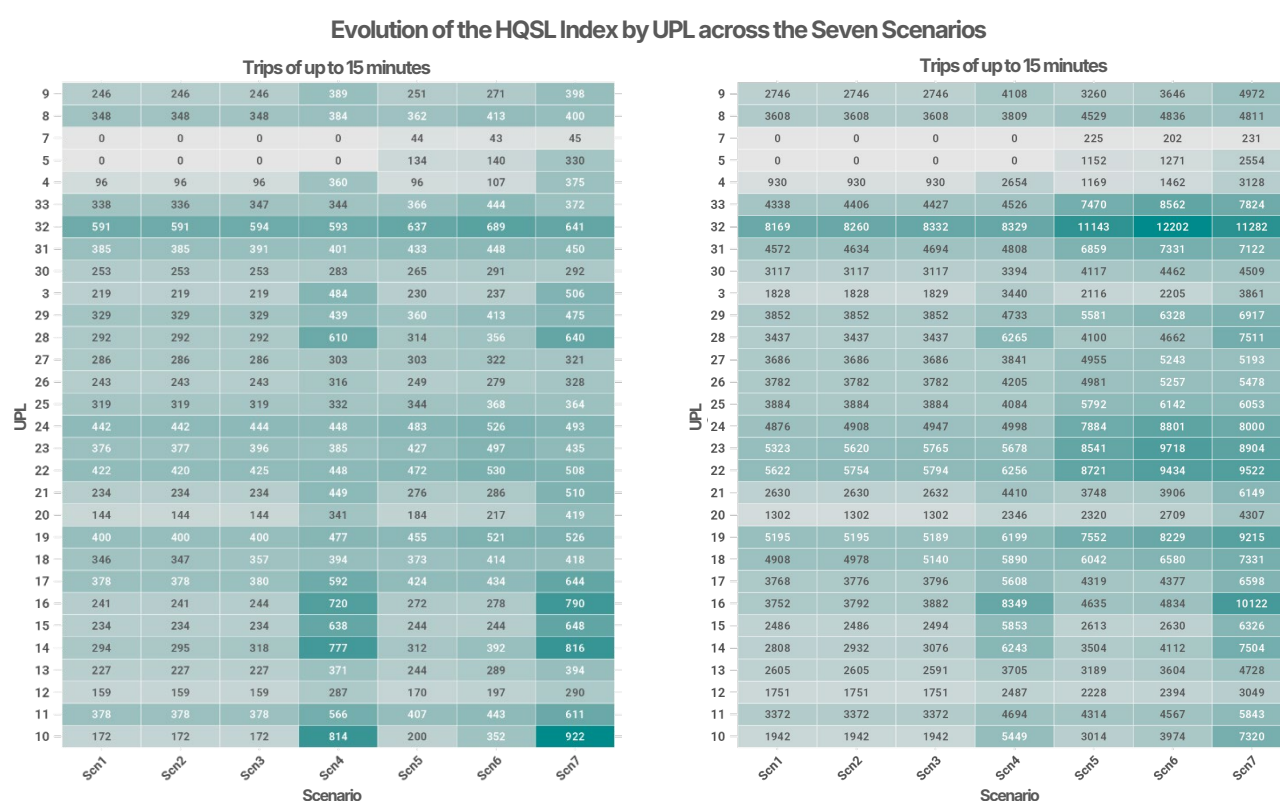


Figure 77. Heatmap of the evolution of the HQSL Index by UPL across the 7 scenarios. Source: Author's own elaboration.

Likewise, **Figure 77** evidences that the greatest positive variations in High Quality of Societal Life are concentrated in those UPLs that presented the lowest accessibility levels in the baseline scenario, which reinforces the redistributive nature of interventions oriented towards proximity and the territorial balance of access to urban opportunities.

With the aim of analysing in greater detail how not only Social Quality of Life evolves as a whole, but also its individual components—the Social Functions (Learning, Supplying, Caring, Enjoying, Working, and Living) as well as the Composite Indicators (Well-being, Sociability, and Sustainability)—and to have a detailed diagnosis for each UPL in relation to the city as a whole, 420 ScoreCards were elaborated. These allow for the systematic examination of each of the 30 urban UPLs of Bogotá for all intervention scenarios and for the two temporal thresholds analysed (15 and 30 minutes). The 420 ScoreCards can be consulted in the GitHub repository.

Figure 78 below presents a comparative example corresponding to UPL 15 *El Porvenir*—one of the units with the lowest levels of accessibility to work, care, and higher education opportunities in the city— for two contrasting scenarios, such as Scenario 1 (baseline) and Scenario 7 (implementation of the urban rail network + implantation of facilities at the UPL level), considering the 15-minute temporal threshold.

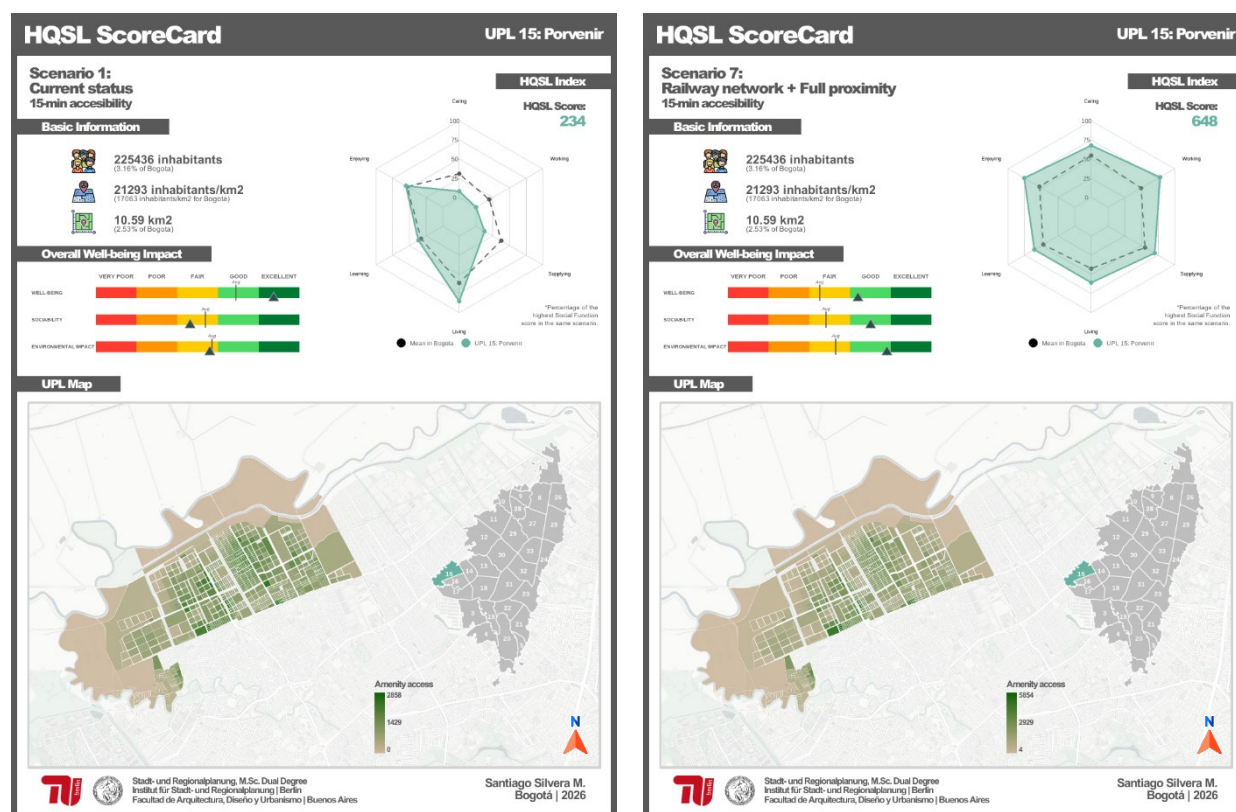


Figure 78. Example of comparative ScoreCard for UPL 15, for 15-minute trips in Scenarios 1 and 7. Source: Author's own elaboration.

Continuing with the example of UPL 15: *El Porvenir*, it is observed that, in those scenarios where rail-based mass transit systems are implemented, the corresponding isochrone extends along the rail axes, increasing spatial coverage for the same temporal threshold. This behaviour reflects the structuring capacity of the rail system to expand the territorial reach of daily displacements.

In practical terms, this implies that a resident of, for example, UPL 15 (the same UPL taken from **Figure 78**) can access the Expanded Centre and its facilities within the same time budget, which implies an effective rupture of their spatial confinement and a substantive improvement in their possibilities of access to urban opportunities. **Figure 79** below shows the expansion of isochrones for the two types of intervention with the implantation of rail systems for Bogotá.

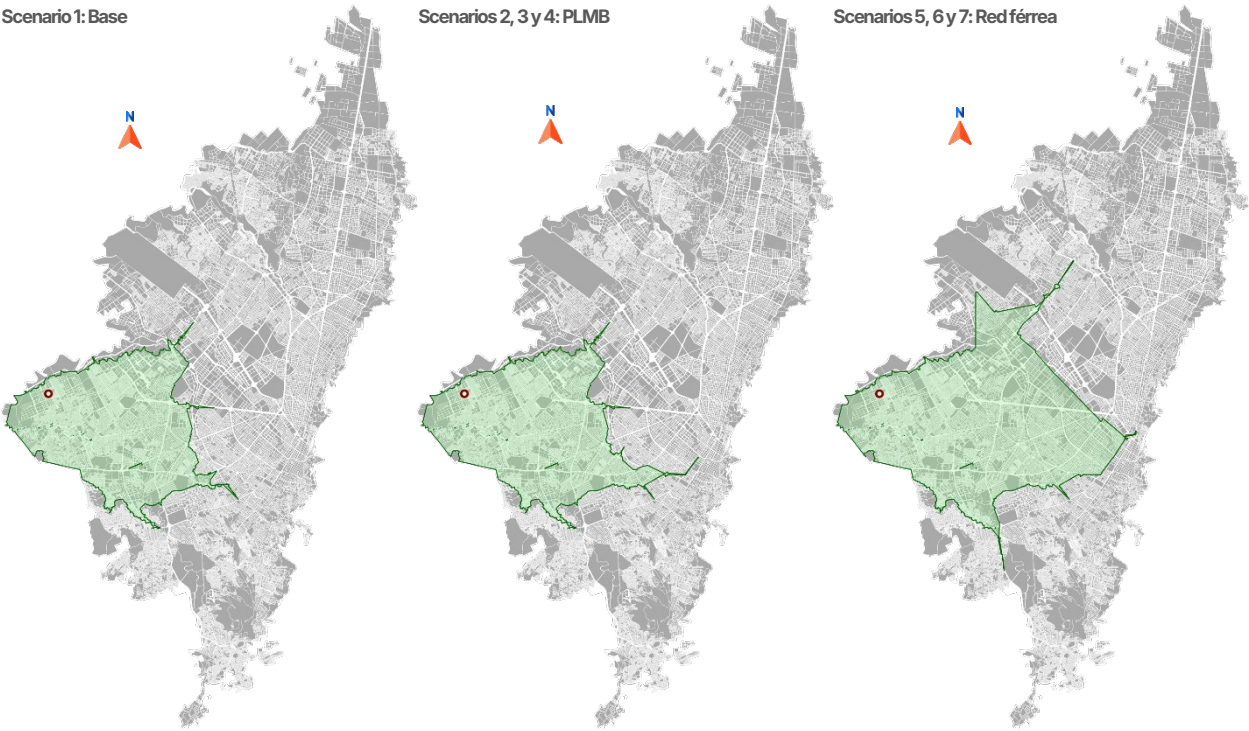


Figure 79. Expansion of isochrones for the different scenarios. **Left:** Scenario 1 (Base). **Centre:** Scenarios 2, 3, and 4 with the implementation of the Metro. **Right:** Scenarios 5, 6, and 7 with the implementation of the Rail Network. Source: Author's own elaboration.

7

Discussion and Implications

This chapter interprets the results obtained in the simulation, placing them in the context of Bogotá's structural challenges. The validation of the posed hypotheses is discussed, contrasting the efficacy of mobility strategies versus land-use strategies, and strategic guidelines for future urban planning public policy are derived.

| 7.1. Synergy between Speed vs. Proximity

The most compelling finding of the research (evidenced in the comparison between Scenarios 2 to 4, and 5 to 7) is that transport infrastructure, on its own, is insufficient to guarantee universal access to the city.

Whilst the First Metro Line (PLMB) and the Rail Network manage to "bring closer" the extremes of the city (space-time compression effect), the 15-minute accessibility maps demonstrate that rapid transport does not resolve the deficit of local services. The model confirms that:

- **Transport (Metro/Trains):** Resolves medium and long-distance accessibility (employment, university), expanding isochrone areas and their corresponding reach to facilities.
- **Land Use (Facilities):** Resolves local accessibility (care, supply, primary health), as opportunities densify within the 15-minute radius in the vicinity of metro or rail network stations.

In this sense, the discussion transcends the dichotomy between "improving transport" or "densifying the city", which, in the particular case of Bogotá, has historically translated into marginal interventions on the transport system. In parallel, urban densification processes have been oriented mainly towards the expansion of residential real estate supply in peripheral areas, rather than towards the densification of services and facilities, which has ended up aggravating general accessibility and mobility conditions in the city.

The results of Scenario 7 show that advancing towards a more equitable city requires the simultaneous implementation of both strategies. In this hybrid model, mass transit fulfills the role of the backbone of the urban system, whilst the capillary distribution of facilities —through an implantation strategy at the UPL level— is what allows for the materialisation of tangible improvements in urban quality of life. Likewise, investment in high-capacity, high-speed, and high-reliability rail infrastructure —traditionally superior to that of the Transmilenio BRT— (Scenarios 5, 6, and 7) contributes significantly to this objective, although on its own it is insufficient to fundamentally resolve structural inequalities in accessibility.

7.2. Spatial Equity and Redistribution of Opportunities

The equity analysis in section 6.2 reveals that the proposed models act almost progressively as mechanisms of spatial justice. Historically, infrastructure investment in Bogotá has tended to reinforce the advantages of consolidated zones, the private vehicle, and some Transmilenio-based mass transit corridors. However, the results produced by this simulation show a reversal of that trend, indicating that the greatest beneficiaries of gained accessibility (both percentage-wise and absolute) are the lowest socioeconomic strata.

This implies a paradigm shift. Accessibility ceases to be a privilege of central location to become a guaranteed right. Observing the case of the UPLs in the city's western arc validates that peripheral zones cease to be synonymous with exclusion. The rail connection, added to the provision of facilities, allows an inhabitant of the south-west to access a basket of services comparable to that of an inhabitant of *Teusaquillo* or *Chapinero*, breaking with the socio-spatial segregation inherited from the poor urban planning of the 20th century.

7.3. Viability of the "City of Proximity" in the Bogotá Context

The results obtained allow for a nuanced application of the "15-Minute City" or "City of Proximity" concept in the context of a Latin American megalopolis. The simulations performed suggest that aspiring to full functional autarky—in which the totality of daily activities can be resolved exclusively on foot—proves unrealistic in the short term for certain metropolitan-scale functions, such as specialised employment or high-complexity health services.

Nevertheless, the model validates an adapted version of this approach, which can be termed the "15+30 City". This proposal combines the principles of the 15-minute city and the 30-minute territory (Chaire ETI - Sorbonne Université Paris 1, 2025), and is structured based on two complementary components:

- **Guaranteeing pedestrian access (15 minutes)** to essential functions of care, supply, and basic education, supported by a strategy of facility implantation at the UPL level.
- **Guaranteeing access via rapid public transport (30 minutes)** to functions associated with employment and higher-complexity services, supported by a high-capacity rail network.

This approach is consistent with the findings presented in section 7.1 and highlights that it is precisely this dual structure that maximises the High Quality of Societal Life (HQSL) Index.

Pretending to resolve the totality of urban needs solely through proximity would ignore the dynamics of the metropolitan labour market; doing so exclusively through rail infrastructure would neglect the basic and care functions that are critical for the most vulnerable population.

| 7.4. Implications for Public Policy and Planning

In light of the findings obtained, three key guidelines for public policy are derived.

7.4.1. From TOD to Integrated Planning

It is not enough to densify housing around Metro stations. The results of Scenarios 3 and 6 indicate that it is imperative to reserve land for public facilities in the partial urban renewal plans associated with future stations. A metro station without a library, kindergarten, jobs, commerce, or health centre in its immediate environment constitutes a missed opportunity in terms of accessibility. In this sense, the Urban Renewal Agency (Empresa de Renovación Urbana - ERU), the Urban Development Institute (Instituto de Desarrollo Urbano - IDU) —entities of the Capital District— as well as the Bogotá Metro Company (which also fulfills functions as a real estate manager), must explicitly incorporate this dimension into their planning and management instruments.

7.4.2. The UPL as an Effective Management Unit

The effectiveness demonstrated by the strategy of facility implantation at the UPL level (Scenarios 4 and 7) validates the commitment of the current Land Use Plan (POT) —Decree 555 of 2021—to functional decentralisation (Secretaría Distrital de Planeación, 2021). Public investment should not be dispersed homogeneously across the territory, but rather focused on completing the missing "service battery" in those deficit UPLs, in order to raise their baseline HQSL Index.

Nevertheless, it is fundamental that the District Planning Secretariat (Secretaría Distrital de Planeación - SDP) and the ERU pay special attention to the implementation and execution of the strategic actions defined in the POT, given that there are risks associated with gentrification processes that could lead to adverse effects, potentially making the cure worse than the disease (Campos, 2021) (Contreras & Mejía, 2024).

7.4.3. Institutional Coordination

The results evidence the need for the Bogotá Metro Company (responsible for Metro lines), the Regional Rail Company (in charge of departmental-scale rail lines), and the District Planning Secretariat (competent in land-use matters) to operate under a common matrix of objectives and intervention criteria. Planning the Metro and the Rail Network without simultaneously articulating the provision of facilities in their areas of influence significantly reduces the social return on investment in infrastructure.

Coordination with national-level agencies proves equally indispensable, even more so considering that, according to the so-called Metro Law (Ley de Metros), rail projects can be financed up to 70% with national resources (Congreso de la República de Colombia, 1996). In this sense, information obtained from a meeting held on 12 December 2026 with the director of the National Real Estate Agency (Agencia Nacional Inmobiliaria) —personal communication— indicates that the lack of articulation between said entity and the Capital District has constituted an obstacle both for the implementation of strategies for the development and rehabilitation of service infrastructures directed at the urban population, and for the formulation of policies oriented towards the promotion of social housing in central sectors of the city, where accessibility to existing facilities is largely guaranteed. This statement aligns with the postulates of Carlos Moreno, who maintains that without social housing there can be no proximity in cities (“Sin vivienda social no puede haber proximidad en las ciudades”) (La Razón, 2026).

8

Conclusions and Future Work

The present research addressed the challenge of urban accessibility in Bogotá through a quantitative approach, modelling and evaluating the impact of the future rail network and strategies for facility implantation and densification. Below are the main conclusions derived from the analysis of the six intervention scenarios, the methodological contributions of the study, and lines for future research.

| 8.1. General Conclusions

The research allows for the conclusion that the materialisation of an equitable and efficient city model in Bogotá does not depend on a single solution, but on the synchronisation of two planning scales: Mobility (understood as urban and regional transport infrastructure) and proximity urbanism.

The results validate that the isolated implementation of mass transit infrastructure (First Metro Line and Rail Network), although effective for reducing extreme travel times and connecting the periphery with the centre, is insufficient to guarantee universal access to basic services at the local scale. This conclusion can be extrapolated to the implementation of the existing BRT-based mass transit in the city. Similarly, "city of proximity" strategies based solely on land use are unviable without a transport system that supports metropolitan dynamics of employment and higher education.

Therefore, it is concluded that Scenario 7 (Rail Network + Facilities per UPL) represents the social optimum, demonstrating that it is possible to transform Bogotá's monocentric structure towards an

interconnected polycentric network, where quality of life (measured by the HQSL index) ceases to depend on the citizen's postcode.

Naturally, the investment scope and time horizon required for an implementation such as that of Scenario 7 may be substantial, even more so when the implantation of facilities in deficit at the UPL level may imply high investments, institutional efforts, and policy continuity amidst administrative changes. Approaches such as that of Scenario 6 (Rail Network + Facilities in proximity to stations) may become possible within shorter timeframes and lower investments, as a preliminary step towards a more ideal scenario, such as Scenario 7.

8.2. Specific Findings

Based on the comparative analysis of indicators, three empirical findings stand out.

- **Reduction of the Centre-Periphery Gap:** The model demonstrated that the rail network acts as a powerful social equaliser. Marginal accessibility benefits are barely notable for high strata (which are already well connected), but strongly increasing for strata 0, 1, 2, and 3. In peripheral zones, such as UPL 15 "El Porvenir", taken as an example in **Figure 78**. Example of comparative ScoreCard for UPL 15, for 15-minute trips in Scenarios 1 and 7. Source: Author's own elaboration., scores for the Social Functions of Working, Caring, Learning, and Supplying increased exponentially, validating the redistributive role of rail transport.
- **The Need for the "15+30 City":** The study quantified the functional difference between walkability and public transport. It was determined that the 15-minute threshold is critical for functions of care, supply, and basic education (dependent on facility density), whilst the 30-minute threshold is the determinant for access to the labour market and specialised services (dependent on the Metro network).
- **The Impact of Territorial Coverage:** It was evidenced that the strategy of facility implantation at the UPL level generates an impact on the HQSL index superior to that of implantation in the immediate vicinity of stations. This suggests that public policy must prioritise covering "service deserts" in neighbourhoods, beyond merely densifying transport corridors.

| 8.3. Research Contributions

This work contributes to the field of urban planning and transport in Bogotá through:

- **Development of Open Tools:** The creation and publication of scripts in R for the generation of synthetic GTFS files, the query of facilities and their classification for the context of urban service supply; the calibration and modelling for the calculation of urban proximity, and the generation and calculation of HQSL indicators, allowing for the reproducibility of the study and its adaptation to other Latin American cities.
- **Adaptation of the HQSL Index:** The calibration of the "15-Minute City" theoretical framework to the data context and dynamics of a Latin American megalopolis, integrating the mass public transport variable as an articulating axis.
- **Evidence for Decision Making:** The generation of a matrix of quantitative results that allows decision-makers to visualise the opportunity cost of not integrating mobility policies with those of habitat and urban quality of life.

| 8.4. Study Limitations

It is important to recognise the restrictions that limit the scope of the obtained results:

- **Static Nature of Demand:** The model assumes a fixed population distribution (based on the 2018 Census). Gentrification dynamics or population displacement that could occur as a consequence of the increase in land value around new Metro stations or the Rail Network were not modelled.
- **Capacity Simplification:** Whilst the GTFS models frequencies, the r5r engine utilised calculates travel times based on itineraries, without simulating overcrowding inside carriages or pedestrian congestion during transfers, factors that could affect the real perception of urban quality of life.
- **Availability of Facility Data:** The synthetic facility inventory based on available open data assumes the physical existence of the building guaranteeing service provision, without evaluating qualitative variables such as service quality or cost for the user, or even the real existence of the facility and its capacity.
- **Reliability of Open and Official Data:** Although the data used for modelling were mostly official data, missing or incompatible information between datasets was evidenced. Thanks to the dedicated data science work which can be consulted in the R scripts available on GitHub, a sufficient level of reliability was achieved so that information could be processed and analysed. Nevertheless, it could occur that some information is not updated from its official sources.

- **Processing Power:** Some scripts and simulation processes for the urban accessibility calculation required days of computational work. The reproducibility of the work could be limited if sufficient computing power is not available.

| 8.5. Recommendations and Future Work

To deepen this line of research, the following paths are suggested:

- **Incorporation of Economic Analysis:** Complementing spatial analysis with cost-benefit models comparing the investment necessary to construct rail infrastructure versus the cost of building the proposed facility network.
- **"Last Mile" Analysis:** Refining the model by incorporating the quality of pedestrian infrastructure (pavements/footpaths, lighting, safety) and integration with micromobility (bicycles, e-scooters) to evaluate real barriers to accessing rail stations.
- **Gentrification Modelling:** Investigating how the improvement in the HQSL index could impact land prices and proposing value capture mechanisms to finance the facility network itself.
- **Scope of Study Area:** Bogotá-Region has a population of 9.3 million inhabitants, of whom 1.8 million do not live in the city, but in 20 other municipalities of the region (Alcaldía de Bogotá, 2023), making it necessary to expand the study area and perform the analysis including that population and their service access needs.
- **Dynamic and Environmental Simulation:** Implementing simulations via Agent-Based Modelling (ABM) software for the proposed scenarios, so as to transcend the static approach. This would allow for the modelling of behavioural decisions to estimate modal shift (substitution of private vehicle trips for public transport or walking), quantify energy savings, emission reductions, and evaluate their repercussion on Bogotá's public health.

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Appendix A: Script Repository on GitHub

The methodological development of this thesis—spatial processing, data science techniques, modelling, and the generation of maps and graphs—was conducted using scripts implemented in the R language, ensuring the reproducibility of the calculations. The set of scripts is available in the `Tesis_MPMU` GitHub repository.

Repository Access

To access the complete repository, any of the following three options may be used:

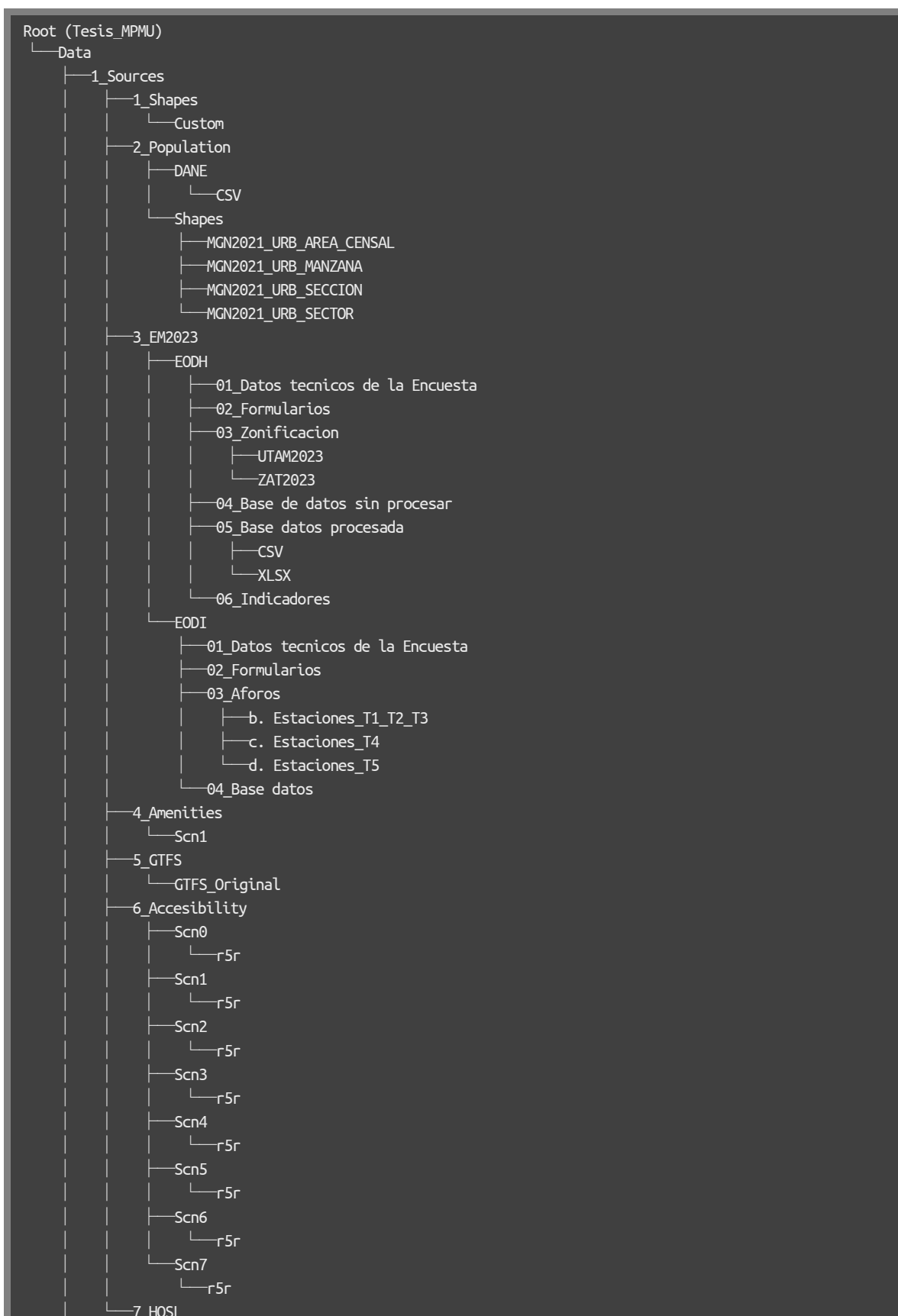
- [Click here](#) (Digital document versions only).
- Go to https://github.com/ssilveram/Tesis_MPMU
- Scan the QR code below:



List of Scripts

1_Empty_Shapes.R	6_2_Accesibility_Scenario_2.R
2_Population.R	6_3_Accesibility_Scenario_3.R
3_EM2023.R	6_4_Accesibility_Scenario_4.R
4_1_Amenities_Scn1.R	6_5_Accesibility_Scenario_5.R
4_3_Amenities_Scn3.R	6_6_Accesibility_Scenario_6.R
4_4_Amenities_Scn4.R	6_7_Accesibility_Scenario_7.R
4_6_Amenities_Scn6.R	7_HQSL.R
5_0_GTFS_Generator.R	8_Red_Ferrea.R
6_0_Model_Calibration.R	9_Misc.R
6_1_Accesibility_Scenario_1.R	

Repository Structure




```

├── 2_Processing
│   ├── 1_Empty_Shapes
│   ├── 2_Population
│   ├── 3_EM2023
│   ├── 4_Amenities
│   │   ├── Scn1
│   │   ├── Scn3
│   │   ├── Scn4
│   │   └── Scn6
│   ├── 5_GTFS
│   │   ├── GTFS_M1-Actual_Generado
│   │   ├── GTFS_M1-Extensiones_Generado
│   │   ├── GTFS_M1-Modificada_Generado
│   │   ├── GTFS_M2-Extensiones_Generado
│   │   ├── GTFS_M2-Modificada_Generado
│   │   ├── GTFS_M3-Boyacá_Generado
│   │   ├── GTFS_M4-Cali_Generado
│   │   ├── GTFS_RE-Norte_Generado
│   │   ├── GTFS_RE-Occidente_Generado
│   │   ├── GTFS_RE-Sur_Generado
│   │   ├── GTFS_T1-K7_Generado
│   │   └── GTFS_T2-ElDorado_Generado
│   ├── 6_Accesibility
│   │   ├── Scn0
│   │   ├── Scn1
│   │   ├── Scn2
│   │   ├── Scn3
│   │   ├── Scn4
│   │   ├── Scn5
│   │   ├── Scn6
│   │   └── Scn7
│   └── 7_HQSL
│       ├── 1_Radares
│       ├── 2_Indicadores
│       ├── 3_Mapas_UPL
│       └── 4_Mapas_Ubicacion_UPL
├── 3_Results
│   ├── 1_Empty_Shapes
│   ├── 2_Population
│   │   ├── 1_Pop
│   │   └── 2_Den
│   ├── 3_EM2023
│   ├── 4_Amenities
│   │   ├── Scn1
│   │   │   ├── 1_Points
│   │   │   │   └── Detailed_Maps
│   │   │   ├── 2_HEX
│   │   │   │   └── 2_HEX
│   │   │   ├── 3_ZAT
│   │   │   │   ├── Detailed_Maps
│   │   │   │   └── 1_Points
│   │   │   ├── 4_UPL
│   │   │   │   └── 3_UPL
│   │   │   └── 5_LOC
│   │   │       └── 4_LOC
│   │   ├── Scn3
│   │   ├── Scn4
│   │   └── Scn6
│   └── 5_GTFS

```

- Scn1
- Scn2
- Scn3
- Scn4
- Scn5
- Scn6
- Scn7
- 6_Accesibility
 - Scn0
 - Scn1
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
 - Scn2
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
 - Scn3
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
 - Scn4
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
 - Scn5
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
 - Scn6
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
 - Scn7
 - 1_MNZ
 - 2_UPL
 - 3_MNZ_UPL
- 7_HQSL
 - 1_ScoreCard
 - 2_Diferencia_Escenarios
- 8_Red_Ferrea
- 9_Misc

Appendix B: Classification Table of Facilities

Classification table of facilities to be queried in OSM. Some facilities were queried under more than one OSM classification due to differences in their keys and values, which were omitted for the sake of brevity. The query procedure can be reviewed in Script `4_1_Amenities.R`, whilst the corresponding function is declared in Script `0_Initialization.R`.

No.	Amenity	Social Function						Composite Indicator			Category								
		Living	Working	Supplying	Caring	Learning	Enjoying	Well-being	Sociability	Sustainability	Education and Culture	Health and Care	Commerce and Supplying	Sport and Public Space	Work and Economic Activities	Administrative and Financial Services	Housing	Transport	Community and Spirituality
1	Apartments	✓						✓									✓		
2	Houses	✓						✓									✓		
3	Residential	✓						✓									✓		
4	Dormitories	✓						✓									✓		
5	Residential	✓						✓									✓		
6	Hotels	✓															✓		
7	Hostels	✓															✓		
8	Offices		✓												✓				
9	Government offices		✓												✓				
10	Company offices		✓												✓				
11	IT offices		✓												✓				
12	Lawyer offices		✓												✓				
13	Architect offices		✓												✓				
14	Industrial use		✓												✓				
15	Commercial use		✓												✓				
16	Crafts		✓												✓				
17	Supermarkets			✓				✓					✓						
18	Conveniences			✓				✓					✓						
19	Marketplaces			✓				✓	✓	✓			✓						
20	Bakeries			✓				✓	✓				✓						
21	Butchers			✓				✓					✓						
22	Greengroceries			✓				✓					✓						
23	Clothes shops			✓									✓						
24	Shoes shops			✓									✓						
25	Hardware shops			✓									✓						
26	Electronics shops			✓									✓						
27	Books shops			✓				✓					✓						
28	Chemist shops			✓				✓				✓							
29	Post Office			✓					✓							✓			

No.	Amenity	Social Function						Composite Indicator			Category									
		Living	Working	Supplying	Caring	Learning	Enjoying	Well-being	Sociability	Sustainability	Education and Culture	Health and Care	Commerce and Supplying	Sport and Public Space	Work and Economic Activities	Administrative and Financial Services	Housing	Transport	Community and Spirituality	
30	Banks			✓												✓				
31	ATMs			✓												✓				
32	Laundries			✓				✓	✓				✓							
33	Dry cleaning			✓				✓	✓				✓							
34	Malls			✓			✓	✓	✓				✓							
35	Hospitals				✓			✓				✓								
36	Doctors				✓			✓				✓								
37	Doctors				✓			✓				✓								
38	Clinics				✓			✓				✓								
39	Clinics				✓			✓				✓								
40	Dentists				✓			✓				✓								
41	Dentists				✓			✓				✓								
42	Pharmacies			✓	✓			✓				✓								
43	Childcare centres				✓			✓				✓								
44	Social facilities				✓			✓				✓								
45	Health laboratories				✓			✓				✓								
46	Veterinaries				✓			✓				✓								
47	Senior centres				✓			✓				✓								
48	Hairdressers				✓			✓				✓								
49	Beauty saloons				✓			✓				✓								
50	Kindergartens					✓		✓			✓									
51	Schools					✓		✓	✓		✓									
52	Colleges					✓		✓	✓		✓									
53	Universities					✓		✓	✓		✓									
54	Libraries					✓		✓	✓		✓									
55	Language schools					✓		✓	✓		✓									
56	Music schools					✓		✓			✓									
57	Training centres					✓					✓									
58	Research institutes					✓					✓									
59	Research institutes					✓					✓									
60	Theatres						✓		✓		✓									
61	Cinemas						✓		✓					✓						
62	Museums						✓		✓		✓									
63	Galleries						✓		✓		✓									
64	Arts centres						✓		✓		✓									
65	Community centres						✓		✓										✓	
66	Restaurants						✓		✓					✓						
67	Coffee shops						✓		✓					✓						
68	Bars						✓		✓					✓						

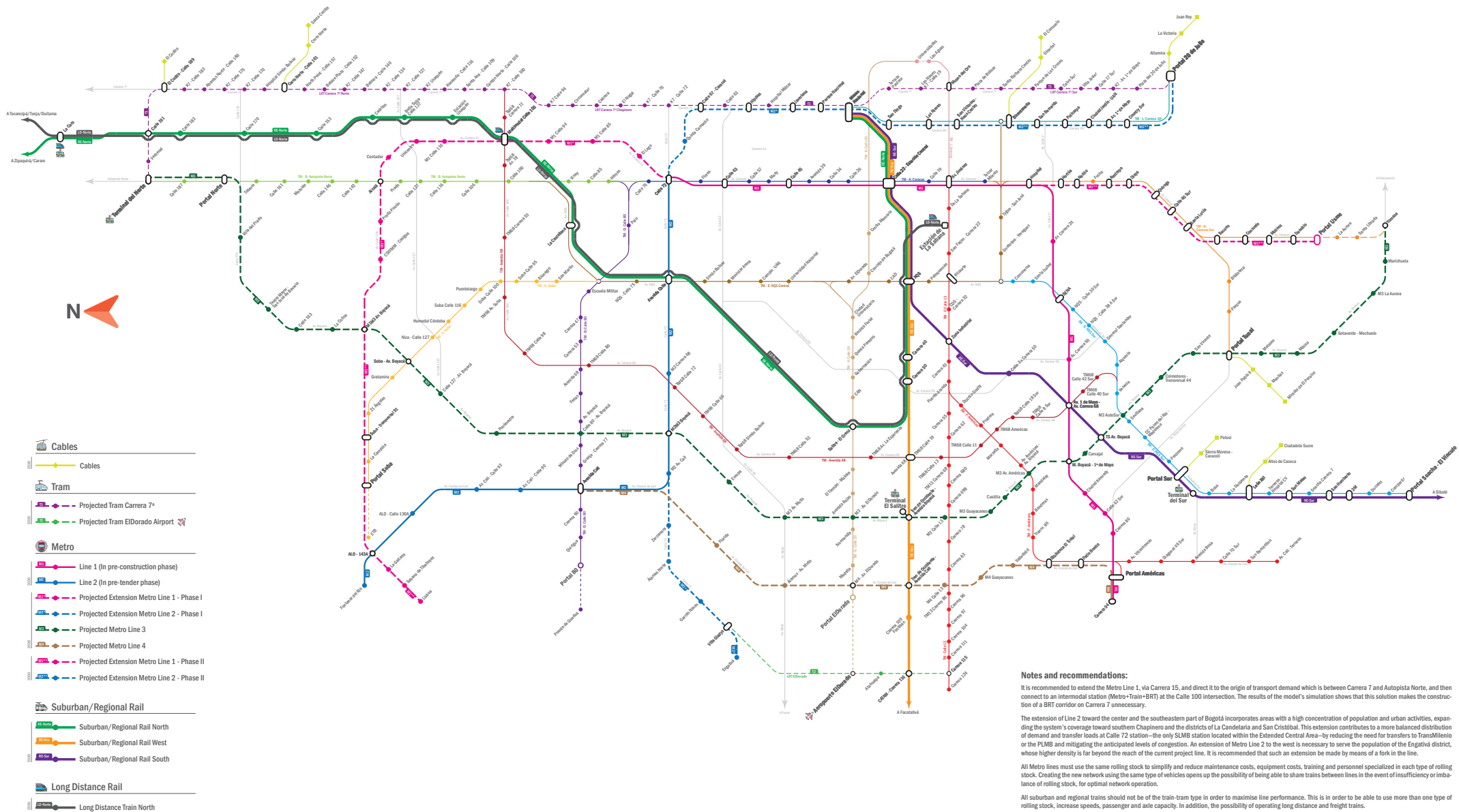
No.	Amenity	Social Function					Composite Indicator			Category									
		Living	Working	Supplying	Caring	Learning	Enjoying	Well-being	Sociability	Sustainability	Education and Culture	Health and Care	Commerce and Supplying	Sport and Public Space	Work and Economic Activities	Administrative and Financial Services	Housing	Transport	Community and Spirituality
69	Pubs						✓		✓				✓						
70	Fast food restaurants						✓	✓	✓				✓						
71	Parks						✓	✓	✓	✓			✓						
72	Playgrounds						✓		✓				✓						
73	Gardens						✓	✓		✓			✓						
74	Sports centres						✓	✓	✓				✓						
75	Stadiums						✓	✓	✓				✓						
76	Pitch fields						✓	✓					✓						
77	Swimming pools						✓	✓					✓						
78	Fitness centres						✓	✓					✓						
79	Recreation grounds						✓		✓				✓						
80	Places of worship						✓		✓										✓
81	Squares						✓		✓				✓						
82	Nature reserves						✓	✓		✓			✓						
83	Forests						✓	✓		✓			✓						
84	Meadows						✓	✓		✓			✓						
85	Bicycle rental						✓			✓			✓					✓	
86	Bicycle parkings						✓			✓								✓	
87	Bicycle shops						✓			✓		✓							
88	Public transport stations	✓								✓								✓	
89	Public transport stations	✓								✓								✓	
90	Recycling points	✓								✓						✓			
91	Train stations	✓								✓								✓	
92	Public transport stations	✓								✓								✓	
93	Public transport stations	✓								✓								✓	
94	Airports									✓								✓	
95	Ferry terminals	✓								✓								✓	
96	Cycleways						✓			✓								✓	
97	Parkings																	✓	
98	Police stations							✓								✓			
99	Fire stations				✓			✓								✓			
100	Embassies															✓			
101	Consulates															✓			
102	Prisons															✓			

Table 3. Classification table of facilities to be queried in OSM by Social Function, Composite Indicator, and Category. Source: Author's own elaboration.

Appendix C

Proposed Public Transit Network for Bogotá | Schematic Map

Long distance rail + Suburban rail + Metro + Tram+ BRT + Cables



Notes and recommendations:

It is recommended to extend the Metro Line 1, via Carrera 15, and direct it to the origin of transport demand which is between Carrera 7 and Autopista Norte, and then connect to an intermodal station (Metro+Train+BRT) at the Calle 100 intersection. The results of the model's simulation shows that this solution makes the construction of a BRT corridor on Carrera 7 unnecessary.

The extension of Line 2 toward the center and the southeastern part of Bogotá incorporates areas with a high concentration of population and urban activities, expanding the system's coverage toward southern Chapinero and the districts of La Candelaria and San Cristóbal. This extension contributes to a more balanced distribution of demand and transfer loads at Calle 72 station—the only SLMB station located within the Extended Central Area—by reducing the need for transfers to TransMilenio or the PLMB and mitigating the anticipated levels of congestion. An extension of Metro Line 2 to the west is necessary to serve the population of the Engativá district, whose higher density is far beyond the reach of the current project line. It is recommended that such an extension be made by means of a fork in the line.

All Metro lines must use the same rolling stock to simplify and reduce maintenance costs, equipment costs, training and personnel specialized in each type of rolling stock. Creating the new network using the same type of vehicles opens up the possibility of being able to share trains between lines in the event of insufficiency or imbalance of rolling stock, for optimal network operation.

All suburban and regional trains should not be of the train-tram type in order to maximise line performance. This is in order to be able to use more than one type of rolling stock, increase speeds, passenger and axle capacity. In addition, the possibility of operating long distance and freight trains.